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Review Article

Modeling and performance analysis of phase change materials in advanced thermal energy storage systems: A comprehensive review

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ABSTRACT

Phase change materials (PCMs) are crucial for efficient energy storage, yet their inherent challenges include low thermal conductivity, limited latent heat capacity, and potential leakage. This review comprehensively examines advanced thermal energy storage (TES), focusing on latent and hybrid systems. Innovative methods to enhance PCM performance are explored, such as incorporating fins, nanoparticles, porous matrices, multiple PCMs, encapsulation, and stabilization techniques. Various numerical simulation methods (Enthalpy, Heat Capacity, Finite Difference, etc.) are detailed for precise modeling of thermal storage systems. The study investigates diverse composite PCMs (NePCM, hybrid, encapsulated, porous), supported by tables summarizing studies on shell-and-tube heat exchanger modifications, hybrid tank investigations, and fin geometries. By critically analyzing recent advancements, this review provides new insights into the effectiveness of enhancement techniques and the role of numerical modeling in optimizing TES performance. Unlike previous reviews, this work integrates a comparative evaluation of hybrid TES systems, highlighting their advantages over conventional latent and sensible storage methods. Additionally, it bridges the gap between experimental studies and numerical modeling, providing engineers with practical guidelines for optimizing PCM-based TES designs. This synthesis of data is essential for researchers and engineers in sustainable energy systems. Emphasizing PCM's pivotal role in energy efficiency, the study underscores the urgent need for innovative solutions. Future research should focus on enhancing thermal conductivity, minimizing leakage, and improving long-term stability to maximize the practicality and efficiency of PCM-based TES systems.

1. Introduction

The rise in living standards and rapid technological advancements has led to an escalating demand for power, resulting in the extensive use of fossil fuels. Unfortunately, fossil fuel consumption contributes to environmental issues, including global warming due to greenhouse gas emissions. Additionally, the depletion of fossil fuel reserves is inevitable, prompting a global shift towards renewable energy sources. Among these, solar energy stands out as a promising solution to address building heating and cooling needs and provide hot water for industrial and domestic purposes [1]. In large-scale industrial thermal systems and power generation applications, thermal energy storage is the predominant choice. This strategic approach addresses the intermittent nature of solar energy and plays a pivotal role in meeting diverse energy needs [2]. The key element in solar energy utilization systems is the hot water storage tank, where heated water is stored during the day and released as required [3,4]. Thermal energy storage offers three primary methods for storing energy: sensible heat storage, latent heat storage, and thermochemical energy storage. This allows for thermal energy storage through sensible heat, latent heat, and chemical reactions, with each method serving distinct purposes in thermal energy storage [5]. Among these technologies, phase change materials (PCMs) stand out as highly efficient techniques in latent thermal energy storage applications [6].

Latent heat thermal energy storage (LHTES) stands as a fundamental method in TES, distinguished by its high energy density and consistent operational temperature range [7]. PCMs leverage latent heat for efficient heat energy storage and release, enhancing solar energy utilization. PCMs, known for their accessibility, wide temperature range, and cost-effectiveness, contribute to increased energy storage capacity in

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Nomenclature

C_P	Heat capacity (J/K)
$C_{P,\ eff}$	Equivalent specific heat capacity (J/kg.K)
d_P	Nanoparticles diameter (m)
f	Liquid fraction (%)
G	Gravity (m/s^2)
H	Enthalpy (J)
h	Sensible enthalpy (J)
k	Thermal conductivity (W/m.K)
L_f	Latent heat capacity (J/kg)
S	Source term
T_{ref}	Reference temperature (K)
T_{fr}	Freezing temperature (K)
u_r	Velocity in the radial direction (m/s)
u_θ	Velocity in theta direction (m/s)
u_ϕ	Velocity in azimuthal direction (m/s)
u_f	PCM velocity

Greek symbols

α	Thermal Diffusivity (m^2/s)
β	Thermal expansion coefficient(1/K)
μ	Dynamic viscosity of water (Pa·s)
ρ	Density(kg/m^3)
ϕ	Solid particles volume fraction
ξ	Constant

Abbreviations

CFD	Computational Fluid Dynamics
DNS	Direct numerical simulation
DHW	Domestic hot water
HTF	Heat transfer fluid
LBM	Lattice Boltzmann method
LHTES	Latent Heat Thermal Energy Storage
LTE	local thermal equilibrium
LTNE	local thermal non-equilibrium
MF	Metal foam

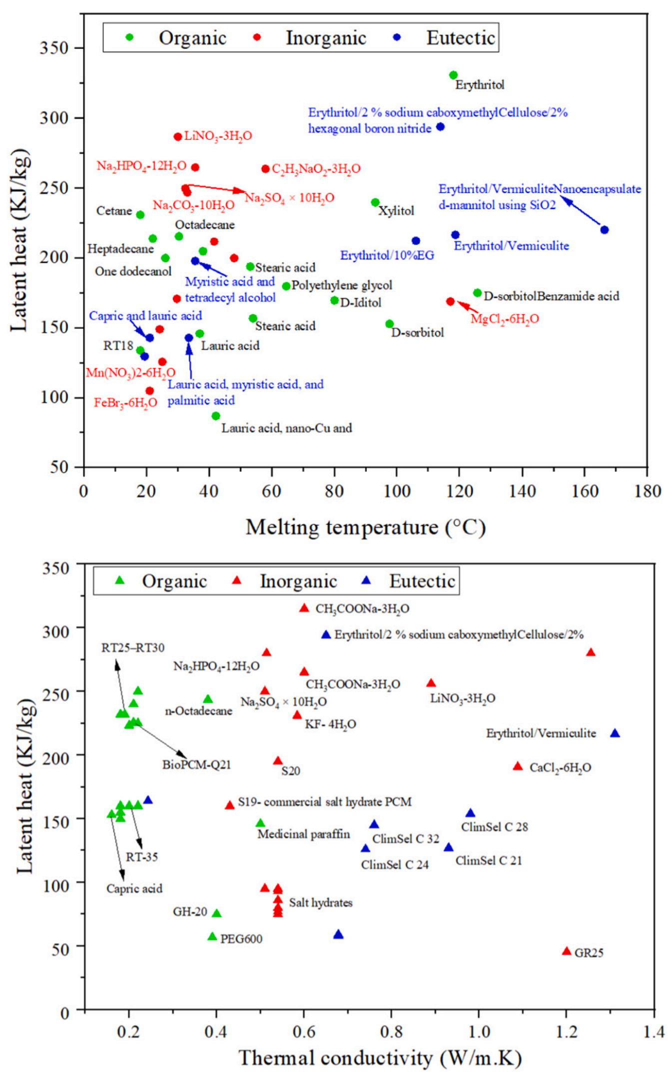


Fig. 1. Variation of specific latent heat of common PCMs:(A)- as a function of melting temperature, (B)- as a function of their solid phase thermal conductivity [18,19,23].

thermal systems. Integrating PCMs into thermal energy storage tanks, particularly in hot water tanks, is pivotal for optimizing energy storage and retrieval rates [8]. The design of hot water tanks holds crucial significance in thermal energy systems [9,10].

PCMs can be categorized into organic, inorganic, and composite types [11–13]. Organic PCMs, which include paraffins, fatty acids, alcohols, and esters, offer advantages such as a broad phase change temperature range, stable chemical properties, and minimal phase separation. However, they also exhibit drawbacks like low thermal conductivity, significant volume change, and high flammability [14,15]. In contrast, inorganic PCMs, comprising inorganic salts, crystalline hydrated salts, molten salts, and metal alloys, are known for their low cost and high latent heat capacity. Nevertheless, their widespread application is hindered by phase separation, corrosion, and poor thermal stability [16,17]. Composite PCMs, combining two or more materials, offer solutions to some limitations. These materials have a lower melting point than their components and, when heated, melt and absorb heat, then solidify and release heat upon cooling. This eutectic composition provides a sharp melting point, enabling more efficient thermal energy storage and release. By integrating different types of PCMs, composite materials can leverage the benefits of each component while mitigating some of their drawbacks. Fig. 1-(A) illustrates the variation of specific latent heat of common PCMs as a function of melting temperature for organic, inorganic, and eutectic PCMs [18,19].

Critical factors in selecting PCMs for any application include thermal conductivity, phase change temperature, high latent heat capacity, and long-term cycle stability [20,21]. Research has highlighted paraffin wax as a particularly advantageous choice due to its excellent thermal properties, which include high latent-heat capacity, low vapor pressure in its molten state, and strong thermal and chemical stability. However, a significant challenge with paraffin wax is its limited thermal conductivity, which ranges from 0.15 to 0.24 W/m.K [22]. Fig. 1-(B) illustrates the variation of specific latent heat of common PCMs as a function of thermal conductivity for organic, inorganic, and eutectic PCMs [18,19,23].

Various strategies have been proposed to enhance the thermal conductivity of PCMs, especially during the melting phase, to improve heat transfer rates. These strategies can be grouped into two main categories. The first involves using extended surfaces such as fins [19] [21], and porous materials like metal foams [27,28], to increase the surface area. The second approach incorporates fine particles, known as nanoparticles, to boost thermal conductivity [25,27]. These methods aim to overcome the inherent limitations of PCMs and enhance their performance in thermal energy storage applications.

Table 1
Literature reviews on PCM-based thermal energy storage.

Authors	Key content	Ref.
Sheikh et al.	This review covers encapsulation methods, shell materials, thermal properties, and applications of EPCMs, offering valuable insights for researchers and developers for different applications, including thermal energy storage and management systems.	[42]
He et al.	This paper reviews the performance and optimization approaches of packed-bed latent thermal energy storage systems (PLTES) using spherical PCM capsules (PLTES-SC)	[43]
Zayed et al.	This literature review examines the advancements in phase change materials (PCMs) and cascaded thermal storage PCMs (CTSPCMs) in solar water collector (SWC) storage tanks, emphasizing their impact on heat transfer improvement and storage efficiency.	[44]
Zhang et al.	The paper reviews recent experimental and numerical investigations on phase change heat transfer in porous shape-stabilized phase change materials (SS-PCMs), highlighting paraffin and metal foam as commonly used materials.	[45]
Rashid et al.	This review explores the widespread applications of phase change materials (PCMs) in various solar energy systems, emphasizing their role in enhancing energy storage efficiency. Analyzing recent literature from 2016 to 2023 highlights the growing maturity of PCM integration into solar technologies and suggests avenues for future research.	[46]
Shirbani et al.	This study reviews numerical simulations of PCMs using the lattice Boltzmann method (LBM), highlighting its suitability for modeling complex fluid flows and thermal interactions in PCM systems.	[47]
Asgharian et al.	This article reviews the fundamentals and applications of PCMs while discussing recent advances in additive manufacturing for latent heat thermal energy storage TES systems. It offers insights into co-optimizing material properties and device geometry for enhanced performance and reduced manufacturing time.	[48]
Jayathunga et al.	The paper reviews PCM's crucial role in CSP plants, highlighting its potential to mitigate greenhouse gas emissions and support UN Sustainable Development Goals.	[49]
Eslami et al.	The paper comprehensively reviews fin designs for LHTES systems, highlighting the importance of geometrical parameters such as length, number, and spacing.	[50]
Mao et al.	The paper reviews geometrical configurations of thermal energy storage tanks for concentrating solar power plants.	[51]
Present study	This study contributes novel insights into latent heat storage and hybrid thermal energy systems, focusing on numerical and experimental approaches. It provides a comprehensive overview of numerical modeling, heat transfer enhancement methods, thermophysical properties, applications, and future aspects of composite PCMs.	–

To efficiently analyze the various factors involved in systems-level implementation, tools powered by artificial intelligence (AI) and machine learning (ML) can offer effective selection methodologies. These technologies can process extensive datasets to enhance the selection process of PCMs and forecast their performance in different applications [32]. Additionally, scaling up manufacturing and automating processes are crucial for implementing thermal storage on an industrial scale. Recent progress in polymer and metal additive manufacturing, particularly when combined with topological optimization (TO), can reveal innovative geometries and intricate structures tailored for specific applications [33].

Numerous research reviews delve into the realm of thermal energy storage utilizing phase change materials. Nazari et al. [34] investigated the nanotechnology's effects on PCMs, revealing enhanced heat transfer rates and reduced charge/discharge times. The study underscored factors like nano-particle concentration and operating conditions impacting nano-PCMs' melting performance. Asgharian et al. [35] conducted a comprehensive review of PCM-based systems in solar energy storage applications, analyzing various mathematical models and numerical methods. They identified research gaps and emphasized the versatile use of PCMs. Kalapala et al. [36] comprehensively analyzed PCM-based heat exchangers, emphasizing shell-and-tube and triple concentric tube types. The review explores operating conditions, design parameters, and heat transfer techniques, offering valuable insights for optimal system design. The review by Xiong et al. [37] delved into recent numerical studies on nano-enhanced PCMs (NePCMs) for TES. It evaluated simulation techniques, explored phase change behavior, and identified research gaps for future directions. Cui et al. [38] presented a review highlighting the potential of metal foam (MF) in enhancing heat transfer for PCMs in LHTES systems. The analysis covered scientometric insights, impact factors, and applications, emphasizing the need for continued research and innovation in this field. The review by Hamidi et al. [39] examined the recent advancements and challenges in computational methods for simulating heat transfer in latent heat thermal energy storage systems using metal foams to enhance PCM thermal conductivity. The analysis encompasses software, geometry, model dimensions, boundary conditions, and porosity considerations. The review underscored the prevalence of the Lattice Boltzmann method (LBM), highlighting its accuracy in capturing complex foam geometries despite higher computational costs. Zhang et al.'s review [40] examined methods for solving phase transition processes, emphasizing the use of CFD software like FLUENT. The impact of natural convection and nanoparticles on melting time was discussed, along with challenges and prospects in numerical simulations. Ghasemi et al. [41] investigated a range of PCMs, covering nano/microencapsulated PCMs, PCM slurries, coating materials, fabrication methods, numerical models, and recent advances. Key findings include environmental considerations in chemical encapsulation, the prevalent use of paraffin wax for affordability, and the specific suitability of inorganic PCMs and eutectics for high-temperature applications and refrigeration. The review noted common models like enthalpy-porous and apparent specific heat for latent heat simulations and identifies research gaps, providing suggestions for future investigations. Table 1 presents summaries of other notable reviews on thermal energy storage systems.

1.1. Analysis focus point

This review aims to fill the existing gaps by providing a comprehensive exploration of advanced TES technologies, with a specific focus on latent heat storage and hybrid systems combining both sensible and latent heat storage. While numerous studies have investigated PCM-based TES, there remains a lack of a unified analysis that systematically compares different enhancement techniques, numerical approaches, and experimental findings. Additionally, existing reviews often focus on either numerical modeling or experimental methods in isolation, without integrating both to provide a holistic perspective. This paper addresses this gap by critically analyzing innovative methods to enhance PCM performance, and examining various numerical and experimental studies to offer a targeted understanding of PCM-based energy storage.

1.2. Novelty of the current analysis

1. This review uniquely integrates theoretical, numerical, and experimental approaches, bridging the gap between research and engineering applications.
2. Detailed investigation of innovative methods to enhance PCM performance, such as fins, nanoparticles, porous matrices, multiple PCMs, encapsulation, and shape stabilization.
3. Emphasis on hybrid TES systems, including their engineering feasibility and real-world implementation strategies.

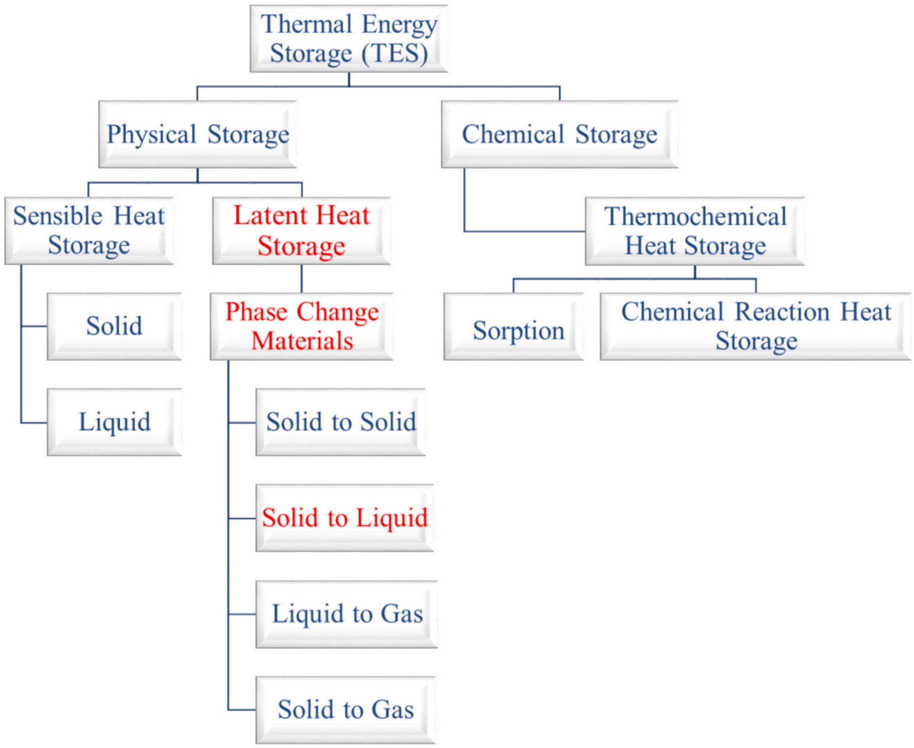


Fig. 2. Classifications of thermal energy storage methods.

4. Discussion on various numerical simulation methods including Enthalpy, Enthalpy-Porosity, Heat Capacity, Finite Difference, Finite Volume, Lattice Boltzmann, Finite Element, and Molecular Dynamics.
5. Examination of thermophysical properties of composite PCMs, such as single NePCM, hybrid NePCM, encapsulated PCM (EPCM), and porous PCM.
6. Comparison with sensible heat storage systems and thermo-chemical energy storage, emphasizing the efficiency and practicality of latent heat storage systems.
7. Emphasis on optimizing PCM behavior and system performance through advanced simulation methods and detailed study of composite PCMs.
8. Inclusion of summarizing tables that provide essential information on various aspects like shell-and-tube heat exchanger modifications, numerical and experimental investigations of hybrid storage tanks, definitions of key dimensionless parameters, and different geometries of fins for PCM storage.
9. Direct application to engineering by providing design insights for optimizing TES configurations, improving system performance, and reducing operational costs.
10. Highlighting the need for innovative solutions to optimize PCM performance and providing a basis for future research aimed at advancing numerical simulation techniques and enhancing PCM performance in sustainable energy systems. Feasible alternatives to our approach include sensible heat storage systems, which store thermal energy through temperature changes without phase transitions, and thermochemical energy storage, which involves reversible chemical reactions. However, latent heat storage systems offer higher energy density and more stable thermal outputs, making them more efficient and practical for space-limited applications.

2. Thermal energy storage

Solar energy is incredibly promising as a renewable resource, given its adaptability to transform into different energy forms like thermal and electrical energy [52]. However, its dependency on daylight and intermittent availability present notable challenges [53]. To address this, TES becomes crucial, storing surplus energy collected during the day for later release at night [54]. Three primary TES types are distinguished by how they absorb and release heat energy through their distinct physical mechanisms: sensible, latent, and thermo-chemical energy storage methods (Fig. 2) [55,56].

2.1. Sensible heat storage

Sensible TES stores the energy by continuously increasing temperature without a phase change [57]. Sensible heat finds storage potential within solid or liquid mediums [58]. Sensible TES relies on materials with high specific heat and substantial size. It requires a significant temperature variance between the heat transfer fluid (HTF) and the storage material for effective operation [16]. The primary limitation of sensible heat storage (SHS) is its reliance on temperature increases to store greater amounts of heat. For higher energy storage capacities, a correspondingly larger temperature rise is required. This non-isothermal storage accumulates an amount of heat proportional to its mass and specific heat capacity, as presented in the Eq. (1) [59]:

$$Q = m.c_p.\Delta T \tag{1}$$

where,

Q represents the stored heat [J], m represents the mass [kg], c_p stands for the specific heat capacity at constant pressure [J/ kg. K], and ΔT denotes the temperature difference [K] between an initial state and a final state.

Three main solutions coexist storage within fluids, storage within solids, and storage within a system composed partially of both.

2.1.1. Thermal storage in fluids

This system works by separating hot and cold fluids in a tank. When hot fluid goes in at the top, it displaces the colder fluid, which is then

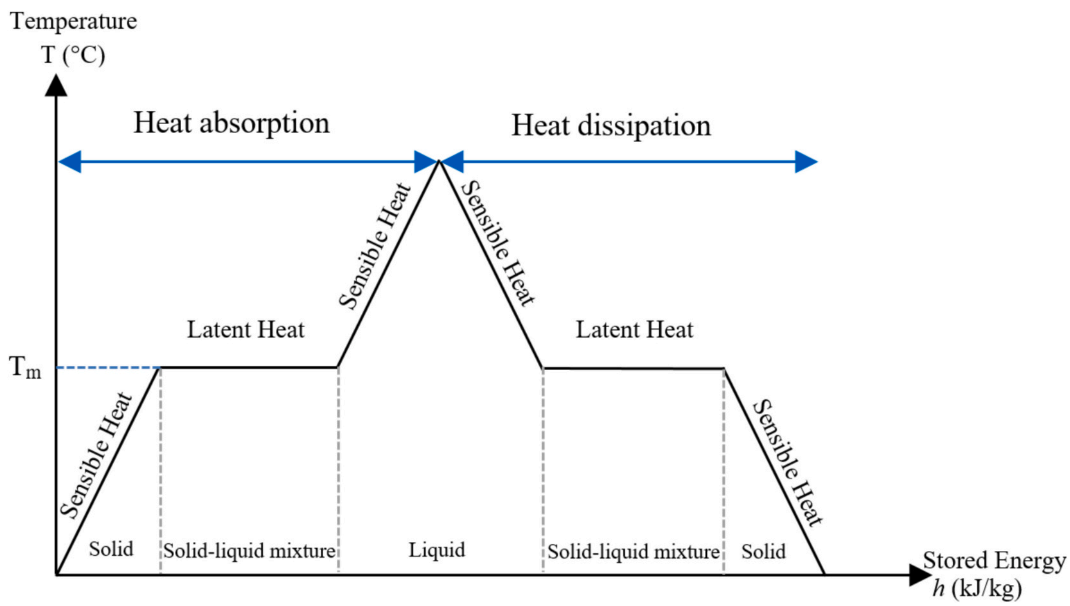


Fig. 3. PCM phase change transition.

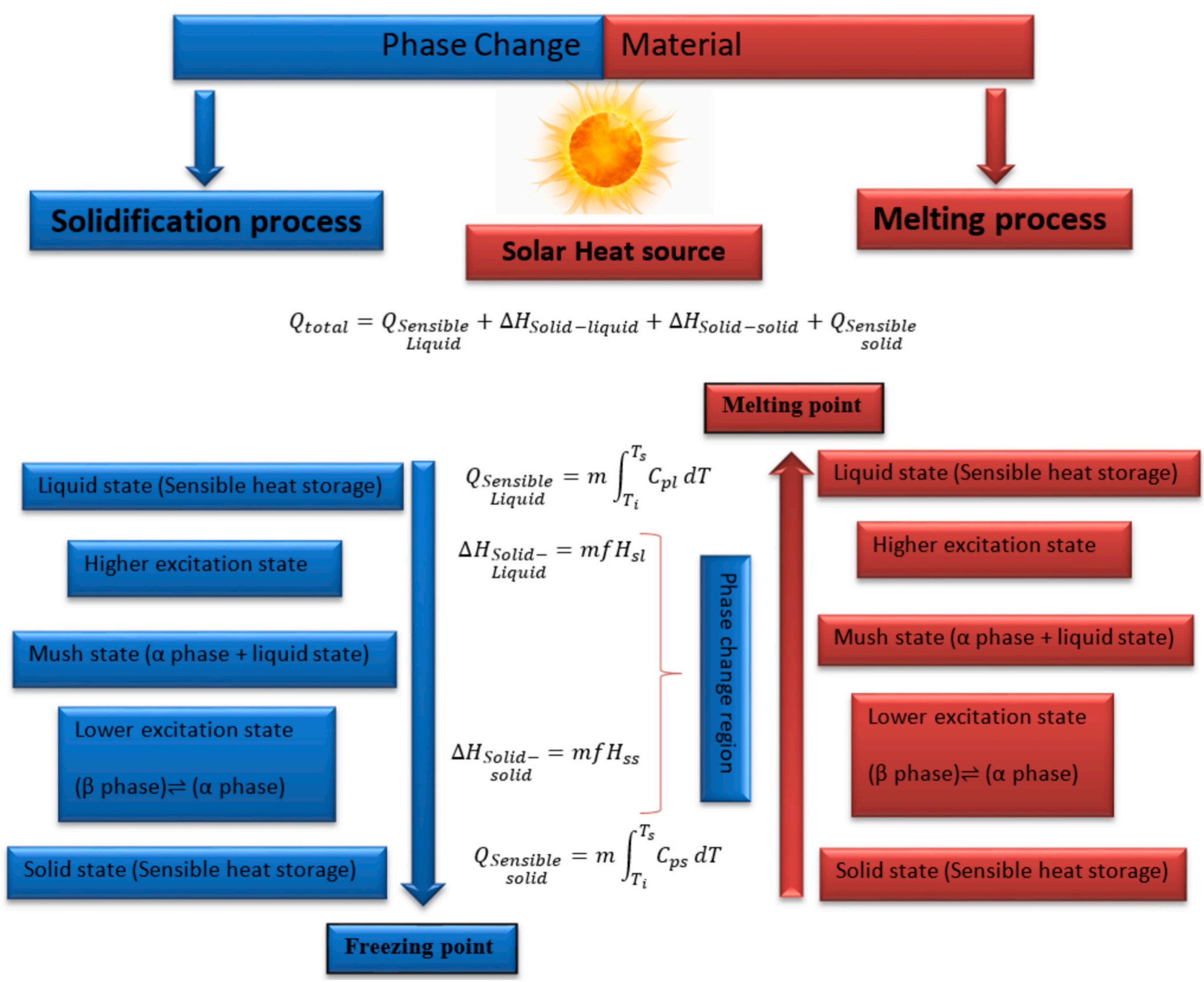


Fig. 4. Optimal phase transition diagram for PCM.

removed from the bottom. Similarly, when cold fluid is introduced from the bottom, it displaces the warmer fluid, which is then collected from the top. The challenge is to keep these layers intact while adding or removing fluid. These setups are known as "*thermocline storage systems*" [60].

2.1.2. Thermal storage in solids

For low-temperature storage, there is a method where energy is stored directly in the ground using geothermal probes or Borehole TES. Alternatively, some systems store energy in rock reservoirs or vitrified waste. The idea is that a hot fluid moves around or through solid materials, transferring its energy to them. A cold fluid is circulated opposite to access the stored energy [60,61].

2.1.3. Hybrid thermal storage

Mixed technologies combine solid and liquid materials for storage, as seen in solar thermal power plants using oil and rock storage [62]. To reduce oil costs, the reservoir is filled with rocks matching the volumetric storage density of oil. The process involves injecting hot oil at the top during charging and cold oil at the bottom during discharging, utilizing the physical phenomenon of the thermocline. Alsolen (Alcen Group) has developed and proposed this direct oil/rock contact technology [63].

2.2. Latent heat storage

Latent heat storage involves heat accumulation during phase transitions from solid to liquid, liquid to gas, or even solid-to-solid transitions. This process requires heating a material until it reaches its phase change temperature, absorbing heat for the latent heat of fusion or vaporization transformation. Conversely, when the material in its liquid or gas form is cooled, it reverts to its solid or liquid phase, releasing its latent heat [64]. PCMs are commonly used for latent heat storage due to their high energy density at constant temperatures; the most common is the solid-liquid PCMs, which have a relatively high melting latent heat and a low volume variation during latent heat exchange (often <10 %) making them ideal for various thermal applications [21]. PCMs, utilized in solid-liquid transitions, offer relatively high melting latent heat and minimal volume variation during heat exchange. For effective latent TES systems, three essential components are required: a suitable PCM with an appropriate melting point, an efficient heat exchange surface, and a compatible container [65].

For a specific PCM, the melting temperature (T_m) is the theoretical temperature at which the material undergoes a phase transition between solid and liquid states. In engineering applications, the phase change is typically considered isothermal for pure PCM [66]. However, in reality, for most PCMs used in engineering, the phase change occurs within a temperature range around the melting temperature (Fig. 3).

Fig. 4 represents the ideal phase change diagram, the lower excitation state marks the point of PCM undergoing a solid-solid phase change, involving structural shifts like the β-phase preceding the α -phase, with temperatures below the melting point. The mush state comprises a fraction of the α-phase and a minimal liquid state portion. In contrast, the higher excitation state contains a marginal α-phase fraction and a predominant liquid state. PCM's total heat storage capacity, crucial for thermal storage systems, encompasses pre-sensible heat, sandwiched total latent heat, and post-sensible heat [67–69]. In brief, the β-phase at low temperatures transforms into the high-temperature α-phase during solid-solid phase conversion [29,70]. Consequently, the mush zone is expected to include fractions of the α -phase and the liquid phase until the melting point is reached, completing the transition from solid to fully liquid states, known as the heat of fusion. To simplify, the heat of transformation (latent heat during solid-solid phase change) and the heat of fusion are combined, resulting in the overall thermal effect and total latent heat storage/release. In summary, the phase change process of PCM involves two temperature regimes: the transition temperature initiating solid-solid phase interconversions and the melting temperature concluding solid-liquid phase interconversions, leading to a standalone liquid state [71].

Experimental studies using differential scanning calorimetry (DSC), such as those by Rolka et al. [72], have focused on measuring latent heat in PCMs used for low-temperature thermal energy storage. They investigated materials like RT15 and RT22 HC, studying varying heating/cooling rates and their effects on measured latent heat capacity, which are crucial for practical PCM applications. Furthermore, numerical simulations conducted by Meng et al. [73] explored the impact of specific heat capacity, thermal conductivity, and latent heat of phase change on shell and tube heat storage units via numerical simulations. They found that higher specific heat capacity slows PCM temperature rise and melting rate while increasing overall heat storage. A 50 % increase in specific heat capacity raised the average heat storage rate by 4 %. Increasing latent heat by 50 % improved average heat storage by 6 %, and a 1.5-fold increase in thermal conductivity nearly doubled the average heat storage rate, though total heat storage was minimally affected.

2.3. Thermochemical heat storage

Thermochemical storage uses energy from reversible chemical reactions, storing the resulting heat, usually derived from splitting chemical components. This stored heat can be retrieved when those components are synthesized again. For efficient and compact storage, the reactants must be solid or liquid. The outcomes can be gases, liquids, or solids. If the reaction involves a gas, the storage system might use a fluidized bed reactor. This method allows for storing the obtained products for later use, making it useful for storing energy across seasons [74]. Let A, B, and C be three distinct chemical compounds, and Q a quantity of thermal energy, the reaction is written as:

$$A + Q \rightleftharpoons B + C \tag{2}$$

The products resulting from the reaction can be kept at room temperature and, if properly isolated, experience minimal chemical degradation. Unlike "sensible" and "latent" storage, where thermal losses bring the system back to its discharge, thermochemical storage is not as limited by storage duration. This makes it particularly well-suited for long-term (seasonal) storage. Schmidt et al. [75] employed a thermochemical reaction to store seasonal thermal energy by converting calcium hydroxide into calcium oxide and water. The reaction can be described as follows:

$$Ca(OH)_2 \leftrightarrow CaO + H_2O \tag{3}$$

2.4. Comparison of storage systems

In a broad context, thermochemical systems boast an energy density roughly five times greater than phase change systems. Additionally, the heat capacity of latent heat is around 5 to 14 times higher than that of sensible heat [76], further emphasizing the storage densities, which are two to three times greater than sensible systems. Thermochemical storage presents numerous advantages, including almost indefinite storage time, simplified transport, and superior storage density compared to alternative technologies. However, a significant challenge remains in establishing a continuous process for storage or retrieval, as current operations are conducted in discrete units. Industrially, only sensible enthalpy systems have achieved maturity, proving beneficial particularly when substantial temperature differences exist between working temperatures (heating/cooling). To minimize thermal losses, these systems necessitate substantial mass, resulting in a low surface-to-volume ratio. While phase change storage has not reached full maturity, advancements in laboratory-scale development have resulted in the creation of initial prototypes. Phase change storage becomes more appealing in scenarios with lower temperature variances compared to

Table 2
Comparison of thermal storage systems and energy densities [77–80].

Sensible TES	Specific heat capacity Cp (kJ/kg.k)	Energy density (kWh/m^3)	Advantages	Disadvantages
	1.3	61	• Inexpensive • Simple engineering	
Sands - rock	2.6	56		• Low energy density
Mineral oil	4.18	70		
Water	0.85	52		• Insulation is essential
Reinforced concrete				
Latent TES	**Latent heat L (kJ/kg)**	**Energy density (kWh/m^3)**	**Advantages**	**Disadvantages**
			• Medium to high energy density	• Insulation is essential
Maleic acid	235	103		
Erythritol	340	137	• Absorption and release of energy at constant temperature	• Complex at a technical level
$NaNO_3$	172	108		
Thermochemical TES	**Enthalpy of reaction ΔH_r (kJ/mol)**	**Energy density (kWh/m^3)**	**Advantages**	**Disadvantages**
			• Possesses an exceptionally high energy density, surpassing both sensible and latent heat	
$MgH_2 + \Delta H_r \leftrightarrow Mg + H_2$	−75 (for a charging pressure changing from 380 to 1 bar, and for discharge, it changes from 230 to 4 bar)			
		430	• No insulation is needed	• Expensive
	−178 (for a charging pressure of 700 bar, and discharge of 650 bar)			• Restricted reversibility
$CaCO_3 + \Delta H_r \leftrightarrow CaO + CO_2$				• Complicated engineering
		110		

sensible energy systems. Integrating thermal storage within a heat network offers various potential advantages, but effective utilization requires strategic consideration of the type and placement of storage to determine system functionality and operational conditions. Table 2 summarizes the advantages and disadvantages of various thermal storage systems. It also presents the energy densities each storage system achieves for select materials.

3. Phase change materials

Phase Change Materials (PCMs) serve as crucial elements in LHTES applications, enabling the storage of heat efficiently. PCMs, whether pure substances or mixtures, play a pivotal role in this process. However, the range of pure compounds available is limited, which poses a challenge in matching specific melting temperatures required for different applications. For instance, meeting the storage temperature range of 60–90 °C necessary for heating water and spaces in buildings is particularly challenging due to the scarcity of compounds melting within this interval [81]. These materials exhibit their latent heat storage capacity during phase transitions, such as solid-to-liquid or liquid-to-gas transformations. PCMs are valued for their energy-saving attributes and fall within the realm of clean energy sources. They are broadly categorized based on their physical transitions, delineating their heat absorption and desorption characteristics. PCMs encompass four primary groups: solid-to-solid, solid-to-liquid, gas-to-solid, and liquid-to-gas transitions. Among these, solid-to-liquid PCMs are extensively utilized in thermal applications due to several advantages, including high energy density, minimal volume change, availability, environmental friendliness, and affordability [56].

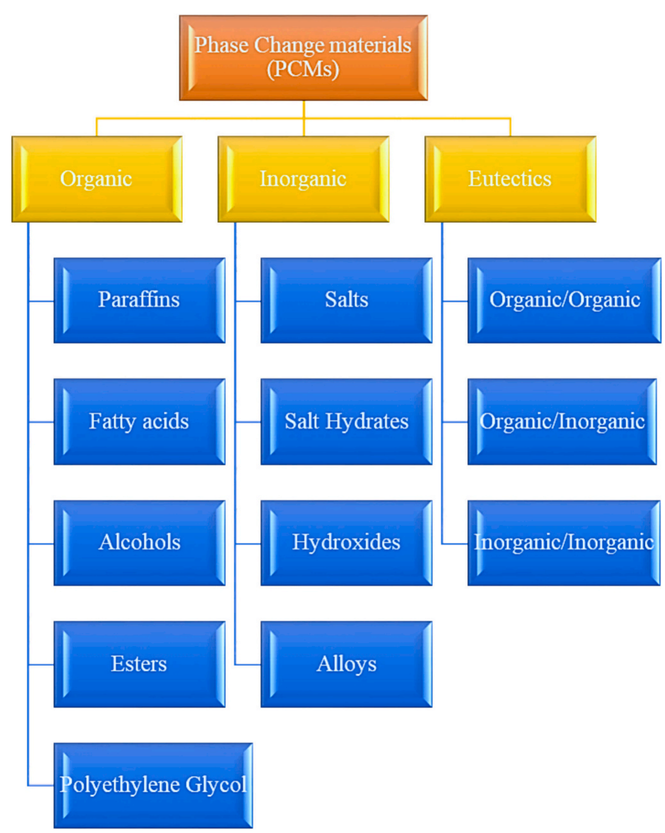


Fig. 5. Classification of phase change materials.

Table 3
List of organic PCM.

Material	Latent heat (J/g)	Temperature range (°C)	Ref
Paraffin	~ 200	20–60	[83]
Lauric acid	146	37	[84]
Myristic acid	168	50–70	[84]
Stearic acid	181	52–58	[84]
Palmitic acid	206	59–64	[85]
Sebacic acid	222	128–133	[86]
Hexadecane	236	28	[87]
Heptadecane	214	22	[87]
Cetane	231	18	[88]
Paraffin RT18	134	18	[89]
n-Nonadecane	243.5	27–32	[90]
n-Tetradecane (C-14)	229	6	[91–93]
n-Pentadecane (C-15)	206	10	
n-Octadecane (C-18)	28.4	244	
n-Eicozane (C-20)	36.5	247	
Heneicozane (C-21)	40.2	213	
n-Docozane (C-22)	44	249	
n-Tetracozane (C-24)	50.6	255	
n-Pentacozane (C-25)	53.5	238	
n-Hexacozane (C-26)	56.3	256	
n-Heptacozane (C-27)	58.8	236	
n-Oktacozane (C-28)	41.2	254	

3.1. PCM classifications

Heat can be stored in various forms, including sensible heat, stored in inert materials; latent heat, utilizing PCMs that store energy during phase transitions; and heat from reactions, involving thermochemistry and absorption processes. Fig. 5 illustrates a classification of PCMs, and can present this classification with details as follows:

3.1.1. Organic PCMs

Paraffin-based organics, like Paraffin waxes recognized with an empirical formula as C_nH_{2n+2}, where 'n' represents the number of carbon atoms [8], and non-paraffin-based organics, encompass polymeric materials such as polyethylene glycols and fatty acids. These materials, particularly paraffin wax and fatty acids, are extensively studied and applied due to their specific melting point ranges. They present the most practical and best-performing materials for PCMs in solar hot water applications. These PCMs exhibit solid-liquid phase changes, enabling fixed-temperature transitions and significant latent heat absorption or release. Despite their advantages of no phase separation and subcooling, organic PCMs encounter challenges like low thermal conductivity, limited latent heat of melting, and potential leakage. Hydrocarbon-based PCMs, inclusive of paraffin waxes and fatty acids, provide high heat of fusion, ensuring substantial TES upon melting or freezing. While generally safe in terms of toxicity and corrosiveness [82], their flammability can restrict usage in specific contexts. Additionally, their slow heat transfers due to low thermal conductivity contrasts with their wide availability across various melting temperatures, contributing to their versatility across diverse applications, as outlined in Table 3 showcasing various organic PCMs and their respective melting temperatures.

3.1.2. Inorganic PCMs

Inorganic PCMs are derived from non-carbon-based substances, encompassing metals, salts, and ceramics [94]. These materials, comprising salts or metals, offer substantial TES capabilities owing to their elevated heat of fusion and excellent thermal conductivity, facilitating rapid heat absorption and release. Despite these advantages, inorganic PCMs encounter several limitations. One notable issue is the occurrence of supercooling in many inorganic PCMs, leading them to cool below their freezing point before solidifying, posing challenges in certain applications. Moreover, corrosiveness and potential phase segregation are inherent concerns in these PCMs, where components may separate over time, impacting their efficiency and reliability. Salt hydrates, a subset of inorganic PCMs, consist of salt and water molecules bound together, offering high thermal conductivity and significant latent heat storage capacity at a relatively low material cost. However, their drawbacks include reduced thermal stability, corrosiveness, and potential recrystallization issues after multiple cycles, rendering them unable to store further heat efficiently. In contrast, metallic PCMs like gallium, known for their low melting points, serve as effective inorganic PCMs, offsetting some of the negative traits associated with salt hydrates. These metals exhibit rapid TES and release due to their high heat diffusivity. Compared to organic PCMs, metallic PCMs offer superior heat conductivity and minimal volume change. Nonetheless, their primary drawback lies in their low heat of fusion per unit weight, resulting in higher overall weight for devices and thus limiting their consideration for PCM use.

3.1.3. Eutectic PCMs

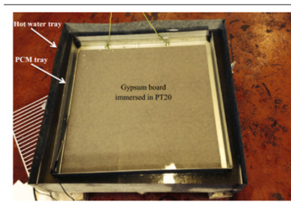
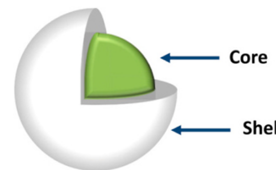
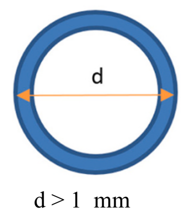
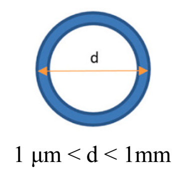
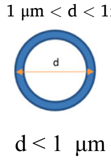
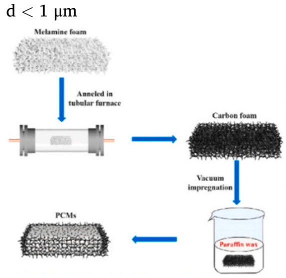
Eutectic PCMs combine two or more substances to achieve a lower melting point than the individual materials. This property efficiently absorbs and releases heat upon melting and solidification. The specific eutectic composition provides a well-defined melting point, enhancing the PCM's ability to effectively store and release thermal energy. By leveraging eutectic PCMs, research aims to overcome the limitations of both organic and inorganic PCMs while combining their respective advantages. Adjusting the mass fractions of components enables customization of phase change temperatures, thermal conductivity, and latent heat values, thereby expanding the applicability of PCM technology. However, challenges persist, including limited thermal properties at high temperatures, reduced latent heat values post-eutectic formation, material leakage issues, and high production costs. To address these limitations, ongoing research focuses on leveraging carbon-based materials to enhance thermal conductivity, mitigate leakage problems, and improve overall performance. Carbon-based materials offer promising solutions due to their improved environmental impact and better dispersion in both solid and liquid states. Despite current limitations, leveraging carbon-based materials is a promising avenue for enhancing eutectic PCM technology for broader applications [95]. PCMs can be integrated into materials through diverse methods to enhance TES efficiency, as detailed in Table 4.

Jebasingh et al. [96] formulated a ternary eutectic mixture composed of caprylic acid, myristic acid, and palmitic acid (CA-MA-PA) with a mass ratio of 64.8:22.6:12.6. To enhance its thermal properties for energy storage, they incorporated exfoliated graphite (xG) composites into the PCM. The addition of 10 % xG improved the thermal conductivity of the composite PCM from 0.149 W/(m.K) to 0.180 W/(m·K), reflecting a 20.8 % increase. The composite also demonstrated melting and curing temperatures of 17.1 °C and 9.7 °C, respectively, and latent heats of 142.2 J/g and 139.5 J/g, with relatively low fabrication costs.

3.2. PCM selection criteria and applications across temperature ranges

These materials are available across various transition temperature ranges, each presenting distinct thermo-physical characteristics impacting their TES capacity. Hence, carefully selecting PCMs remains pivotal in optimizing the thermal performance of any TES system. The selection process for PCMs involves a comprehensive evaluation of several factors, including thermophysical, kinetic, chemical, environmental, and economic considerations. Key parameters like phase change temperature, thermal conductivity, latent heat of phase change, compatibility with encapsulation materials, and material flammability play vital roles in this assessment [56]. Among these parameters, the phase change temperature emerges as the primary criterion, dictating PCM selection. Choosing PCMs aligned with specific working temperature ranges for different applications is crucial; for instance, domestic refrigerators operate between −20 °C to 5 °C [105,106], and buildings and electronics typically require 5 °C to 40 °C [24,25,107], and solar water heating functions around 40 °C to 80 °C [108]. This variance in

Table 4
Summary of incorporation methods.

Incorporation method	Brief description	Examples	Examples ref
-Impregnation method	✓ The procedure involves placing solid PCM, metal foam, and a supporting mesh in a container, evacuated using a vacuum pump. Heating causes the metal foam to immerse into the molten PCM until the support is saturated. Cooling solidifies the PCM, allowing for the removal of excess PCM after slightly reheating the container for specimen retrieval [45].		[97]
- Encapsulation method	✓ Encapsulation of PCMs involves enclosing them within a protective shell or capsule, addressing concerns such as leakage, environmental reactivity, and toxicity when directly incorporated into thermal systems. This method effectively coats the PCM, preventing potential leakage that could compromise system efficiency and performance. Encapsulation stands as the preferred approach for integrating PCMs into thermal systems, ensuring their containment and safeguarding their functionality.		[42]
• Macroencapsulation	✓ Macroencapsulation involves enclosing a PCM within a shell material that exceeds 5 mm in size [98]. This method is favored for PCM encapsulation primarily due to its cost-effectiveness. When macroencapsulated PCM is utilized in various applications, it contributes to improving the system's latent heat storage capacity, consequently impacting the overall temperature control.	 d > 1 mm	[102]
• Microencapsulation	✓ Microencapsulation involves utilizing a PCM as the core for energy storage, encased within inorganic and/or organic polymers forming a protective shell (supporting materials) [99,100] and forming 0.5–5000 μm particles. This technique serves to prevent PCM leakage, bolster thermo-mechanical performance, and offer heightened defense against external environmental factors [101].	 1 μm < d < 1mm	[102]
• Nanoencapsulation	✓ Nano-encapsulation involves enclosing a substance within a nano-sized shell, typically ranging between 1 and 100 nm in size, and it finds diverse applications across various fields. The encapsulation process has gained significant traction, especially in PCM applications, due to the enhancement it offers in thermal properties.	 d < 1 μm	[102]
-Shape stabilization method	✓ Shape-stabilized PCM resides within a carrier matrix, presenting several advantages: leak-proof properties, enhanced thermal conductivity, high specific heat, and the capability to retain its shape across multiple phase transition cycles. Primarily advocated for organic PCMs, this method significantly bolsters thermal conductivity while mitigating leakage concerns associated with organic PCM applications [103].		[104]

working temperature highlights the necessity of tailoring PCM selection to suit distinct operational temperatures across various applications. The summarized criteria for selecting PCMs are encapsulated in Fig. 6, delineating the pivotal parameters guiding PCM choice across diverse applications. Additionally, Fig. 7 illustrates the classification of reviewed PCM applications across four distinct temperature ranges, providing a comprehensive overview of their varied utilization in different thermal scenarios [109].

4. Heat transfer enhancement methods of PCMs

While PCM boasts numerous advantages and versatile applications within thermal systems, it grapples with three notable limitations: inadequate thermal conductivity [114], phase transition leakage [115], and suboptimal solar-to-thermal conversion [116]. These challenges restrict PCM's widespread use across various thermal systems. However, addressing these concerns is feasible through several techniques. Enhancing thermal conductivity and preventing leakage is pivotal. Numerous methods exist to mitigate these drawbacks (Fig. 8), including the incorporation of fins [37,38,59–61], utilization of multiple PCM arrangements [120], integration of nanoparticles [41,63–65], employing porous metal matrices [66,67], encapsulation [56,115], and Shape stabilization of PCMs [124,125]. Implementing these techniques not only counteracts the limitations but also expands the potential applicability of PCMs across diverse thermal systems.

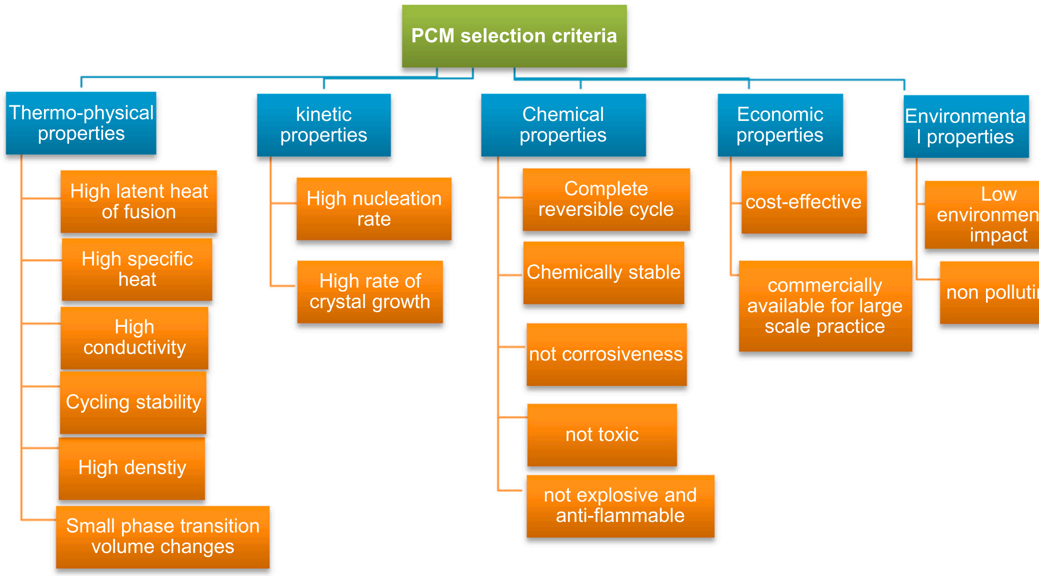


Fig. 6. Criteria for PCM selection in TES systems [110–113].

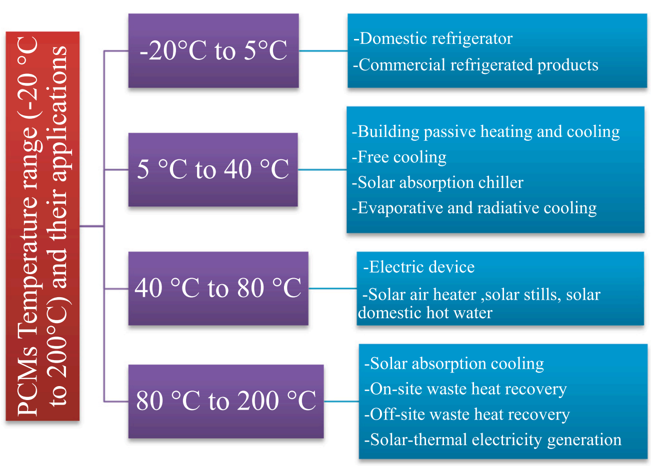


Fig. 7. Categorization of examined PCM applications across four temperature ranges [109].

4.1. Fins

PCMs boast high energy density and a consistent phase change temperature. However, their drawback lies in inefficient heat transfer during melting and freezing cycles, particularly noticeable in organic PCMs. To address this, research extensively explores the use of fins to augment contact area, thereby accelerating PCM responsiveness for heat transfer [127–129]. This method has garnered significant attention due to its promising outcomes compared to scenarios without fins [130]. Various studies by different authors have delved into this technique, highlighting its efficacy in improving heat transfer rates during PCM phase transitions [36,107,119]. While the use of fins significantly enhances heat transfer rates during PCM phase transitions, there remains a need to optimize fin design and configuration to maximize effectiveness across different operational conditions and PCM types. Notably, multitudes of fin types are detailed in Table 5, offering diverse options for implementation.

Kirincic et al. [131] study on a shell-and-tube latent heat storage system employing paraffin and water, augmented with longitudinal tube fins, demonstrates significant reductions in melting and solidification times 52 % and 43 %, respectively, compared to plain tubes. Despite fin-induced changes in PCM mass and liquid-phase convection, overall thermal performance during charging and discharging cycles remains largely unaffected. Nie et al.'s [132] research investigates PCM phase changes in a double-tube heat storage system with fins, favoring straight and fewer, longer fins for superior heat transfer enhancement in scenarios with low thermal conductivity. Mhood et al. [133] investigated PCM melting in horizontal LHTS, emphasizing the need to optimize fin configurations for enhanced thermal efficiency. Their study highlighted that taller fins and specific angles notably reduced melting time by up to 50 %, improving heat transfer in critical zones of the system. Song et al. [134] innovatively introduced a multi-tube latent heat storage (MTLHS) unit employing tree-shaped fins to enhance PCM thermal efficiency. Their study extensively compared this design with conventional ones,

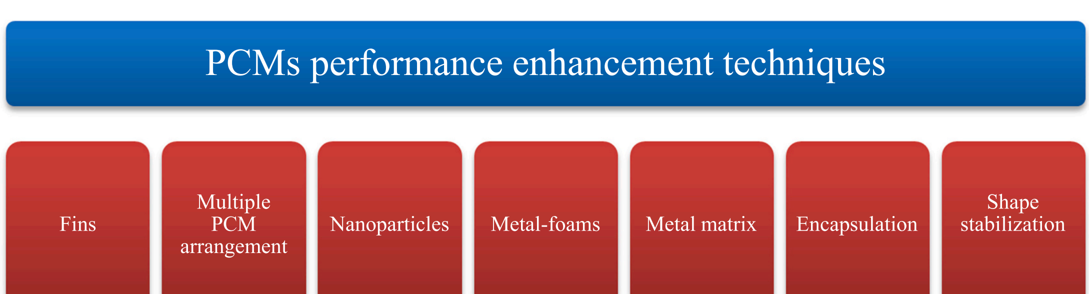
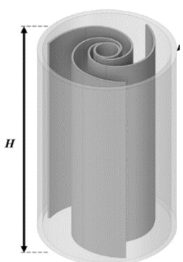
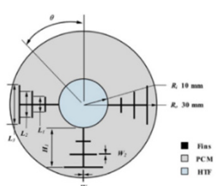
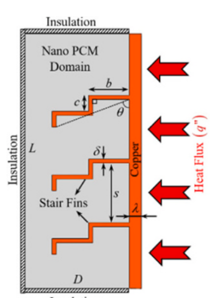
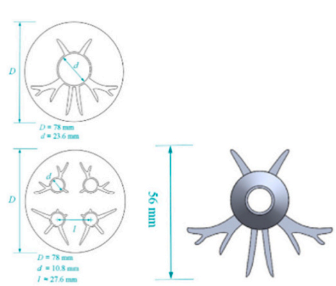


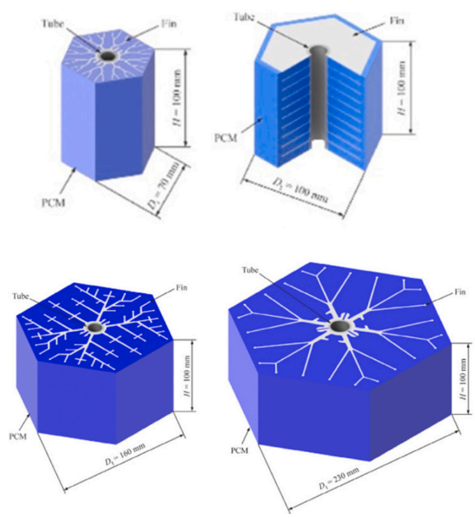
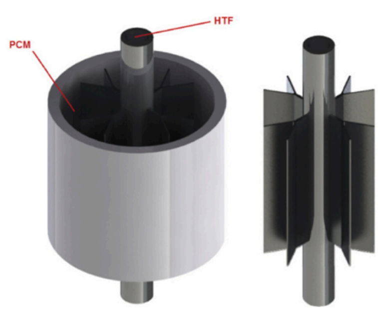
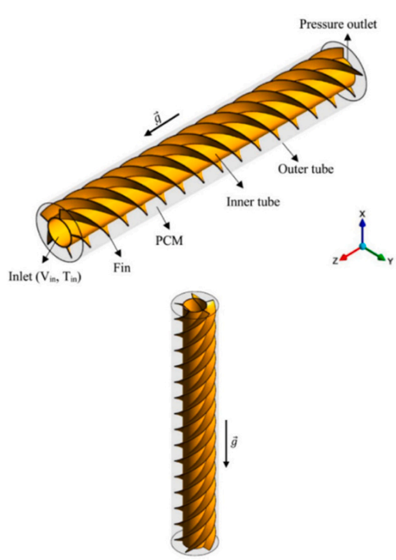
Fig. 8. Improving PCM performance [126].

Table 5
Various geometries of fins for PCM storage.

Authors	The model	Dimension type	Study	Fins names	Principal investigation
Wu et al. [136]		2D	Numerical	Spiderweb-like fins	Enhancing solidification efficiency with spiderweb-like fins in LHTES
Oskouei et al. [137]		3D	Experimental and numerical	Fibonacci-inspired fins (tree-like fins)	Enhancing LTES efficiency with Fibonacci sequence-based fins
Liu et al. [138]		2D	Numerical	Gradient fins	Enhancing LHTES efficiency with gradient fins and natural convection analysis
Nakhchi et al. [139]		2D	Numerical	Stair fins	Enhancing Latent Heat Storage with Nano-Enhanced PCM and Stair Fins
Ge et al. [140]		2D	Experimental and numerical	Topology optimized fins	Optimizing LHTES Efficiency through Topology-Optimized Fins Analysis

(*continued on next page*)

Table 5 (*continued*)

Authors	The model	Dimension type	Study	Fins names	Principal investigation
Vogel et al. [141]		3D	Numerical	Organic shape, Plate shape, Snow-flake shape, Eco shape	Natural Convection Impact on Extended Fins in Shell-and-Tube Storage Systems
Sciacovelli et al. [142]		2D	Numerical	Tree-shaped	Optimizing LHTES Efficiency via Y-Shaped Fins
Ghalambaz et al. [143]		3D	Numerical	Twisted fins	Enhancing PCM Charging Efficiency via Twisted-Fin Array Design

revealing an 80.2 % reduction in complete melting time, improved temperature uniformity, and insights into melting convection stages. Exploring HTF temperature and PCM types provided valuable insights for optimizing heat storage performance in practical applications. In their numerical investigation, Kim et al. [135] analyzed a latent heat thermal storage system with angled fins, revealing significant effects on heat transfer and energy storage performance across varying fin inclination angles ($-40^\circ \le \theta_f \le 40^\circ$). The study identifies $\theta_f = -20^\circ$ as the optimal angle, enhancing average power by up to 19.3 % compared to horizontal fins.

4.2. Multiple PCMs

Multi-PCMs LHTS units utilize multiple PCMs or a variety of them, presenting a similarity to conventional single PCM LHTS systems, except these systems differ from conventional single PCM units by employing multiple PCMs with varying melting temperatures or properties. Single PCM units, despite their simplicity, suffer from prolonged melting or freezing times, which is a significant drawback compared to systems utilizing multiple PCMs [144]. However, research indicates that employing more than three stages yields diminishing additional

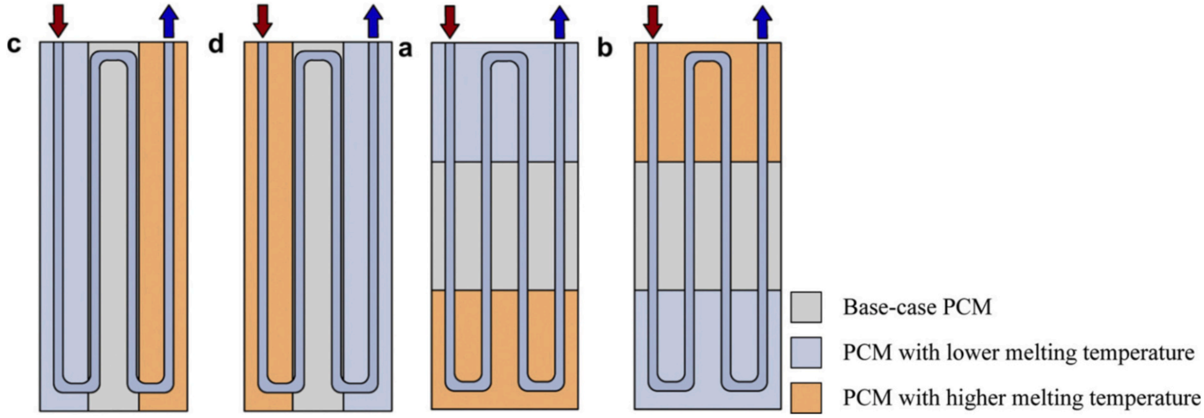


Fig. 9. Configurations of multi-PCM thermal energy storage. Reused with permission from Elsevier [147], with license number 5837011453408.

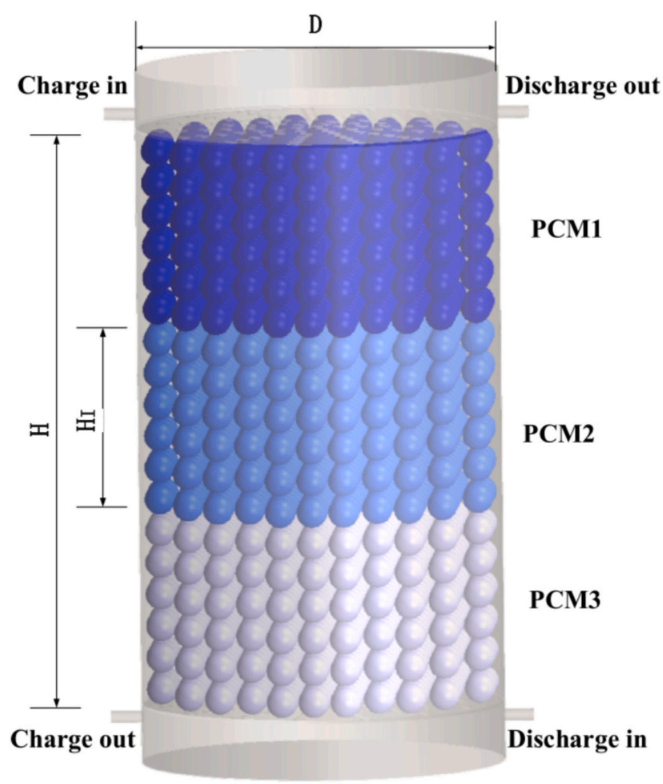


Fig. 10. Packed cascaded structure for thermal energy storage. Reused with permission from Elsevier [148], with license number 5836991091943.

improvements in system efficiency [145]. Multiple PCM latent heat thermal storage (LHTS) systems enhance heat transfer in heat exchangers by layering PCMs; authors vary the number of PCMs and the heat exchanger's orientation [146].

Kurnia et al. [147] analyze various PCM thermal storage designs, showcasing a novel festoon channel with superior heat transfer and exploring multiple PCM configurations for enhanced efficiency (Fig. 9). Gao et al. [148] devised a three-stage PCM CPBTES (cascaded packed TES) for solar heating, establishing a model via thermal resistance analysis and validating it through experiments. Their multi-objective optimization improved exergy efficiency by 5 % and reduced TES capacity by 4 %, highlighting CPBTES's superior temperature distribution and optimized parameters for enhanced performance, offering valuable insights for similar engineering challenges (Fig. 10).

4.3. Nanoparticles

Nanoparticles were employed to amplify the thermal conductivity of PCMs owing to their significant surface-to-volume ratio advantage. The enhancement in thermal conductivity of shape-stabilized PCM, integrated with nanoparticles, is contingent upon various factors: nanoparticle concentration, particle size, intermolecular interactions, aspect ratio, Brownian motion, clustering, purity, and interfacial thermal resistance [74,149,150]. Fig. 11 provides an overview of commonly employed nanoparticles in combination with PCMs. The incorporation of metals, metal oxides (such as ZnO, Al_2O_3, and TiO_2 [151,152]), and carbon-based nanomaterials into PCM composites has proven effective in substantially enhancing heat transfer rates [153–156]. Khodadadi and Hosseinizadeh [157] were the first to employ nanoparticle infusion in LHTES systems to enhance their effectiveness. They achieved significant improvement by dispersing copper nanoparticles into the water-based PCM during the solidification process, known as nanoparticle-enhanced PCM (NePCM). This method notably boosts thermal conductivity, expediting both melting and solidification rates compared to standard PCMs. Despite efforts to enhance the latent heat of phase PCMs using nanoparticles, research consistently reports degraded latent heat in experimental studies [158,159] and molecular dynamics simulations [160]. Table 6 presents a comprehensive overview of the thermal properties of nanoparticles utilized in numerical studies involving PCM, including thermal conductivity, size, density, and specific heat capacity.

Hosseinizadeh et al. [161] explored enhancing PCM solidification rates using fins and nanoparticles. Their study showcases individual technique efficacy and the combined impact, favoring fins' superior performance over nanoparticle use alone. Das et al. [162] investigate melting behavior in a shell-and-tube energy storage unit, finding higher heat transfer fluid temperatures accelerate natural convection, and graphene nanosheets notably reduce melting times by up to 41 % at 60 °C and 37 % at 70 °C with 2 vol% loading. Nakhchi et al. [139] explore CuO nanoparticles and stair-shaped fins, boosting energy storage by 9.1 % and achieving 474.1 kJ capacity. Descending stair fins, coupled with nanoparticles, enhance melting performance through improved thermal conductivity and convection effects. Singh et al. [163] conducted a numerical analysis of the melting process in a finned latent heat storage system with varying Graphene nanoplate (GNP) volume fractions. They investigated the impact of fins, GNP, and their combination on effective thermal conductivity, considering factors like aspect ratio and anisotropy. Experimental tests on sugar alcohol provided phase change parameters, and transient variations were studied for designing medium-temperature storage systems. The results showed a significant 68 % reduction in total melting time for the optimized finned system with 5 % GNP (Fig. 12).

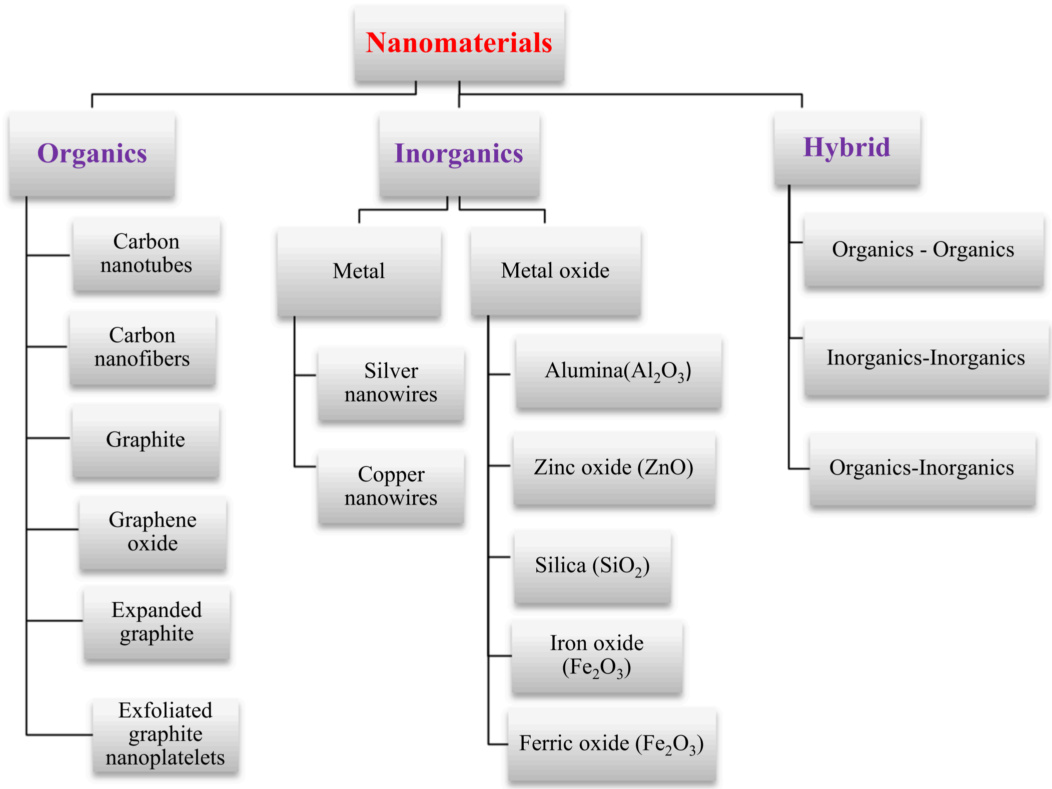


Fig. 11. Widely employed nanomaterials in composite PCM [164–169].

Table 6
Thermal properties of nanoparticles used in numerical studies with PCM [122,170–175].

Nanoparticles Materials	Symbol	Size	Density (kg/m^3)	Thermal conductivity (W/m.K)	Specific heat capacity (J/kg.K)
Aluminum Oxide	Al_2O_3	20 nm	3980	38.493	778
Silver	Ag	–	10,500	429	235
Carbon nanofiber	CNF	D:400 nm; L:50 μm	2100	–	–
Cerium(IV) Oxide	CeO_2	15–30 nm	6100	11.715	352
Cobalt Oxide	CoO	30 nm	6460	10.042	703
Copper	Cu	90 nm	8954	400	383
Copper Oxide	CuO	40–80 nm	6500	17.991	536
Gadolinium Oxide	Gd_2O_3	20–80 nm	7640	10.042	290
Graphene	GN	–	2200	5000	790.1
Graphene oxide	GO	–	1800	5000	717
Graphite nanopowder	Grp	APS:50 nm	2200	–	–
Graphene nano-platelets	GNPs	Thickness: 7 nm APS: 15 μm	100	-	-
Iron oxide	Fe_2O_3	20–40 nm	5240	12.552	628
Multi-walled carbon nanotubes	MWCNTs	–	1600	3000	796
Magnesium Oxide	MgO	35 nm	3580	61.923	921
Nickel Oxide	NiO	20 nm	6400	12.970	603
Single-walled carbon nanotube	SWCNT	–	2600	6600	425
Silicon Oxide	SiO_2	10–20 nm	2650	11.715	753
Strontium Titanate	$SrO.TiO_3$	100 nm	5110	05.858	536
Tin Oxide	SnO_2	100 nm	5560	31.380	343
Titanium Oxide	TiO_2	100 nm	4250	08.954	686
Yttrium Oxide	Y_2O_3	30–50 nm	5000	14.226	448
Zinc Oxide	ZNO	100 nm	5630	27.196	494

4.4. Porous metal matrix

Enhancing PCM thermal conductivity involves adding high thermal-conductivity fillers [176]. The incorporation of highly conductive fillers into composite PCMs offers appealing prospects, including nanoparticles [177,178], fins [179], and porous structures [27,176,177]. Among these materials, metal foam (MF) stands out for its high porosity, continuous structure, and superior thermal conductivity [180]. MF's porous nature accommodates PCMs effectively, making it a promising solution for mitigating PCM's low thermal conductivity [181]. However, their complex, tortuous geometry poses significant challenges for accurate numerical simulation. Future research should prioritize improving the realism of these simulations to better understand and optimize the performance of metal foams in TES systems [39].

Ghalambaz et al. [182] analyzed seven latent heat storage systems, finding geometry changes impact PCM behavior minimally. The frustum tube consistently outperforms the conical shell, with slight improvements in melting times for both pure PCM and composite metal foam/

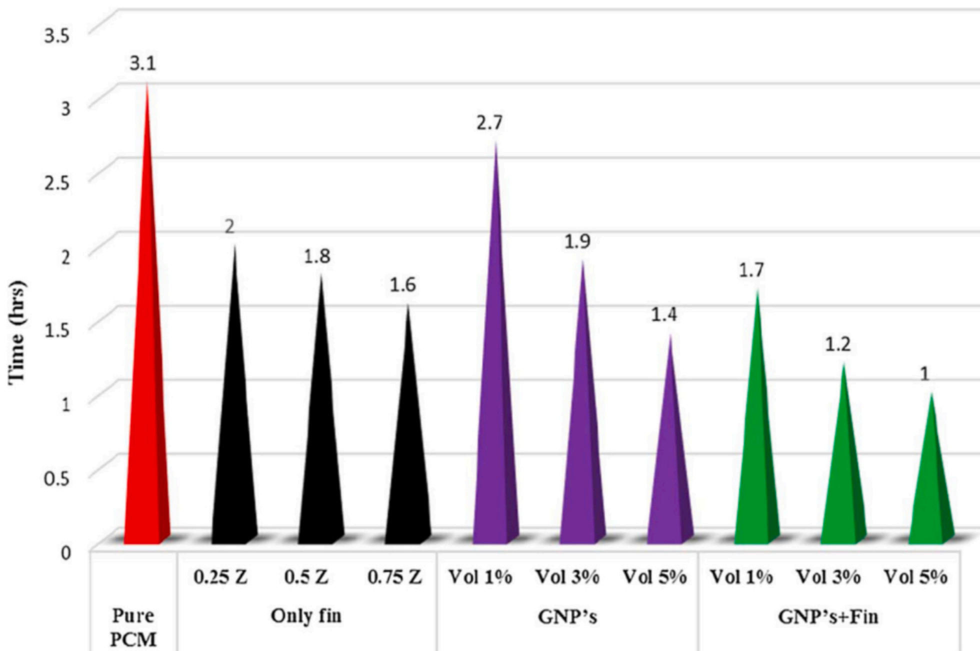


Fig. 12. Overall Melting Time with Various Heat Transfer Enhancement Techniques. Reused with permission from Elsevier [163], with license number 5837000239021.

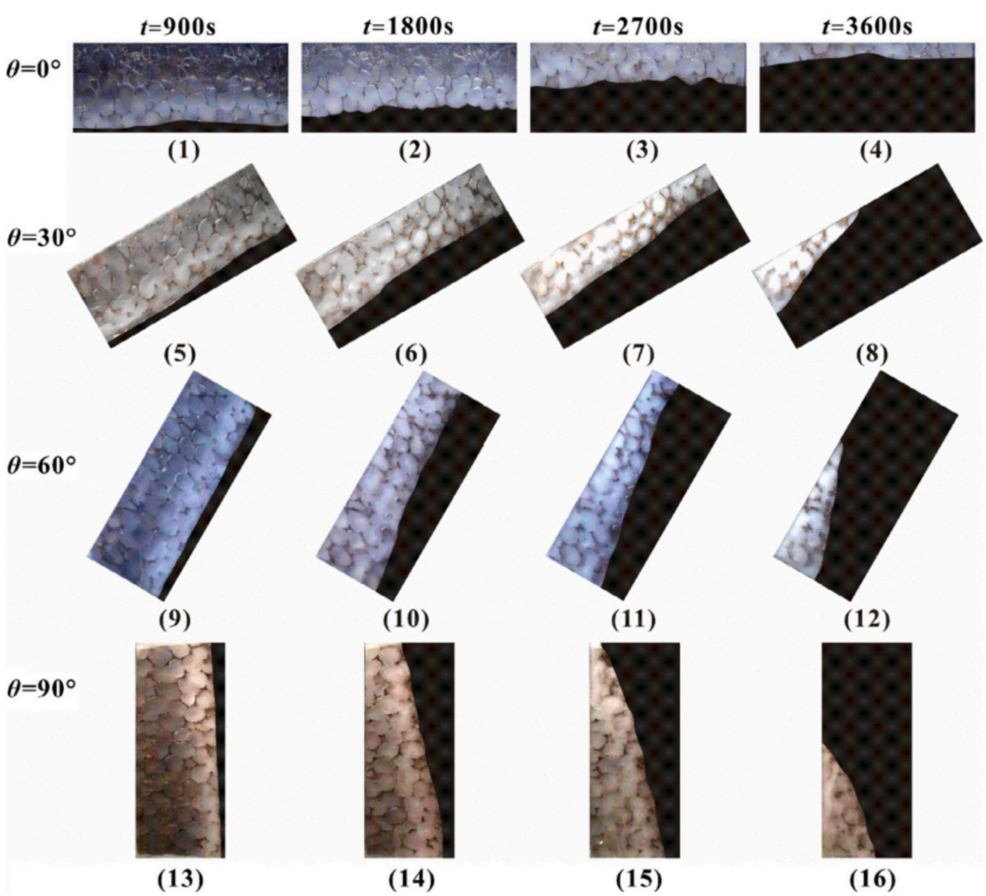


Fig. 13. Evolution of melting front in paraffin-embedded within metal foam at different time intervals and inclination angles. Reused with permission from Elsevier [185], with license number 5837000559542.

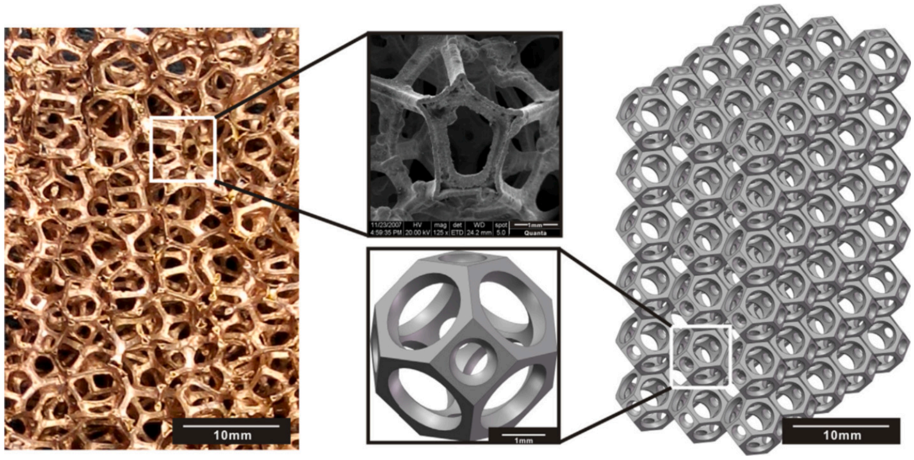


Fig. 14. Copper foam with open-cell structure and reconstructed tetrakaidecahedron-shaped cut. Reused with permission from Elsevier [186], with license number 5837000761348.

PCM. Abandani et al. [183] investigated PCM melting within a triplex heat exchanger with three-layer PCM and Al6061 metal foam. Despite initial lower rates, the metal foam significantly boosts the melting rates, and lower porosity leads to increased heat transfer. Parida et al. [184] presented a pore-scale numerical model for PCM melting in PCM-metal foam systems, capturing metal foam geometry's impact on localized heat transfer accurately. The model considers convection effects and reveals higher convection impact with increased metal foam porosity and larger pores in the composite system. Yang et al. [185] explored paraffin's melting behavior in metal foam at angles (0°, 30°, 60°, 90°). Lower angles notably decrease full melting times by 12.28 % (0°), 22.81 % (30°), and 34.21 % (60°) due to enhanced natural convection, contrasting the minimal impact of angles on melting behavior within the metal foam (Fig. 13). Yang et al. [186] conducted a comprehensive comparison of melting heat transfer in PCMs with embedded metal foam using the volume-averaged method (VAM) and direct numerical simulation (DNS). Integrating a two-temperature model in VAM, the study successfully predicted macroscopic phenomena like interface evolution and temperature distribution, emphasizing the role of natural convection. Despite losing pore-scale details, the research sheds light on the practical and economic viability of utilizing metal foam in waste heat recovery systems for residential heating or hot water (Fig. 14).

4.5. Encapsulation

PCM usage faces issues due to reactivity and volume changes, impacting effectiveness. Encapsulation counters these challenges by shielding PCMs in a protective shell, mitigating reactivity, and curbing mechanical stress during phase transitions, ensuring enhanced durability and functionality [42]. Encapsulation techniques offer solutions for PCM-related challenges like leakage, stability enhancement, and improved thermal conductivity. These methods include macro-encapsulation (above 1 mm), offering varied shapes like spherical [187], rectangular [188], tubular [189], or cylindrical [190]. Micro-encapsulation (0–1000 μm) boasts a higher surface-to-volume ratio, enhancing heat exchange [91], while nano-encapsulation (0–1000 nm) exhibits superior structural stability, showing promise in thermal storage [56,191]. The reduction in energy storage capacity is an inevitable outcome of encapsulating PCMs, leading to a decrease in their latent heat capacity and consequently affecting their TES efficiency. To counteract this challenge, researchers have explored methods such as employing thinner or lighter encapsulation materials to mitigate latent heat loss. However, opting for thinner materials raises concerns about potential PCM leakage, underscoring the need for robust solutions that balance energy retention with structural integrity [42]. However, more efforts should focus on developing encapsulation processes that are easy to operate, widely adopted, feature strong sealing, low cost, and long life. These attributes are crucial for enhancing the practical application and scalability of PCM encapsulation technologies [40].

Fig. 15. The top view of the tank incorporates macro-encapsulated PCM. Reused with permission from Elsevier [193], with license number 5837000983587.

Memon et al. [192] innovatively crafted paraffin-loaded macro-capsules within lightweight aggregates (LWA) via vacuum incorporation. Achieving remarkable outcomes, the study demonstrated a maximum absorption of 70 % paraffin within LWA. Furthermore, by infusing 15 % graphite powder, the thermal conductivity of PCM surged,

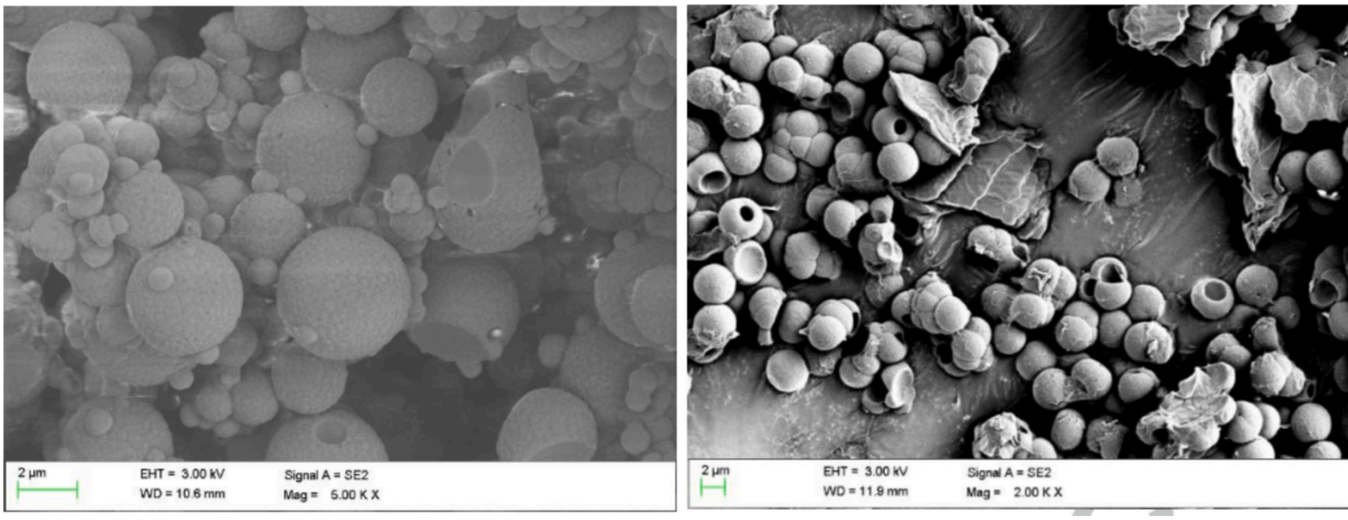


Fig. 16. SEM analysis of Micro-encapsulated PCM (MEPCM) samples. Reused with permission from Elsevier [194], with license number 5837001192573.

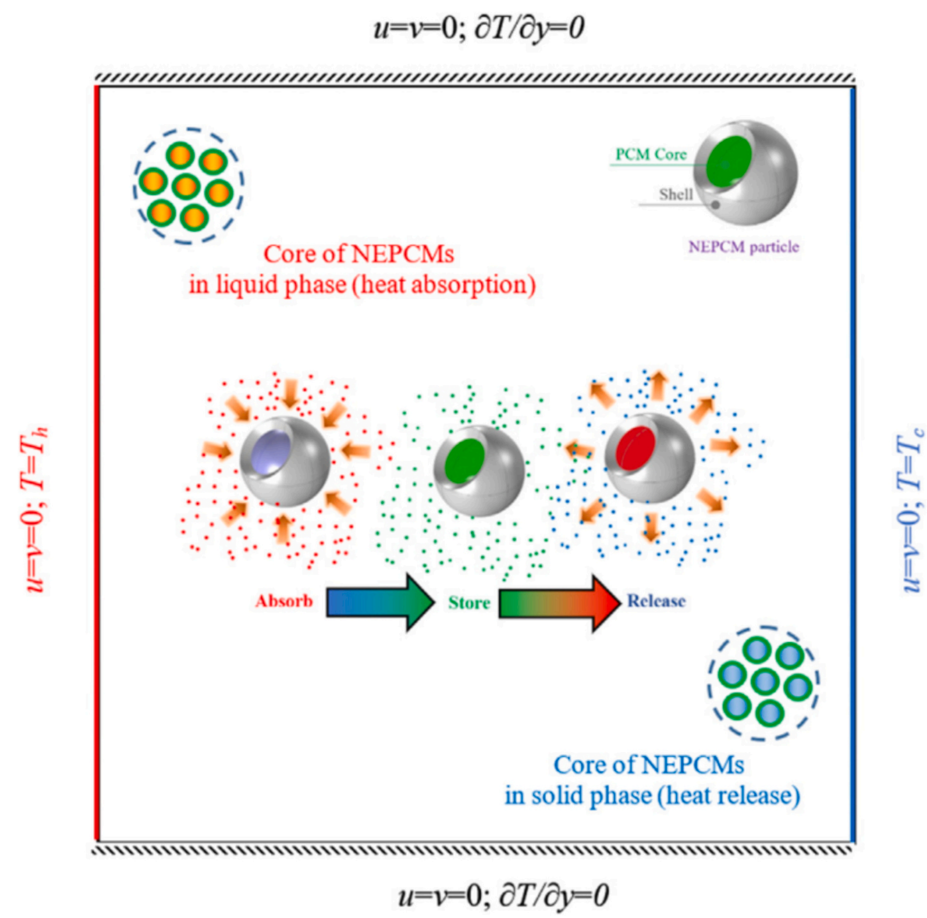


Fig. 17. Schematic representation of the physical model of nano-enhanced PCM (NEPCM). Reused with permission from Elsevier [195], with license number 5837010114916.

exhibiting a remarkable 162 % increase compared to epoxy. Neri et al. [193] studied a hybrid latent-sensible heat storage system by integrating a hot water tank with macro-encapsulated PCMs (Fig. 15). Employing three numerical models, validated through experiments, the research aims to boost overall thermal capacity. The results demonstrate that only 40 % of PCM's latent heat potential is utilized due to limitations in thermal transport properties affecting heat transfer. Su et al. [194] employed Melamine-formaldehyde (MF-3) resin to microencapsulate paraffin wax (Fig. 16), creating five distinct encapsulation types. Their study revealed improved thermal stability, particularly due to the role of binary emulsifiers, assessed via thermogravimetric analysis. The potential application of these materials in thermal storage systems remains an area for future exploration. In their study, Ghalambaz et al. [195] investigated the convective flow and heat transfer of nanoencapsulated

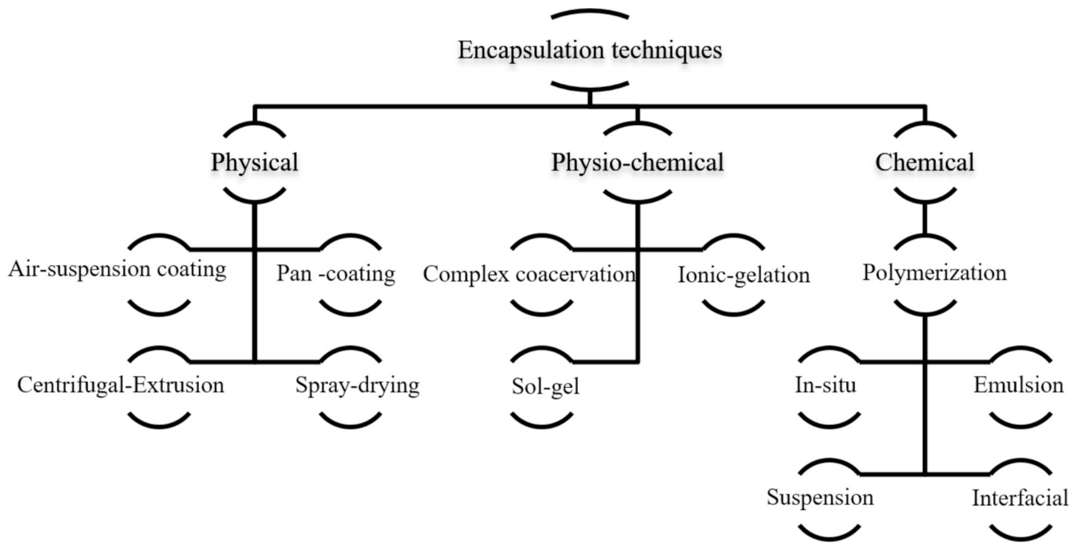


Fig. 18. PCM encapsulation methods [98].

PCM (NEPCMs) within an insulated square cavity (Fig. 17).

4.5.1. Encapsulation methods

Encapsulation methods for PCMs are typically classified into four categories: physical-mechanical, physical-chemical, chemical-mechanical, and chemical. The synthesis process of the EPCMs determines these categories. Fig. 18 illustrates the overarching classification of encapsulation methods.

4.5.1.1. Physical-mechanical methods. Typically, physical-mechanical methods do not entail chemical reactions leading to the creation of relatively large-sized microcapsules.

1. Pan Coating

Involves mixing solid particles with a dry coating material and heating them to melt the coating material, which encloses the core material (e.g., PCMs). The mixture is then solidified by a cooling medium [196].

2. Air-Suspension Coating

Solid particles are suspended in an upward air stream, and the shell material (usually polymer or inorganic material) is repeatedly sprayed onto the core particles in a cyclic process until the desired thickness is achieved for encapsulation [197].

3. Centrifugal Extrusion

Utilizes centrifugal force to produce microcapsules. The core material flows through a central tube, while the coating material flows through the annular around the tube. With rotation or vibration of the tube, the core and coating materials extrude from the orifice, forming spherical capsules due to surface tension forces [197].

4. Vibration Nozzle

Microgranulation or matrix encapsulation is carried out through a vibrating nozzle under a laminar flow regime. The vibration helps in forming uniform capsules, although some studies reported achieving stable PCM composites without vibration [198].

5. Spray Drying

Involves preparing a dispersion/emulsion of the wall material, homogenizing the dispersion/emulsion, atomizing the in-feed dispersions, and dehydrating/evaporating the atomized particles. This technique is widely used in the food and pharmaceutical industries for encapsulating heat-sensitive materials [199–201].

6. Solvent Evaporation/Extraction

Uses a liquid-liquid emulsification system where a polymer is emulsified in a volatile solvent in water, followed by solvent removal. The hydrophilicity or hydrophobicity of the core material determines the specific method used for microencapsulation [202].

7. Vacuum Impregnation

Frequently used to remove air from encapsulation materials, this technique involves impregnating the core material with the shell material under vacuum conditions. It has been used in the food industry and for preparing TES materials [192].

4.5.1.2. Physical-chemical methods. Physical-chemical methods encompass various physical reactions like phase separation, condensation, boiling, and complexation, which are utilized in three main techniques: Ionic gelation, Coacervation, and the Sol-gel method.

1. Ionic Gelation

Gel-forming solutions are transformed into droplets and immersed in a gelation bath, resulting in the formation of hydrogel beads. This process relies on ionic bonding, such as when an alginate solution is dropped into a calcium bath, leading to the formation of calcium alginate microcapsules or by cooling, as in the case of an agarose solution [203].

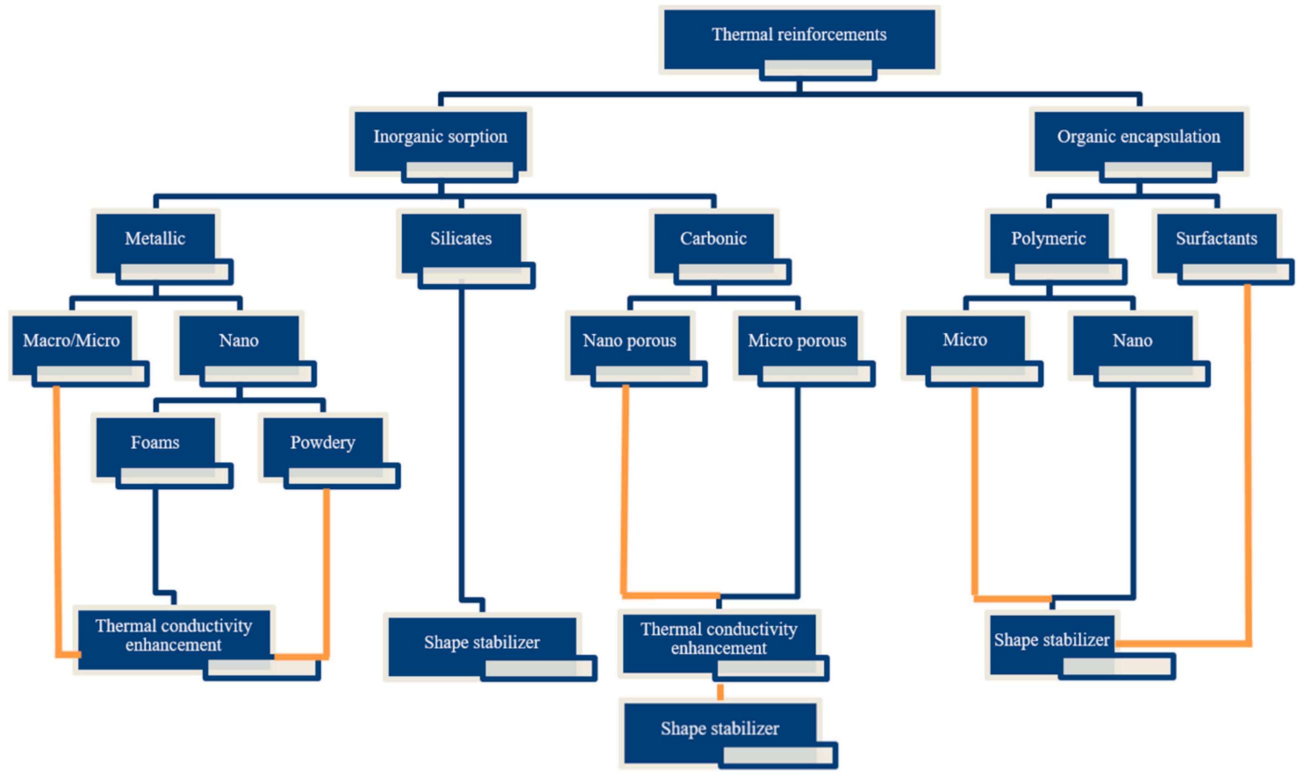


Fig. 19. Classification of additives for PCM [219].

2. Coacervation

The coacervation method involves two main types: simple coacervation and complex coacervation. In simple coacervation, a low-molecular substance interacts with a dissolved polymer, while in complex coacervation, two polymers with opposite charges interact. This method yields smaller microcapsules of spherical shape with greater stability, though it involves processes like cross-linking, desolvation, and thermal treatment, which can be more complex and costly [204,205].

3. Sol-Gel Method

The sol-gel process involves the polycondensation reactions of a molecular precursor in a liquid phase to form a colloidal solution (sol), which then undergoes conversion to an oxide network (gel). This method is commonly used to synthesize inorganic nanocapsules and is favored for its cost-effectiveness and mild processing conditions [56].

4.5.1.3. Chemical-mechanical methods. In chemical-mechanical methods, a mechanical machine is employed to facilitate the chemical reactions involved in encapsulating PCMs. This category generally comprises two main methods:

1. Microfluidic Technique

Utilizes capillary microfluidic devices to generate monodisperse double emulsion droplets. Typically applied in medical and pharmaceutical fields, this technique involves the formation of double emulsion droplets, such as water-oil-water (W-O-W) emulsions, by controlling flow motion geometry within microchannels. The resulting micro-PCMs are obtained through subsequent washing and drying processes [206,207].

2. Melt-Coaxial Electrospray Technique

A modification of the spray drying technique, this method introduces a chemical reaction during the process. Initially proposed by Loscertales et al. [208], it involves generating steady coaxial jets of immiscible liquids with micrometer/nm diameter using electrospraying. The microcapsules are formed by applying this technique with appropriate core and shell materials, leading to the encapsulation of PCMs [209].

4.5.1.4. Chemical methods. Chemical methods focus on producing smaller-sized capsules, such as nanocapsules, using organic and inorganic materials as shell materials or precursors for shell formation. The main techniques under chemical methods include:

1. Interfacial Polymerization/Polycondensation

Involves preparing oil-water (O/W) or water-oil (W/O) emulsions with appropriate emulsifiers, followed by interfacial polymerization at the interface between two phases containing suitable reaction monomers. This results in the formation of polymer capsules around the core materials (OPCMs), which are then separated from the oil or water phase [56].

2. Suspension-Like Polymerization

Accomplished in a system with two phases: the discontinuous or dispersed phase containing core material reagents and monomers, and the continuous phase containing shell material reactants and solvent. The process involves dissolving polymer monomers into core materials, forming a stable oil-water (O/W) emulsion, and then initiating polymerization reactions at a constant high temperature to produce encapsulated PCMs [210].

3. Emulsion Polymerization

Polymerization occurs in the presence of emulsifiers in an oil-based system. Insoluble monomers with emulsifiers are dispersed in the

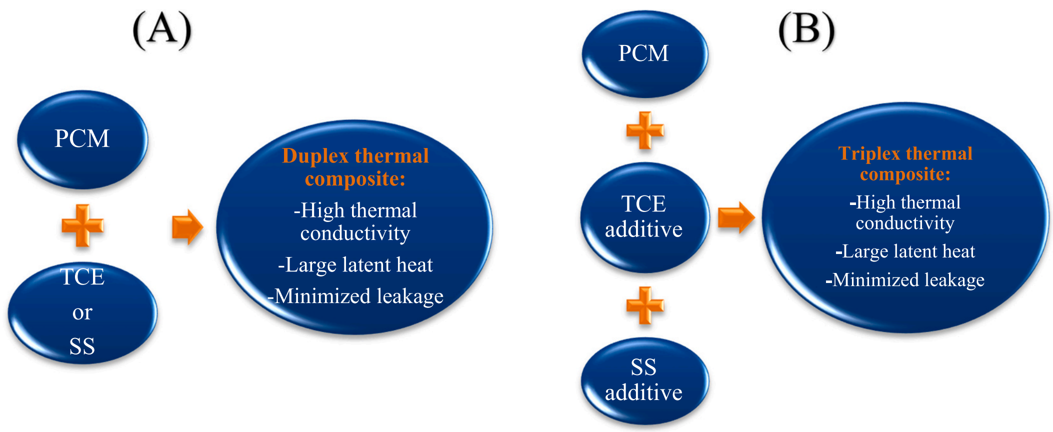


Fig. 20. (A) Duplex thermal composite **and** (B) Triplex thermal composite.

reaction medium, followed by the addition of initiators to initiate polymerization reactions and form a polymer membrane on the surface of the core material. The resulting micro−/nano-PCMs are obtained after washing and removing the oil [210].

4. Miniemulsion Polymerization

This method produces nano-PCMs by polymerizing within small droplets, which requires less input energy compared to emulsion polymerization. The process involves forming stable nanometer-sized droplets containing emulsifiers, water, monomers, surfactants, initiators, and dispersed and continuous phases. Polymerization occurs within these droplets without changing their chemical composition [211].

5. In-Situ Polymerization

Chemical reactions occur in a continuous phase of two immiscible liquids (water-soluble and oil-soluble phases). Monomers are dissolved in the continuous phase, while polymers are not soluble, leading to polymerization reactions on the surface of the core materials. The process involves forming oil-water (O/W) emulsions, preparing the prepolymer mixture liquid, mixing the emulsion and prepolymer liquid, and finally washing and drying the microcapsules/nanocapsules [212].

Each method offers unique advantages and limitations depending on the specific requirements of the encapsulation process and the properties of the core and shell materials being used.

4.6. Shape stabilization

Various additives, known as thermal reinforcements, contribute to creating an optimal thermal composite for PCMs. They can be categorized as thermal conductivity enhancers (TCEs), shape stabilizers (SS), or a combination of both. These reinforcements, like graphene, serve dual roles, resulting in duplex thermal composites. They are divided into organic and inorganic types based on their natural structures (Fig. 19). Effective reinforcements possess high thermo-physical properties and large, thermally robust bodies to encase PCM molecules for leak-proof, shape-stable PCMs. However, using triplex composites, while advantageous, can reduce latent heat. When mixed with PCM, supporting materials like porous and nano-materials form shape-stabilized composite PCMs (SS-CPCMs), overcoming thermal conductivity limitations and leakage issues (Fig. 20). Nevertheless, their TES capacity might decrease due to the limited participation of supporting material molecules in thermal storage [213–215].

Zhang et al. [213] developed an RT100/EG composite PCM containing 80 wt% RT100 as a form-stable PCM for medium-temperature latent heat storage. The composite exhibited enhanced thermal conductivity compared to RT100, increasing linearly with packing density. The RT100/EG composite PCM also displayed excellent photothermal properties, maintained thermal conductivity during melting, and demonstrated structural stability and thermal reliability over 200 heating-cooling cycles. Given its enhanced thermal conductivity, stability, and formability, this composite shows promise for medium-temperature TES. Innovatively utilizing Lewis acid catalysis, Liu et al. [216] developed a method to craft hyper-crosslinked polymers (HCPs) encapsulating PCMs, achieving up to 55 % enhanced thermal conductivity. This strategy incorporates waste polystyrene foam as support for PCM, presenting a promising approach for efficient TES. Xiao et al. [217] formulated a shape-stabilized composite PCM with 72.5 wt% $CH_3COONa\cdot 3H_2O$, 0.4 wt% Na_2HPO_4, 17.1 wt% expanded graphite and 10 wt% CuS. The addition of Na_2HPO_4 facilitated SAT crystallization, while EG prevented PCM leakage, and CuS notably boosted light-to-thermal conversion efficiency from 66.9 % to 94.1 %. This promising composite retained 194.8 J/g phase change enthalpy with minor reductions (5.9 % in enthalpy and 2.6 % in efficiency) after 150 thermal cycles, making it a viable candidate for solar-based TES. Zou et al. [218] explored the impact of different expanded graphite (EG) sizes (50, 80, and 100 mesh) on composite PCMs (CPCMs). EG-50 exhibited superior shape-stabilization and adsorption due to larger micropores and specific area, enhancing thermal conductivity and latent heat compared to EG-80 and EG-100. The study underscores the significance of carbon-based supporting materials with suitable sizes for improving thermal performance in shape-stabilized CPCMs.

5. Numerical simulation methods of PCMs

Numerical simulations are a cornerstone in PCM investigations, valued for their efficiency, stability, and cost-effectiveness. In exploring the thermal characteristics of PCMs within TES tanks, it becomes imperative to account for solid-liquid interface dynamics, solid and liquid phase heat conduction, and the impact of liquid PCM flow on heat transfer dynamics [35]. Several numerical methodologies have been utilized in PCMs within solar energy systems. These approaches, including the enthalpy method, enthalpy porosity method, finite difference, and finite volume techniques, cater to diverse PCM applications like photovoltaic systems, building envelopes, and solar thermal collectors [220].

5.1. Enthalpy method

The enthalpy method, introduced by Voller et al. [221] in 1987, employs enthalpy and temperature as core functions in solving the energy conservation law, incorporating latent heat within the enthalpy definition [222]. This approach establishes a unified energy equation across the entire heat transfer domain, enabling the determination of enthalpy distribution and subsequent identification of the two-phase interface without the necessity of tracking phase interfaces. Widely adopted due to its suitability for multi-dimensional cases, this method has found extensive application in diverse phase change heat transfer problems. Additionally, it has been favorably utilized in various studies [223,224] for simulating PCM behavior, particularly in predicting temperature distribution during phase change, convection, and diffusion processes in TES systems. Moreover, its transient form is instrumental in accurately modeling unsteady melting and solidification processes [225].

The enthalpy method exhibits several advantages: it operates swiftly with the right scheme, effectively accommodating abrupt and gradual phase changes. It offers ease of scalability and simplicity, facilitating its extension to various applications. However, drawbacks include challenges in accurately simulating phase changes involving supercooling, potential oscillations of temperature at specific grid points over time, and unclear phase change interfaces on coarse meshes. Uncertainty in physical parameters within the mushy zone poses a challenge, as does the method's limitation to incorporating only heat conduction and latent heat absorption, neglecting the impact of melted PCM flow behavior on material thermal performance [41,223,226]. According to Voller, the energy conservation during a phase-change process (melting or freezing) is articulated as per reference [227]:

$$\frac{dH}{dt} = \nabla(k(\nabla T)) \tag{4}$$

Here, H symbolizes the total volumetric enthalpy, and k denotes the thermal conductivity. H is a temperature-dependent function, described concerning the latent heat L and the local liquid fraction, represented as $f(T)$ [227]:

$$H(T) = h(T) + \rho f(T)L \tag{5}$$

Where $h(T)$ is a sensible enthalpy and is defined as:

$$h(T) = \int_{T_m}^{T} \rho c dT \tag{6}$$

In this equation, "T_m" represents the reference isothermal phase change temperature specific to a PCM. The key element in Eq. (5) is $f(T)$, establishing the connection between the liquid fraction of the PCM and its temperature. Assuming an isothermal phase change (where T equals T_m and no temperature fluctuations occur during the phase change), this function is calculated using the Heaviside step function:

$$f(T) = \begin{cases} L, T > T_m \\ 0, T < T_m \end{cases} \tag{7}$$

Based on the equations provided, when PCM transitions to a liquid phase, the total volumetric enthalpy equals the latent and sensible heat combination. Conversely, if the PCM remains in a solid state, the enthalpy is zero. Employing Eqs. (4) and (5), an alternative formulation for the heat transfer within the PCM can be expressed as:

$$\frac{\partial h}{\partial t} = \nabla.(\alpha \nabla h) - \rho L \frac{\partial f}{\partial t} \tag{8}$$

where α is the thermal diffusivity.

Fang et al. [228] modeled a shell and tube latent thermal energy storage (LTES) system using multiple PCMs via the enthalpy method, highlighting the significance of PCM fractions and melting temperatures on energy storage and outlet fluid temperature, emphasizing the need for precise PCM selection to boost LTES performance. Basal et al. [229] explore a novel triple concentric-tube thermal storage system with an annulus-shaped PCM layer, leveraging numerical investigation via the enthalpy method. Their analysis underscores the system's heightened efficiency and underscores the PCM annulus' critical influence on design parameters and overall performance. Santos et al. [230] explore solidification enhancement around radial finned tubes and devise correlations to predict their thermal performance. They conduct experiments, validate a numerical code based on the enthalpy method, and develop correlations for interface properties, aiding quick estimations of phase change solidification around finned tubes.

The enthalpy method is commonly used in various commercial software to analyze PCMs for solar applications. Xie et al. [231] employed this method in MATLAB to study the thermal performance of a PCM wall exposed to solar irradiation, demonstrating that a graphite-modified micro-encapsulated PCM wall can store thermal energy for two hours during the day and release it gradually over 10.3 h at night. Al-Saadi et al. [232] utilized TRNSYS software to simulate latent heat storage walls, employing the enthalpy method. They examined various thermal properties of PCMs to identify the optimal properties suitable for different climates in the United States. Their study provided valuable insights into the selection of PCM characteristics that maximize thermal efficiency in varying environmental conditions. Additionally, Liu et al. [233] simulated photovoltaic/thermal collectors with micro-encapsulated phase change slurry as the cooling fluid using ANSYS FLUENT 14. Luo et al. [234] explored the effects of a high thermal conductive PCM for cooling solar photovoltaic panels using ANSYS FLUENT, showcasing the diverse applications and benefits of PCMs in solar energy systems.

These studies collectively demonstrate the effectiveness of using the enthalpy method in different simulation software to optimize PCM applications in solar energy systems, tailored to specific climate conditions and architectural designs.

5.2. *Enthalpy-porosity method*

The enthalpy-porosity method, widely utilized in PCM heat transfer analyses, employs a fixed grid rather than explicitly tracking the melting front, streamlining the process [39]. Unlike the traditional method of directly monitoring the phase interface position, this approach introduces the concept of the liquid phase ratio [235]. This ratio represents the proportion of liquid material in the control volume during both melting and solidification. Indirectly determined by this ratio, the variation in the phase interface position is tracked. Calculated through an enthalpy balance, the liquid fraction during phase change spans between 0 % and 100 % [236,237]. However, as multiple phases are present, employing a stationary mesh technique complicates the treatment of velocity gradients between the stationary solid phase and the mobile liquid phase. This method is more suitable for conduction-dominated phase change scenarios. The method's flexibility in defining source terms allows for modeling a wide range of phase change situations [238–240]. The governing equations encompass the continuity, momentum, and energy equations. This can be expressed as follows:

Continuity

$$\nabla.\left(\rho \vec{V}\right) = 0 \tag{9}$$

Momentum

$$\frac{\partial\left(\rho \vec{V}\right)}{\partial t} + \nabla.\left(\rho \vec{V}\right) = -\nabla p + \mu \nabla^2 \vec{V} - \rho_0 \beta\left(T - T_{ref}\right) + S \tag{10}$$

Energy

$$\frac{\partial(\rho H)}{\partial t} + \nabla.\left(\rho \vec{V} H\right) = k \nabla^2 T - \rho L_f \frac{\partial f}{\partial t} \tag{11}$$

5.2.1. *For non-porous media*

5.2.1.1. *Cartesian.* The 3D Cartesian form of the governing equations, be expressed as:

1. Continuity

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z}=0 \tag{12}$$

2. Momentum

For the x-axis:

$$\rho\left(\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}+w\frac{\partial u}{\partial z}\right)=\mu\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}\right)-\frac{\partial P}{\partial x}+A_m\frac{(1-f)^2}{(f^3+\delta)}u \tag{13}$$

For the y-axis:

$$\rho\left(\frac{\partial v}{\partial t}+u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}+w\frac{\partial v}{\partial z}\right)=\mu\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}+\frac{\partial^2 v}{\partial z^2}\right)-\frac{\partial P}{\partial y}+A_m\frac{(1-f)^2}{(f^3+\delta)}v-\rho g\beta\left(T-T_{ref}\right) \tag{14}$$

For the z-axis:

$$\rho\left(\frac{\partial w}{\partial t}+u\frac{\partial w}{\partial x}+v\frac{\partial w}{\partial y}+w\frac{\partial w}{\partial z}\right)=\mu\left(\frac{\partial^2 w}{\partial x^2}+\frac{\partial^2 w}{\partial y^2}+\frac{\partial^2 w}{\partial z^2}\right)-\frac{\partial P}{\partial z}+A_m\frac{(1-f)^2}{(f^3+\delta)}w \tag{15}$$

3. Energy

$$\frac{\partial T}{\partial t}+u\frac{\partial T}{\partial x}+v\frac{\partial T}{\partial y}+w\frac{\partial T}{\partial z}=\alpha\left(\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial y^2}+\frac{\partial^2 T}{\partial z^2}\right)-\rho L_f\frac{\partial f}{\partial t} \tag{16}$$

5.2.1.2. Cylindrical. The 3D cylindrical form of the governing equations can be expressed as:

1. Continuity

$$\frac{1}{r}\frac{\partial(r\rho u_r)}{\partial r}+\frac{1}{r}\frac{\partial(\rho u_{\theta})}{\partial \theta}+\frac{\partial(\rho u_z)}{\partial z}=0 \tag{17}$$

2. Momentum

The expression for the radial projection of the momentum conservation equations is as follows:

$$\frac{\partial u_r}{\partial t}+u_r\frac{\partial u_r}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_r}{\partial \theta}-\frac{u_\theta^2}{r}+u_z\frac{\partial u_r}{\partial z}=-\frac{1}{\rho}\frac{\partial P}{\partial r}+\nu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_r}{\partial r}\right)-\frac{u_r}{r^2}+\frac{1}{r^2}\frac{\partial^2 u_r}{\partial \theta^2}-\frac{2}{r^2}\frac{\partial u_\theta}{\partial \theta}+\frac{\partial^2 u_r}{\partial z^2}\right]+A_m\frac{(1-f)^2}{(f^3+\delta)}u_r \tag{18}$$

The momentum conservation equations are projected in the theta (θ) direction as follows:

$$\frac{\partial u_\theta}{\partial t}+u_r\frac{\partial u_\theta}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_\theta}{\partial \theta}+\frac{u_\theta u_r}{r}+u_z\frac{\partial u_\theta}{\partial z}=-\frac{1}{\rho r}\frac{\partial P}{\partial \theta}+\nu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_\theta}{\partial r}\right)-\frac{u_\theta}{r^2}+\frac{1}{r^2}\frac{\partial^2 u_\theta}{\partial \theta^2}+\frac{2}{r^2}\frac{\partial u_r}{\partial \theta}+\frac{\partial^2 u_\theta}{\partial z^2}\right]+A_m\frac{(1-f)^2}{(f^3+\delta)}u_\theta \tag{19}$$

The axial projection of the momentum conservation equations is as follows:

$$\frac{\partial u_z}{\partial t}+u_r\frac{\partial u_z}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_z}{\partial \theta}+u_z\frac{\partial u_z}{\partial z}=-\frac{1}{\rho}\frac{\partial P}{\partial z}+g\beta\left(T-T_{ref}\right)+\nu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_z}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2 u_z}{\partial \theta^2}+\frac{\partial^2 u_z}{\partial z^2}\right]+A_m\frac{(1-f)^2}{(f^3+\delta)}u_z \tag{20}$$

3. Energy

$$\frac{\partial T}{\partial t}+u_r\frac{\partial T}{\partial r}+\frac{u_\theta}{r}\frac{\partial T}{\partial \theta}+u_z\frac{\partial T}{\partial z}=\frac{\lambda}{\rho C_p}\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2 T}{\partial \theta^2}+\frac{\partial^2 T}{\partial z^2}\right]-\rho L_f\frac{\partial f}{\partial t} \tag{21}$$

5.2.1.3. Spherical. The 3D spherical form of the governing equations can be expressed as:

1. Continuity

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(\rho u_r r^2\right)+\frac{1}{r sin\theta}\frac{\partial}{\partial \theta}\left(\rho u_\theta sin\theta\right)=0 \tag{22}$$

2. Momentum

The equation governing the conservation of momentum in the radial direction is as follows:

$$\rho\left[\frac{\partial u_r}{\partial t}+u_r\frac{\partial u_r}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_r}{\partial \theta}-\frac{u_\theta^2}{r}\right]=-\frac{\partial P}{\partial r}+\mu\left[\nabla^2 u_r-\frac{2u_r}{r^2}-\frac{2}{r^2}\frac{\partial u_\theta}{\partial \theta}-\frac{2u_\theta cot\theta}{r^2}\right]-\rho g_r\beta\left(T-T_{ref}\right)+A_m\frac{(1-f)^2}{(f^3+\delta)}u_r \tag{23}$$

The equation governing the conservation of momentum in the polar direction is as follows:

$$\rho\left[\frac{\partial u_\theta}{\partial t}+u_r\frac{\partial u_\theta}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_\theta}{\partial \theta}-\frac{u_r u_\theta}{r}\right]=-\frac{1}{r}\frac{\partial P}{\partial r}+\mu\left[\nabla^2 u_\theta-\frac{2u_r}{r^2}-\frac{2}{r^2}\frac{\partial u_r}{\partial \theta}-\frac{u_\theta}{r^2 sin^2\theta}\right]-\rho g_\theta\beta\left(T-T_{ref}\right)+A_m\frac{(1-f)^2}{(f^3+\delta)}u_\theta \tag{24}$$

where,

$$\nabla^2=\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)+\frac{1}{r^2 sin\theta}\frac{\partial}{\partial \theta}\left(sin\theta\frac{\partial}{\partial \theta}\right) \tag{25}$$

3. Energy

$$\frac{\partial h}{\partial t}+\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 u_r h\right)+\frac{1}{r sin\theta}\frac{\partial}{\partial \theta}\left(u_\theta sin\theta h\right)=\alpha\nabla^2 h-\frac{1}{\rho C_p}\frac{\partial \lambda}{\partial t}-\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 u_r \Delta H\right)+\frac{1}{r sin\theta}\frac{\partial}{\partial \theta}\left(u_\theta sin\theta \Delta H\right)-\rho L_f\frac{\partial f}{\partial t} \tag{26}$$

The PCM phase transitions are determined through the utilization of enthalpy as a key parameter [241]:

$$\frac{\partial(\rho H)}{\partial t}+\nabla.(\rho u H)=k\nabla^2 T-\rho L_f\frac{\partial f}{\partial t} \tag{27}$$

The enthalpy function is derived by combining the temperature-dependent latent heat of phase transition with the sensible heat capacity, which can be written as:

$$\begin{cases} H=h+\Delta H \\ h=h_{ref}+\int_{T_{ref}}^{T_{solid}} C_P dT \\ \Delta H=fL_f \end{cases} \tag{28}$$

$$f=\begin{cases} 0, & T<T_{solid} \\ \frac{T-T_{solid}}{T_{Liquid}-T_{solid}}, & T_{solid}\le T\le T_{Liquid} \\ 1, & T\ge T_{Liquid} \end{cases} \tag{29}$$

In this equations, the symbols represent various parameters: h_{ref} denotes the sensible enthalpy of PCM at the reference temperature T_{ref}, f represents the liquid fraction, ρ signifies density, k denotes thermal conductivity, μ stands for dynamic viscosity, u signifies velocity, T represents temperature, P stands for pressure, H denotes specific enthalpy, g represents gravitational acceleration, and δ is an infinitesimal value to prevent division by zero.

Lee et al. [242] utilized the enthalpy porosity method in their study, assessing PCM melting in a cold thermal energy storage (CTES) tank with interior fins. Through numerical and experimental comparisons, the study explored fin variations, proposing a stratified fin design that substantially augmented mean power by 156.3 %, signaling the potential for enhanced CTES systems. Bouzennada et al. [243] examined RT-27 PCM in a rectangular capsule with varying fin inclinations (0°, 45°, and 90°), utilizing the enthalpy porosity method. Reduced inclinations accelerated melting and energy storage rates, cutting melting time by 1.28 % to 20.52 % and enhancing energy storage by 14.75 % to 36.88 %. The study concluded that a horizontal fin at 0° inclination demonstrated optimal efficiency. Xu et al. [244] investigated triplex-layer PCMs in a horizontal LTES unit using the enthalpy-porosity method. Metal fins aided heat transfer, with a novel CSDE criterion evaluating melting performance. Their findings optimized fin arrangements, reducing melting time by up to 71 % and guiding enhanced LTES design.

5.2.2. For porous media

When assumptions of local thermal equilibrium (LTE) or local thermal non-equilibrium (LTNE) between PCM and metal foam ligaments the volume-averaged Navier-Stokes and energy equations, presented in a general vector form, are expressed as follows [245–247]:

5.2.2.1. Cartesian.

1. Continuity

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial v}{\partial y}=0 \tag{30}$$

2. Momentum

Momentum in the x-direction:

$$\begin{aligned}\frac{\rho_f}{\varepsilon}\left(\frac{\partial u}{\partial t}+\frac{u}{\varepsilon}\frac{\partial u}{\partial x}+\frac{v}{\varepsilon}\frac{\partial u}{\partial y}+\frac{w}{\varepsilon}\frac{\partial u}{\partial z}\right) &= -\frac{\partial P}{\partial x}+\frac{\mu_f}{\varepsilon}\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{dy^2}+\frac{\partial^2 u}{dz^2}\right)\\ &\quad -A_m\frac{(1-f)^2}{f^3+\delta}u-\left(\frac{\mu_f}{K}u+\frac{C_f}{\sqrt{K}}\rho_f u\sqrt{u^2+v^2+w^2}\right)\end{aligned} \tag{31}$$

Momentum in the y-direction:

$$\begin{aligned}\frac{\rho_f}{\varepsilon}\left(\frac{\partial v}{\partial t}+\frac{u}{\varepsilon}\frac{\partial v}{\partial x}+\frac{v}{\varepsilon}\frac{\partial v}{\partial y}+\frac{w}{\varepsilon}\frac{\partial v}{\partial z}\right) &= -\frac{\partial P}{\partial y}+\frac{\mu_f}{\varepsilon}\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{dy^2}+\frac{\partial^2 v}{dz^2}\right)\\ &\quad -(\rho\beta)_f g\varepsilon\left(T-T_{ref}\right)-A_m\frac{(1-f)^2}{f^3+\delta}v\\ &\quad -\left(\frac{\mu_f}{K}v+\frac{C_f}{\sqrt{K}}\rho_f v\sqrt{u^2+v^2+w^2}\right)\end{aligned} \tag{32}$$

Momentum in the z-direction:

$$\begin{aligned}\frac{\rho_f}{\varepsilon}\left(\frac{\partial v}{\partial t}+\frac{u}{\varepsilon}\frac{\partial w}{\partial x}+\frac{v}{\varepsilon}\frac{\partial w}{\partial y}+\frac{w}{\varepsilon}\frac{\partial w}{\partial z}\right) &= -\frac{\partial P}{\partial z}+\frac{\mu_f}{\varepsilon}\left(\frac{\partial^2 w}{\partial x^2}+\frac{\partial^2 w}{dy^2}+\frac{\partial^2 w}{dz^2}\right)\\ &\quad -A_m\frac{(1-f)^2}{f^3+\delta}w-\left(\frac{\mu_f}{K}w+\frac{C_f}{\sqrt{K}}\rho_f w\sqrt{u^2+v^2+w^2}\right)\end{aligned} \tag{33}$$

3. Energy -For the LTE model

$$(\rho C_P)\frac{\partial T}{\partial t}+(\rho C_P)_f\left(u\frac{\partial T}{\partial x}+v\frac{\partial T}{\partial y}+w\frac{\partial T}{\partial z}\right)=k_e\left(\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial y^2}+\frac{\partial^2 T}{\partial z^2}\right)-\varepsilon\rho_f L_f\frac{\partial f}{\partial t} \tag{34}$$

4. Energy -For the LTNE model

$$\varepsilon\rho C_P\left(\frac{\partial T}{\partial t}+V.\nabla T\right)+\varepsilon\rho L\frac{\partial f}{\partial t}=k_e\left(\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial y^2}+\frac{\partial^2 T}{\partial z^2}\right)+h_s A_s\left(T_f-T_s\right) \tag{35}$$

5.2.2.2. Cylindrical.

1. Continuity

$$\frac{1}{r}\frac{\partial(r\rho u_r)}{\partial r}+\frac{1}{r}\frac{\partial(\rho u_\theta)}{\partial\theta}+\frac{\partial(\rho u_z)}{\partial z}=0 \tag{36}$$

2. Momentum

Momentum in the r-direction:

$$\begin{aligned}\frac{\rho_f}{\varepsilon}\left(\frac{\partial u_r}{\partial t}+\frac{1}{\varepsilon}(V.\nabla u_r)\right) &= -\nabla P+\frac{u_f}{\varepsilon}\nabla^2 u_r+(\rho\beta)_f g sin\theta\left(T-T_{ref}\right)\\ &\quad -A_m u_r\frac{(1-f)^2}{f^3+\delta}-\left(\frac{\mu_f}{K}-\frac{\rho_f C_f|u_r|}{\sqrt{K}}\right)u_r\end{aligned} \tag{37}$$

Momentum in the θ-direction:

$$\begin{aligned}\frac{\rho_f}{\varepsilon}\left(\frac{\partial u_\theta}{\partial t}+\frac{1}{\varepsilon}(V.\nabla u_\theta)\right) &= -\nabla P+\frac{u_f}{\varepsilon}\nabla^2 u_\theta+(\rho\beta)_f g sin\theta\left(T-T_{ref}\right)\\ &\quad -A_m u_\theta\frac{(1-f)^2}{f^3+\delta}-\left(\frac{\mu_f}{K}-\frac{\rho_f C_f|u_\theta|}{\sqrt{K}}\right)u_\theta\end{aligned} \tag{38}$$

Momentum in the z-direction:

$$\begin{aligned}\frac{\rho_f}{\varepsilon}\left(\frac{\partial u_z}{\partial t}+\frac{1}{\varepsilon}(V.\nabla u_z)\right) &= -\nabla P+\frac{u_f}{\varepsilon}\nabla^2 u_z+(\rho\beta)_f g sin\theta\left(T-T_{ref}\right)\\ &\quad -A_m u_z\frac{(1-f)^2}{f^3+\delta}-\left(\frac{\mu_f}{K}-\frac{\rho_f C_f|u_z|}{\sqrt{K}}\right)u_z\end{aligned} \tag{39}$$

3. Energy -For the LTE model

$$\left[\varepsilon\left((\rho C_P)_f+(1-\varepsilon)(\rho C_P)_s\right]\frac{\partial T}{\partial t}+(\rho C_P)_f(V.\nabla T)=k_e\nabla^2 T-\varepsilon(\rho L)_f\frac{\partial f}{\partial t}\right. \tag{40}$$

4. Energy -For the LTNE model

$$\varepsilon\rho C_P\left(\frac{\partial T}{\partial t}+V.\nabla T\right)+\varepsilon\rho L\frac{\partial f}{\partial t}=k_e\nabla^2 T+h_s A_s\left(T_f-T_s\right) \tag{41}$$

5.2.2.3- Spherical.

1. Continuity

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(\rho u_r r^2\right)+\frac{1}{rsin\theta}\frac{\partial}{\partial\theta}\left(\rho u_\theta sin\theta\right)+\frac{1}{rsin\theta}\frac{\partial}{\partial\phi}\left(\rho u_\phi\right)=0 \tag{42}$$

2. Momentum

In the r-direction

$$\begin{aligned}\frac{\rho_f}{\varepsilon}\left(\frac{\partial u_r}{\partial t}+\frac{1}{\varepsilon}(V.\nabla u_r)\right) &= -\frac{\partial P}{\partial r}+\frac{u_f}{\varepsilon}\nabla^2 u_r+(\rho\beta)_f g sin\theta\left(T-T_{ref}\right)\\ &\quad -A_m u_r\frac{(1-f)^2}{(f^3+\delta)}-\left(\frac{\mu_f}{K}-\frac{\rho_f C_f|u_r|}{\sqrt{K}}\right)u_r\end{aligned} \tag{43}$$

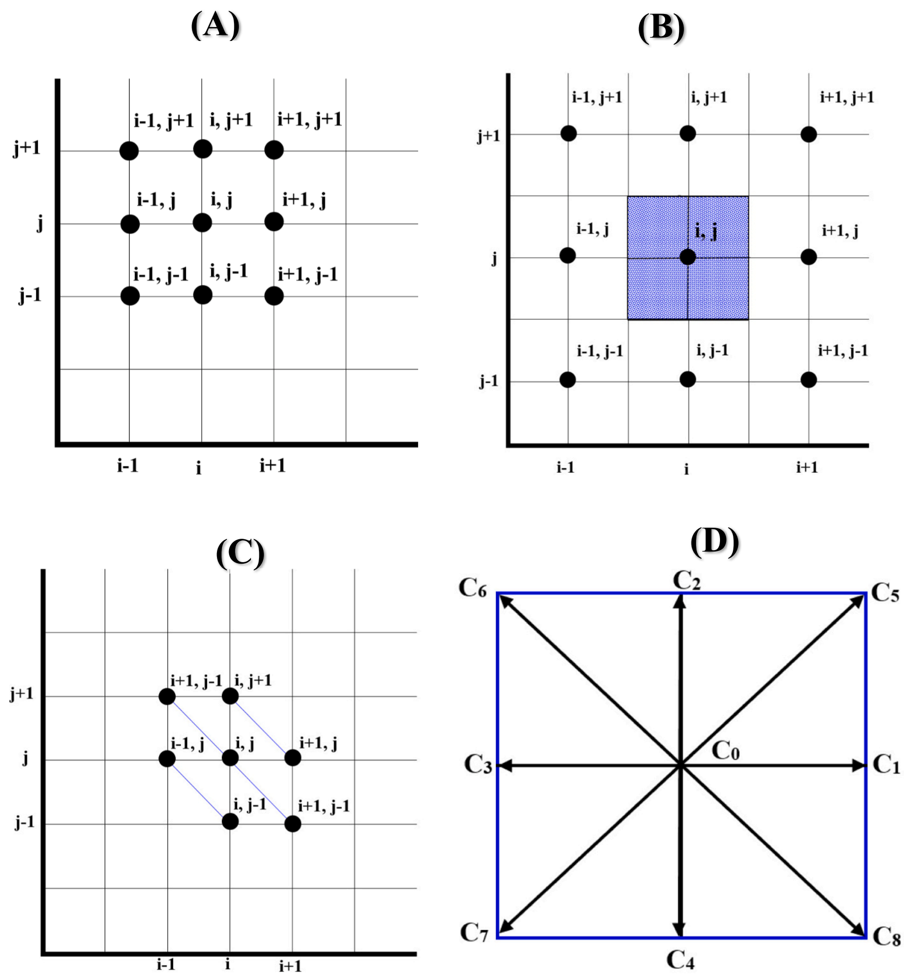


Fig. 21. CFD numerical approaches: (A)- Finite difference method; (B) Finite volume method; (C)- Finite element method; (D)- LBM method (D2Q9 model).

In the θ-direction

$$\frac{\rho_f}{\varepsilon}\left(\frac{\partial u_\theta}{\partial t}+\frac{1}{\varepsilon}(V.\nabla u_\theta)\right) = -\frac{1}{r}\frac{\partial P}{\partial \theta}+\frac{u_f}{\varepsilon}\nabla^2 u_\theta-\frac{u_\phi^2}{r} + (\rho\beta)_f g cos\theta cos\phi(T-T_{ref}) - A_m u_\theta\frac{(1-f)^2}{(f^3+\delta)}-\left(\frac{\mu_f}{K}-\frac{\rho_f C_f|u_\theta|}{\sqrt{K}}\right)u_\theta \tag{44}$$

In the φ-direction

$$\frac{\rho_f}{\varepsilon}\left(\frac{\partial u_\varphi}{\partial t}+\frac{1}{\varepsilon}(V.\nabla u_\varphi)\right) = -\frac{1}{rsin\theta}\frac{\partial P}{\partial \varphi}+\frac{u_f}{\varepsilon}\nabla^2 u_\varphi+\frac{u_\theta u_\varphi}{r} + (\rho\beta)_f g cos\theta sin\varphi(T-T_{ref}) - A_m u_\varphi\frac{(1-f)^2}{(f^3+\delta)} - \left(\frac{\mu_f}{K}-\frac{\rho_f C_f|u_\varphi|}{\sqrt{K}}\right)u_\varphi \tag{45}$$

3. Energy -For the LTE model

$$\left[\varepsilon\left((\rho C_P)_f+(1-\varepsilon)(\rho C_P)_s\right]\frac{\partial T}{\partial t}+(\rho C_P)_f(V.\nabla T) = k_e\nabla^2 T-\varepsilon(\rho L)_f\frac{\partial f}{\partial t} \tag{46}$$

4. Energy -For the LTNE model

$$\varepsilon\rho C_P\left(\frac{\partial T}{\partial t}+V.\nabla T\right)+\varepsilon\rho L\frac{\partial f}{\partial t} = k_e\nabla^2 T+h_s A_s(T_f-T_s) \tag{47}$$

The subscript f denotes the PCM, regardless of its state (solid or liquid), while the subscript s pertains to the metal foam. Here, ε represents the porosity of the metal foam, u_r signifies the velocity component in the r-direction, u_θ indicates the velocity component in the θ-direction, u_φ indicates the velocity component in the φ-direction, T denotes temperature, μ stands for dynamic viscosity, P represents effective pressure, C_p signifies specific heat, and L represents latent heat. Additionally, A_m denotes the mushy zone constant commonly utilized to regulate velocity damping during PCM solidification.

5.3. Heat capacity method

In this method, the heat capacity is treated as a temperature-dependent function throughout the phase change process. The apparent heat capacity approach considers the latent heat of the PCM as a significantly large apparent heat capacity within a narrow temperature range. This transforms the phase change problem into a nonlinear thermal conduction problem within a single region. The entire problem is then solved by establishing an energy equation across the entire region, with temperature being the sole parameter to be determined. This approach simplifies the solution process and renders the calculations more straightforward [223,248,249]. However, this method is difficult in handling cases where the phase-change temperature range is small, as it can easily distort. Additionally, it is challenging to obtain convergence since it tends to underestimate the latent heat. The method is only suitable for gradual phase change conditions and only incorporates heat conduction and latent heat absorption, omitting the effects of the flow behavior of the melted PCM on material thermal performance [41]. The effective heat capacity, $C_{P,eff}$, may be expressed as a function of temperature T in the following manner:

$$C_{P,eff}(T) = \frac{1}{\rho}\frac{\partial h}{\partial T} \tag{48}$$

where "$C_{P,eff}$" represents the heat capacity encompassing both sensible and latent heat.

Utilizing Eq. (48), it is possible to formulate a heat transfer equation for the effective heat capacity, addressing the phase change in the heat transfer problem:

$$\rho C_{P,eff}\frac{\partial T}{\partial t} = \nabla.(k\nabla T) \tag{49}$$

Han et al. [250] utilize numerical simulations and the apparent heat capacity method to scrutinize heat storage and release in multi-cavity-structured phase-change microcapsules. Their findings emphasize that elevating the number of cavities expedites heat storage and release rates, pinpointing the crucial role of the cavity interlayer in enhancing overall heat transfer efficiency. Artinov et al. [251] exposed the influence of latent heat on the temperature field through the implementation of the sensible heat capacity method. Zhang et al. [252] presented refined computational methods for predicting transient heat transfer in building envelopes with PCMs, emphasizing an effective heat capacity model. Experimental validation demonstrates superior accuracy, with deviations below 0.25 °C, and relative errors under 2.5 % and 4.0 %.

5.4. Finite difference method

The finite difference method divides the calculation area into a series of unit bodies, taking several points as nodes on each unit body to obtain a discrete equation by integrating the governing equations. Despite its adaptability to irregular regions, the method often involves a large computational volume, and some treatment methods for solving flow and heat transfer problems may lack maturity compared to the finite volume method [40]. However, the finite difference method is a conventional and straightforward approach to solving partial differential equations. It discretizes the continuous problem domain, considering dependent variables solely at discrete points, and converting PDEs into algebraic equations by approximating derivatives as differences. Renowned for its high accuracy, stability, and rapid convergence [253], the finite difference method presents an effective alternative, particularly suitable for scenarios demanding computational efficiency (Fig. 21-**(A)**).

Jin et al. [254] conducted a numerical analysis using the finite difference method to study PCM systems integrated into building walls. Their study focused on optimizing the placement of a thin PCM layer to mitigate peak heat fluxes through the wall and enhance thermal mass. They employed an implicit method for time discretization and central difference for spatial discretization, with a time step of 20 s and a spatial grid size of 0.5 mm. Mazo et al. [255] investigated a radiant floor system incorporating PCM through numerical simulations. They utilized a one-dimensional finite difference method and compared its thermal response with a more complex two-dimensional model, finding comparable results between the two approaches. Their study demonstrated that using PCM to store and release radiant energy during peak electricity demand periods reduced costs by 18 % compared to conventional heating methods.

5.5. Finite volume method

The finite volume method is a numerical technique used to express partial differential equations in the form of algebraic equations. It is adaptable to unstructured meshes, with the calculation area divided into control volumes, where each node represents a control volume. The discrete equations obtained through this method maintain conservation properties, and the coefficients hold precise physical significance. This approach, particularly effective for phase change processes, enhances calculation speed without introducing errors, making it the prevalent numerical solution method. The computational core of the FLUENT software employs the finite volume method for numerical simulations [40,256] (Fig. 21-**(B)**).

Tao et al. [257] conducted a numerical study on heat transfer in PCMs within solar dish collectors. This complex problem involves various heat transfer modes, including conduction, radiation, convection, and phase change. To address this, they employed a coupled model combining the Monte Carlo Ray Trace Method (MCRTM) and the Finite Volume Method. Mahdaoui et al. [90] utilized ANSYS FLUENT software to simulate PCM bricks using the finite volume method. They modeled the phase change process with the enthalpy-porosity method, employing

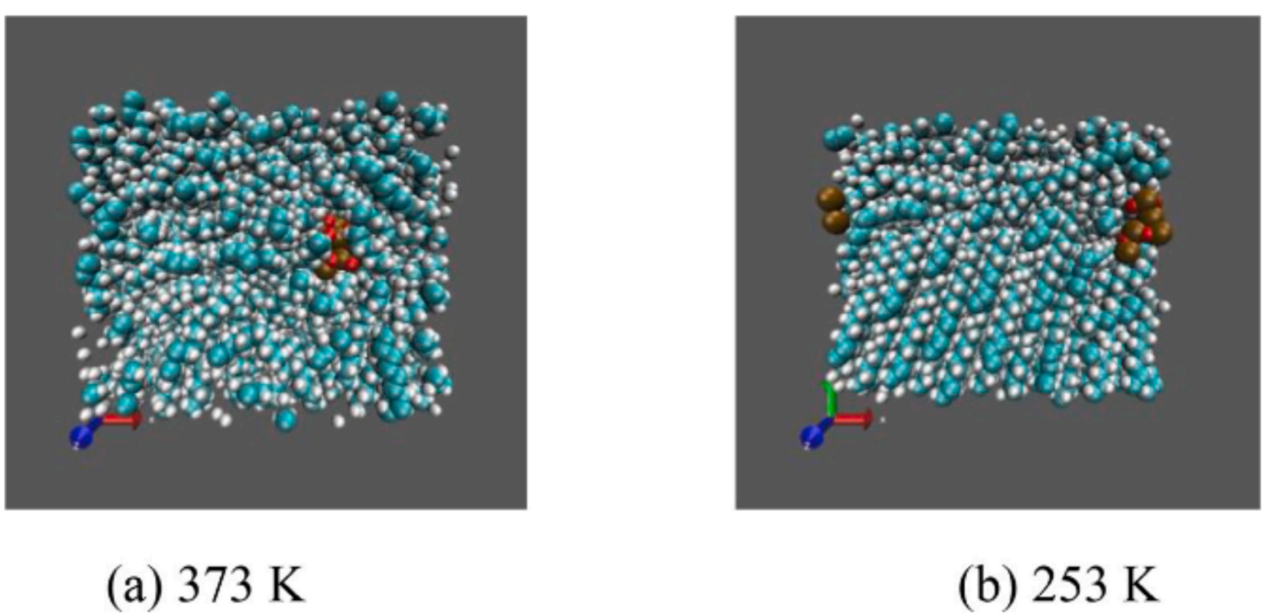


Fig. 22. Molecular distribution states in PCM simulation systems: Visualization of carbon and hydrogen atoms (cyan and white, respectively), along with copper and oxygen atoms (ochre and red, respectively). Reused with permission from Elsevier [268], with license number 5837011000280. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

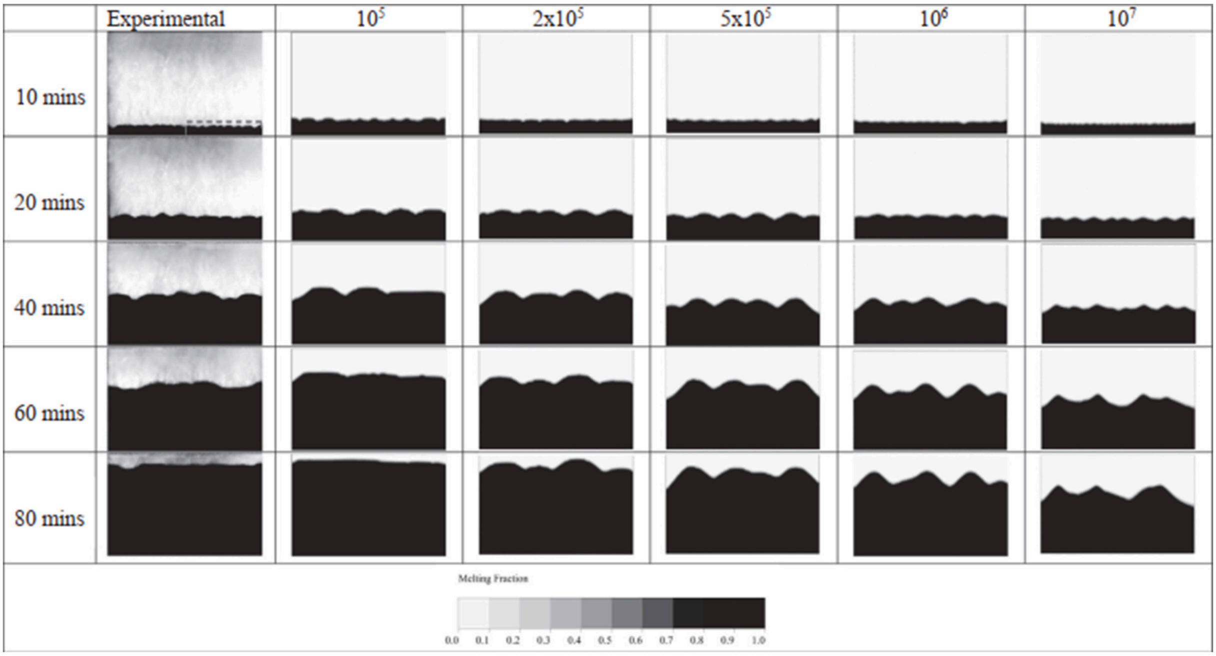


Fig. 23. Solid/liquid interface comparison for different A_{mush} values [274].

thermal lag and the damping factor to describe changes in the internal surface temperature of the brick.

5.6. Finite element method

The finite element method divides the calculation area into a series of unit bodies, using several points as nodes on each unit body to obtain discrete equations through the integration of governing equations. While well-suited for irregular regions, the computational volume of the finite element method is generally large, and some treatment methods for solving flow and heat transfer problems are not as mature as the finite volume method. Despite these challenges, the finite element method is extensively employed in solving various engineering problems related to structural analysis, heat and mass transfer, fluid flow, and electromagnetic problems [35,40] (Fig. 21-**(C)**).

COMSOL software, which incorporates the finite element method, is frequently employed for a variety of simulations. Kant et al. [258] used this software to investigate the impact of convection heat transfer within melted PCM on the temperature regulation of photovoltaic (PV) panels. Similarly, Klemm et al. [259] used COMSOL Multiphysics 5.2 to study the effect of embedding PCMs within a metallic fiber structure behind PV modules. This setup improved thermal conductivity and reduced module temperatures by 20 K. Among the tested PCM thicknesses (10–70 mm), 50 mm was found to be optimal.

5.7. Lattice Boltzmann method

The Lattice Boltzmann Method (LBM) represents a modern CFD approach that has gained prominence in recent years. Unlike traditional CFD methods, LBM does not directly solve the Navier-Stokes equation. Instead, it simulates fluid behavior by computing the interactions between microscopic particles through streaming and collision processes. The lattice model, initially proposed by Qian et al. [150] in 1992, is commonly referred to as the DdQm model, where ‘d’ denotes the dimensional space and ‘m’ signifies discrete velocities. A visual representation of a typical two-dimensional lattice model is depicted in Fig. 21-**(D)**. The fundamental equation of the LBM can be expressed as follows [260]:

$$g_i\left(x+\vec{c_i}\Delta t, t+\Delta t\right) = g_i(x,t) - \frac{1}{\tau_T}\left(g_i(x,t) - g_i^{eq}(x,t)\right) + R_i\Delta t \tag{50}$$

Here, τ_T represents the relaxation time selected for the temperature field, while R_i denotes the source term of the temperature distribution function.

The LBM offers advantages such as solving linear equations, ease of programming, and managing intricate boundary conditions. Its explicit scheme and local interactions render it well-suited and stable for parallel computing [261]. Nonetheless, challenges persist, including the generation of deviation terms in many LBM-based phase change models during the recovery of corresponding macroscopic equations [262,263].

5.8. Molecular dynamics simulation

Molecular dynamics (MD) simulation is a computational technique used to investigate the behavior and interactions of atoms and molecules at the atomic scale. Initially introduced by Alder and Wainwright in 1950, MD simulation involves solving Newton’s equations of motion to track the positions and velocities of individual particles within a simulated system [37]. Various properties and phenomena can be explored by simulating molecular motions, including molecular structure, thermodynamics, chemical reactions, and material behavior [264]. Given the brief time frame involved, MD simulations typically employ time steps on the scale of 1 femtosecond (10^{-15} s). These simulations utilize Newton’s equations of motion to track the trajectories of atoms and molecules, enabling the derivation of desired physical quantities. Molecular mechanics force fields or interatomic potentials are then employed to compute the forces between particles and their corresponding potential energies. This approach is fundamental for achieving precise MD simulations of the target physical phenomenon [265]. The thermal properties of NePCMs may be subject to various factors, leading to potential uncertainties in the obtained results. MD simulation offers a means to explore the thermophysical characteristics of NePCMs, providing insights into their behavior and properties [37].

Zhang et al. [266] conducted a non-equilibrium MD study to simulate ethylene–vinyl acetate paraffin PCMs containing varying concentrations of graphene dispersion. The simulation outcomes revealed that the sample containing 0.7 wt% of graphene exhibited superior

Table 7
Skewness values and cell quality [276].

Value of skewness	Cell quality
1	Degenerate
0.9 - <1	Bad
0.75–0.9	Poor
0.5–0.75	Fair
0.25–0.5	Good
>0–0.25	Excellent
0	Equilateral

enhancement in thermal conductivity compared to those containing higher concentrations (1.5 wt%, 3.6 wt%, and 7.0 wt%). Yu et al. [267] employed an optimized MD approach to model the impact of SiO_2 nanoparticles on the thermophysical characteristics of NaCl molten salt PCM. Their findings indicated that the addition of 2.4 vol% SiO_2 nanoparticles led to a substantial enhancement in the thermal conductivity of NaCl, increasing it by up to 44.2 %. Additionally, the shear viscosity of molten NaCl was observed to decrease by up to 23.6 %. Moreover, the introduction of SiO_2 nanoparticles resulted in an elevation of the melting point of NaCl, with a direct correlation observed with the volume fraction of SiO_2 nanoparticles. Zhao et al. [268] investigated the impact of CuO nanoparticles on paraffin PCM melting enthalpy using MD simulations (Fig. 22). They found that increasing nanoparticle mass fractions led to a significant reduction in melting enthalpy, by about 51.5 % for NPCM with 19.72 wt% nanoparticles. The interactions between nanoparticles and PCM, as well as the formation of a dense paraffin phase around nanoparticles, were identified as key factors contributing to the observed enthalpy reduction.

6. Mushy zone and mesh quality

6.1. Mushy zone

The mushy region, situated between the molten and solid states of PCMs, defines a semi-solid zone [269]. A_{mush}, depicting the rate of change in material velocity during solidification, quantifies how fluid velocity gradually drops to zero as it solidifies [270]. Higher A_{mush} values indicate a swifter decline in velocity during solidification, notably influencing heat transfer accuracy in both solidification and melting phases [271].

Soliman et al. [272] explored paraffin wax melting in variously shaped storage capsules at 27 °C, studying the mushy zone parameter (A_{mush}) effect. Results showed A_{mush} impact on heat transfer, validating models: circular (2×10^6), vertical oval (1×10^5), and horizontal oval (1×10^6) capsules. The study linked A_{mush} to fluid dynamics, revealing its shape-dependent influence on PCM melting. Hameter et al. [273] analyzed latent heat thermal storage systems, varying the mushy zone constant (C) from 10^5 to 10^8, impacting the melting and solidification of PCM. The study showcases how higher C values extend PCM phase transitions, affecting the chronology of sodium nitrate ($NaNO_3$) liquid fraction and shifting the melting front towards a vertical orientation. Fadl et al. [274] investigated Lauric acid's phase change in vertical and horizontal enclosures, scrutinizing the effect of the mushy zone parameter (A_{mush}) on accurate LHTES simulations (Fig. 23). Their study highlighted the pivotal role of A_{mush} in solid-liquid interface morphology, heat transfer rates, and energy storage, emphasizing the necessity of choosing the right A_{mush} value for precise LHTES modeling.

6.2. Mesh quality

To optimize the utilization of the researcher's computational resources and time, it is crucial to employ a high-quality mesh that ensures both accuracy and cost-effectiveness. This can be assessed through three key indicators measured on a scale of 0–1 [275]:

1. **Skewness**, indicating the distortion of cell shapes, is advised to be maintained at a very low level, typically below 0.7 for the average value. Illustration of skewness values and cell quality in Table 7.
2. **Orthogonal quality**, the scale for orthogonal quality spans from 0 to 1, with 0 denoting the poorest quality and a value of 1 indicating the highest quality.
3. **Element quality**, similar to the two indicators mentioned earlier, provides a measure of cell quality, and it is expected to be high, ideally 0.9 or higher.

In a triangle mesh, optimal shapes are observed with equilateral triangles for 2D studies, and for a quadrilateral mesh, squares are preferred. In 3D studies, regular tetrahedra and hexahedra are recommended.

Ansys-FLUENT emerges as a prevalent choice in PCM studies, exhibiting commendable agreement with validation experiments. Single-phase PCM systems are often explored using a pressure-based solver, with the solidification/melting model activated in Ansys. Additionally, Ansys facilitates the incorporation of user-defined functions written in C++, enabling the customization and augmentation of properties and models within the physical system.

7. Thermophysical properties of composite PCMs

The thermophysical properties of composite PCMs encompass a range of characteristics crucial for their performance in TES applications. These properties include thermal conductivity, specific heat capacity, latent heat of fusion, density, and porosity. Composite PCMs often exhibit enhanced thermal properties compared to pure PCMs, owing to the addition of nanoparticles, microcapsules, or other additives. Understanding and optimizing these thermophysical properties are essential for maximizing the efficiency and effectiveness of composite PCMs in various energy storage and thermal management applications.

7.1. For the single NePCM

Examining the integration of nanoparticles into PCMs necessitates a thorough investigation into how the thermo-physical characteristics of these nanoparticles affect those of the PCM [277]. Enhancing the thermal efficiency of PCMs is achieved through the dispersion of thermally conductive nanostructures, such as nanocarbons, nanometals, and nanometal oxides. Consequently, NePCMs are developed, showcasing distinctive thermal properties [37]. The incorporation of nanoparticles alters the properties of NePCMs, and their density, specific heat, latent heat, viscosity, thermal conductivity, and thermal expansion coefficient can be determined through the following theoretical models [278,279]:

1. Effective dynamic viscosity

(A) Einstein model

Einstein proposed a model grounded in kinetic theory to describe mixtures of solids and liquids, applicable when the volume fractions are below 1 %. This model provides the effective viscosity of the mixture as:

$$\mu_{NePCM} = \mu_{PCM}(1 + 2.5\,\phi) \tag{51}$$

(B) Brinkman model

$$\mu_{NePCM} = \frac{\mu_{PCM}}{(1-\phi)^{2.5}} \tag{52}$$

Brinkman [280] introduced a model, building upon Einstein's Eq. [281], aimed at predicting the viscosity of particle suspensions with moderate loadings. This model accounts for the dispersion of a solute molecule within a continuous solution. Although experiments

[282–284] have confirmed the validity of the Brinkman model in estimating viscosity for low particle volume fractions ($\phi < 3$ %), it tends to underestimate the viscosity of nanofluids for higher particle concentrations.

(C) **Krieger-Dougherty (K—D) model** [285–288]

$$\mu_{NePCM} = \mu_{PCM}\left(1 - \frac{\phi}{\phi_{max}}\right)^{-A\phi_{max}} \tag{53}$$

Where ϕ_{max} represents the maximum packing factor, and A denotes the intrinsic viscosity.

(D) **Vajjha's model** [289–291]

$$\mu_{NePCM} = \mu_{PCM}\left(0.983e^{12.959\phi}\right) \tag{54}$$

(E) **Corcione's model** [282]

$$\mu_{NePCM} = \frac{\mu_{PCM}}{1 - 34.87\left(\frac{d_n}{d_{P,e}}\right)^{-0.3}\phi^{1.03}} \tag{55}$$

Where, d_n represents the nanoparticle diameter and $d_{P,e}$ denotes the equivalent diameter of the base fluid molecule, defined as:

$$d_{P,e} = 0.1\left(\frac{6M}{N\pi\rho_{P,0}}\right)^{\frac{1}{3}} \tag{56}$$

2. Effective thermal conductivity

The Maxwell model, as described by [292] and based on effective medium theory (EMT), assumes no slip between the base fluid and particles, making it a "static" model. Hamilton and Crosser [293] expanded upon the Maxwell model to accommodate non-spherical particles. However, the Hamilton-Crosser (HC) model not only underestimated the enhancement in thermal conductivity due to dispersing nanosized particles but also overlooked the effects of nanoparticle size and temperature dependency. Additionally, factors such as thermal dispersion, thermal interface resistance, Brownian motion, and nanoparticle morphology play significant roles. Consequently, several extensions were developed by incorporating a "dynamic" component into the Maxwell-HC model [294]. The thermal conductivity of NePCM is computed using the following correlations:

(A) The Maxwell model

The Maxwell model [292,295] is probably the first model presented for the thermal conductivity k_{NePCM} of solid-liquid dispersions, where we set

$$k_{NePCM} = \frac{k_{np} + 2k_{PCM} - 2\phi\left(k_{PCM} - k_{np}\right)}{k_{np} + 2k_{PCM} + \phi\left(k_{PCM} - k_{np}\right)}k_{np} \tag{57}$$

The thermal conductivities of the base fluid and the nanoparticles, denoted as k_{PCM} and k_{np}, respectively, are parameters in this model. It applies specifically to systems with spherical particles and low concentrations ($\varphi \ll 1$).

(B) Bruggeman model

Bruggeman [295,296] introduced a model that takes into account the interactions among spherical particles.

$$\frac{k_{NePCM}}{k_{PCM}} = \frac{(3\phi - 1)\frac{k_{np}}{k_{PCM}} + \{3(1-\phi) - 1\} + \sqrt{\Delta}}{4} \tag{58}$$

Where;

$$\Delta = \left[(3\phi - 1)\frac{k_{np}}{k_{PCM}} + \{3(1-\phi) - 1\}\right]^2 + 8\frac{k_{np}}{k_{PCM}} \tag{59}$$

(C) Hamilton and Crosser

Hamilton and Crosser (1962) [293] extended the Maxwell model by incorporating a shape factor into their formulation.

$$K_{NePCM} = \frac{k_{np} + (n-1)k_{PCM} + (n-1)\phi\left(k_{np} - k_{PCM}\right)}{k_{np} + (n-1)k_{PCM} - \phi\left(k_{np} - k_{PCM}\right)}k_{PCM} \tag{60}$$

The empirical shape factor n is determined by $= \frac{3}{\psi}$, where ψ represents the particle sphericity. For spherical particles, the sphericity parameter is 1, while it decreases to 0.5 for cylindrical particles.

(D) Xuan and Roetzel model

Xuan and Roetzel [297] initially proposed the formulation of dispersion thermal conductivity, subsequently adopted by Khanafer et al. [298] in their Cu nanofluid modeling. However, it is essential to highlight that this model doesn't account for temperature dependency, potentially leading to deviations in simulation outcomes. Can be expressed as:

$$k_{NePCM} = \frac{k_{np} + 2k_{PCM} - 2\phi\left(k_{PCM} - k_{np}\right)}{k_{np} + 2k_{PCM} + \phi\left(k_{PCM} - k_{np}\right)}k_{np} + C'\left(\rho C_P\right)_{np}|u|\,\phi d_n \tag{61}$$

Where u represents the fluid velocity, and C' is an empirical constant typically determined based on the research conducted by Wakao and Kaguei [299]. It's noteworthy that the specific values of C' were only explicitly provided in studies [300,301] among those focusing on NePCMs. The latter part of Eq. (61) accounts for the thermal dispersion effect, commonly known as dispersion thermal conductivity.

(E) Vajjha's model

The thermal conductivity enhancement in NePCM, as described by Vajjha's model, is captured in the second part of Eq. (62), where the influence of temperature on the Brownian motion of nanoparticles is considered. This model, proposed by the authors [302], specifies the applicable ranges for nanoparticle diameter (d_{np}) and temperature (T) as 29–77 nm and 25–90 °C, respectively.

$$k_{NePCM} = \frac{k_{np} + 2k_{PCM} - 2\phi\left(k_{PCM} - k_{np}\right)}{k_{np} + 2k_{PCM} + \phi\left(k_{PCM} - k_{np}\right)}k_{PCM} + \gamma\varsigma Z\phi\rho_{PCM}\sqrt{\frac{BT}{\rho_{np}d_{np}}}f(T,\phi) \tag{62}$$

The first part of Eq. (62) is derived from the Maxwell model, while the second part represents Brownian motion, explaining the temperature-dependent thermal conductivity. In the equation, d_{np} stands for nanoparticle diameter, Z is the Brownian motion constant (5 $\times$ 10^4), B is the Boltzmann constant (1.381 $\times$ 10^{-23}), and γ is a correction factor. The values for these parameters are determined based on specific formulas, and the overall equation contributes to the understanding of thermal conductivity dependence on temperature. The symbol ς represents an empirical correlation within the Brownian motion term and is defined as follows:

$$\varsigma = 8.4407\left(100\phi^{-1.07304}\right) \tag{63}$$

The additional correction function $f(T,\phi)$ is defined as follows:

$$f(T,\phi) = 10^{-3}[28.217\phi + 3.917)T/T_{ref} - 30.669\phi - 3.91123] \tag{64}$$

(F) Corcione's model

Corcione [282] formulated a semi-empirical model through regression analysis, which incorporates nanoparticle size, fluid temperature, and base fluid properties. This model demonstrated a standard deviation of 1.86 % when compared against experimental data involving water, ethylene glycol (EG), and propylene glycol (PG) dispersed with copper (Cu), aluminum oxide (Al_2O_3), copper oxide (CuO), and titanium dioxide (TiO_2) nanoparticles. The model's applicable ranges for nanoparticle diameter (d_n), volume fraction (ϕ), and temperature (T) were 10–150 nm, 0.002–0.09, and 21–51 °C, respectively. Additionally, Chon et al. [303] presented a similar equation for Al_2O_3 nanofluids, albeit with a distinct definition of nanoparticle Reynolds number compared to Eq. (66). Can be expressed Corcione model with Eq. (65):

$$k_{NePCM} = k_{PCM}\left[1 + 4.4Re^{0.4}Pr^{0.66}\left(\frac{T}{T_{fr}}\right)^{10}\left(\frac{k_{np}}{k_{PCM}}\right)^{0.03}\phi^{0.66}\right] \tag{65}$$

Pr represents the Prandtl number of the base fluid, while T_{fr} denotes the freezing point of the base fluid. R_e stands for the nanoparticle Reynolds number, which is defined as:

$$R_e = \frac{2\rho_P BT}{\pi\mu_P^2 d_n} \tag{66}$$

(G) Nan's model

This effective medium theory (EMT) based model, developed by Nan [304], stands out for its incorporation of interfacial thermal resistance and nanostructure morphology. As a result, Nan's model proves to be applicable to various nanostructures, including aligned continuous fibers, laminated flat plates, spheres, and misoriented ellipsoidal particles.

$$k_{NePCM} = k_{PCM}\frac{3 + \phi[2\beta_{11}(1 - L_{11}) + \beta_{33}(1 - L_{33})]}{3 - \phi(2\beta_{11}L_{11} + \beta_{33}L_{33})} \tag{67}$$

where,

$$\beta_{11} = \frac{k_{11} - k_{PCM}}{k_{PCM} + L_{11}(k_{11} - k_{PCM})}, \beta_{33} = \frac{k_{33} - k_{PCM}}{k_{PCM} + L_{33}(k_{33} - k_{PCM})} \tag{68}$$

where L_{11} and L_{33} represent the geometrical factors of nanostructures, and their expressions are as follows:

$$L_{11} = \frac{a^2}{2(a^2 - 1)} - \frac{a}{(a^2 - 1)^{\frac{3}{2}}}cosh^{-1}a, \text{for } a > 1 \text{ and } L_{33} = 1 - 2L_{11} \tag{69}$$

$$L_{11} = \frac{a^2}{2(a^2 - 1)} + \frac{a}{(a^2 - 1)^{\frac{3}{2}}}cos^{-1}a, \text{for } a < 1 \text{ and } L_{33} = 1 - 2L_{11} \tag{70}$$

$$L_{11} = L_{33} = \frac{1}{3}, \text{for } a = 1 \tag{71}$$

Where a denotes the aspect ratio of nanostructures, while k_{11} and k_{33} represent the equivalent thermal conductivities along different cell directions. These thermal conductivities can be expressed as:

$$k_{ii} = \frac{k_n}{1 + \frac{\theta L_{ii}k_n}{k_p}}, i = 1, 3 \tag{72}$$

where

$$\theta = \begin{cases} \left(2 + \frac{1}{a}\right)\frac{Rk_P}{l_n}, for \quad a \geq 1 \\ (2 + 2a)\frac{Rk_P}{d_n}, for \quad a \leq 1 \end{cases} \tag{73}$$

Where R denotes the thermal boundary resistance between the nanostructures and the base fluid.

3. Effective density

The effective density of NePCM is defined as follows:

$$\rho_{NePCM} = \phi\rho_{np} + (1 - \phi)\rho_{PCM} \tag{74}$$

4. Effective specific heat

The effective specific heat of NePCM is defined as follows:

$$(\rho C_P)_{NePCM} = (1 - \phi)(\rho C_P)_{PCM} + \phi(\rho C_P)_{np} \tag{75}$$

5. Effective latent heat capacity

The latent heat of NePCM is expressed as:

$$L_{NePCM} = \frac{(1 - \phi)(\rho L)_{PCM}}{\rho_{NePCM}} \tag{76}$$

6. Effective thermal diffusivity

The thermal diffusivity of the Nano-enhanced PCM is defined by Eq. (77) [305], which represents a proportion of the thermal conductivity, density, and heat capacity of the NePCM. Consequently, leveraging the high thermal conductivity and reduced heat capacity of the NePCM would result in exceptional thermal diffusivity, enhancing thermal transmission. This can be expressed in Eq. (77) as:

$$\alpha = \frac{k_{NePCM}}{(\rho C_P)_{NePCM}} \tag{77}$$

7. Effective thermal expansion coefficient

The thermal expansion coefficient of NePCM can be expressed as:

$$(\rho\beta)_{NePCM} = (1 - \phi)(\rho\beta)_{PCM} + \phi(\rho\beta)_{np} \tag{78}$$

7.2. For the hybrid NePCM

The incorporation of hybrid nanoparticle additives alters the thermophysical characteristics of the PCM. The thermophysical properties of hybrid NePCM (HNePCM) are determined through theoretical models of mixtures, which can be expressed as follows [153,170,274,306–309]:

1. Effective density

The density of HNePCM is defined as follows:

$$\rho_{HNePCM} = \rho_{PCM}(1 - \phi_2)\left[(1 - \phi_1) + \phi_1\left(\frac{\rho_{np1}}{\rho_{PCM}}\right)\right] + \phi_2\rho_{np2} \tag{79}$$

2. Effective specific heat

The specific heat of HNePCM is defined as follows:

$$(\rho C_P)_{HNePCM} = (\rho_{C_P})_{PCM}(1 - \phi_2)\left[(1 - \phi_1) + \phi_1\left(\frac{(\rho C_P)_{np1}}{(\rho C_P)_{PCM}}\right)\right] + \phi_2(\rho_{CP})_{np2} \tag{80}$$

3. Effective dynamic viscosity

The viscosity of HNePCM can be expressed as:

$$\mu_{HNePCM} = \frac{\mu_{PCM}}{(1 - \phi_1)^{2.5}(1 - \phi_2)^{2.5}} \tag{81}$$

4. Effective latent heat capacity

The latent heat of NePCM is expressed as:

$$(\rho L)_{HNePCM} = (\rho L)_{PCM}(1-\phi_1)(1-\phi_2) \quad (82)$$

5. Effective thermal conductivity

The thermal conductivity of HNePCM is stated as:

$$\frac{k_{HNePCM}}{k_{NePCM}} = \frac{k_{np2} + 2k_{NePCM} - 2\phi_2(k_{NePCM} - k_{np2})}{k_{np2} + 2k_{NePCM} + \phi_2(k_{NePCM} - k_{np2})} \quad (83)$$

where,

$$\frac{k_{NePCM}}{k_{PCM}} = \frac{k_{np1} + 2k_{PCM} - 2\phi_1(k_{PCM} - k_{np1})}{k_{np1} + 2k_{PCM} + \phi_1(k_{PCM} - k_{np1})} \quad (84)$$

6. Effective thermal expansion coefficient

The thermal expansion coefficient of HNePCM can be expressed as:

$$(\rho\beta)_{HNePCM} = \phi_2(\rho\beta)_{np2} + \left[(1-\phi_2)\left(\phi_1(\rho C_P)_{np1} + (1-\phi_1)(\rho\beta)_{PCM}\right)\right] \quad (85)$$

In Eqs. (79) to (85), the variables ϕ_1 and ϕ_2 denote the volume fractions of nanoparticles for type 1 and type 2, respectively. The subscripts HNePCM, np, $np1$, and $np2$ correspond to hybrid NePCM, nanoparticles, PCM, type 1 nanoparticles, and type 2 nanoparticles, respectively.

7.3. For encapsulated PCM

Encapsulated PCM (EPCM) particles, characterized by core-shell configurations, undergo phase transitions, absorbing and releasing heat during circulation. The governing equations, transformed into non-dimensional form and solved using the finite element method, illustrate that the fusion temperature of encapsulated PCM particles profoundly impacts heat transfer enhancement. Specifically, it achieves a significant 10 % enhancement compared to the base fluid. Can be expressed thermophysical properties of EPCM are as follows:

The density of EPCM can be expressed as [310]:

$$\rho_e = \frac{(1+\chi)\rho_P\rho_c}{\rho_P + \chi\rho_c} \quad (86)$$

where,

$$\chi = \frac{L_e}{L_P} \quad (87)$$

The symbols ρ_P, ρ_c, and χ represent the density of the PCM, the density of the shell material, and the ratio of PCM to shell material, respectively. The expression for χ is determined by the latent heat of the PCM (L_P, in kJ/kg) and the latent heat of the encapsulated PCM (L_e).

Another significant thermophysical parameter, specific heat, is determined using the following correlation [41]:

$$C_e = \frac{(C_P + \chi C_c)\rho_P\rho_c}{(\rho_c + \chi\rho_P)\rho_e} \quad (88)$$

In this equation, C_e, C_P, and C_c represent the specific heats of the EPCM, shell, and core materials.

The thermal conductivity of encapsulated EPCM denoted as k_e, is computed as [311]:

$$\frac{1}{k_e d_e} = \frac{1}{k_P d_P} + \frac{d_e - d_P}{k_c d_e d_P} \quad (89)$$

$$\left(\frac{d_e}{d_P}\right)^3 = 1 + \frac{\rho_c}{\rho_c + \chi\rho_P} \quad (90)$$

The symbols d_e and d_P represent the diameters of the encapsulated EPCM and PCM.

Within the literature, various mathematical models have emerged to evaluate the physical and thermal behaviors of EPCM. These models are intricately tied to parameters like the masses of core and shell materials, as well as the quantities of emulsifier and cross-linking agents employed during the encapsulation process. Through these equations, can be estimated both the theoretical loading or core content (C_{th}) and actual loading (C_{act}) or core content of PCM within the capsules [312]:

$$C_{th} = \frac{m_{core}}{m_{core} + m_{shell}} \times 100\% \quad (91)$$

$$C_{act} = \frac{m_{core} - m_{shell}}{m_{core}} \times 100\% \quad (92)$$

The primary assessment of thermal performance in EPCMs typically involves key parameters such as encapsulation ratio (*ER*), encapsulation efficiency (*EF*), thermal energy storage capability (*TESC*), and thermal cycling performance (*TCP*) [313–315]:

$$ER = \frac{\Delta H_{m,EPCM}}{\Delta H_{m,PCM}} \times 100\% \quad (93)$$

$$EF = \frac{\Delta H_{m,EPCM} + \Delta H_{s,EPCM}}{\Delta H_{m,PCM} + \Delta H_{s,PCM}} \times 100\% \quad (94)$$

$$TESC = \frac{\Delta H_{m,PCM}(\Delta H_{m,EPCM} + \Delta H_{s,EPCM})}{\Delta H_{m,EPCM}(\Delta H_{m,PCM} + \Delta H_{s,PCM})} \times 100\% \quad (95)$$

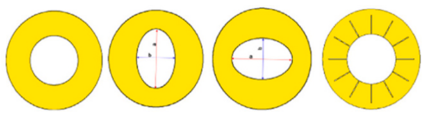
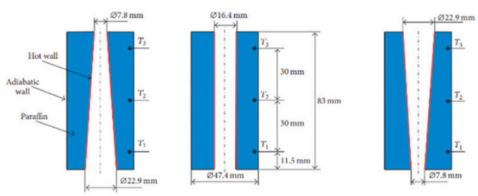
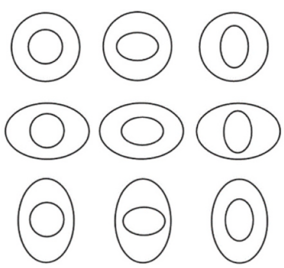
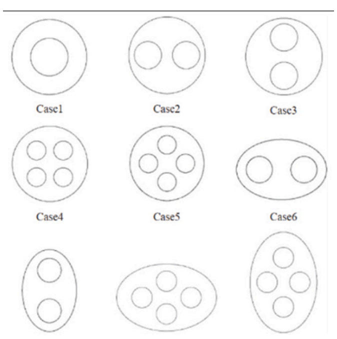
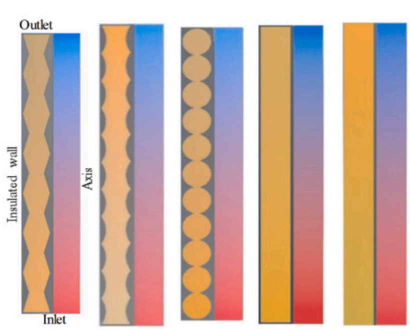
$$TCP = \frac{\Delta H'_{m,EPCM}}{\Delta H_{m,PCM}} \times 100\% \quad (96)$$

In this context, $\Delta H_{m,EPCM}$ and $\Delta H_{s,EPCM}$ represent the alterations in enthalpies associated with the melting and solidification processes of EPCM, while $\Delta H_{m,PCM}$, and $\Delta H_{s,PCM}$ denote the changes in enthalpies corresponding to the melting and solidification phases of PCM. These shifts in enthalpy or latent heat of fusion are typically determined using Differential Scanning Calorimetry (DSC). The determination of core percentage or leakage rate in microcapsules/nanocapsules over different time intervals is commonly employed to characterize their leakage performance [316,317]. The leakage rate (L_r), which reflects the change in mass between the initial mass (m_0) of the capsules and the mass after heating periodically at a specific melting temperature (m_t), is defined as follows:

$$L_r = \frac{m_0 - m_t}{m_0} \times 100\% \quad (97)$$

In this context, it is evident that augmenting the thickness leads to a reduction in the percentage of leakage rate observed in the capsules. However, this adjustment also brings about a simultaneous decrease in the encapsulation ratio (*ER*). This indicates a trade-off between leakage rate and encapsulation efficiency, wherein increasing thickness offers improved containment of the core material but at the expense of a lower overall encapsulation ratio. While EPCM-based TES systems provide improved thermal stability and heat transfer, recent studies [318] highlight the impact of particle size and melting temperature on creep damage reduction in molten salt packed-bed TES tanks. A two-layered EPCM configuration was found to lower mechanical stress and improve tank durability [319]. Additionally, stress analysis revealed that creep damage mainly occurs at the bottom wallboards and fillet welds, while upper sections remain elastic. These findings provide insights for optimizing PCM-packed bed TES designs for long-term stability in high-temperature applications.

Table 8
Summary of studies that show shell-and-tube heat exchanger modifications involving geometrical alterations.

Authors	Geometry	Model Approach	PCM type	Main Research	Main Conclusions
Hu et al. [270]		The enthalpy–porosity formulation	Paraffin	Optimizing frustum-shaped thermal units with multiple PCM arrangements for enhanced melting heat transfer and reduced thermal storage time	Reduced thermal storage time through optimized frustum-shaped units with multiple PCM arrangements
Rabienataj Darzi et al. [324]		The enthalpy–porosity technique	N-eicosane	Enhancing PCM heat transfer using various annulus configurations and nanoparticle additions for improved melting and solidification rates	Improved heat transfer rates in PCM melting and solidification via annulus configurations and nanoparticle incorporation
Dwi Korawan et al. [325]		The enthalpy-porosity technique	Paraffin	Comparative study of paraffin melting in different thermal storage models through 3D numerical simulations	The nozzle-and-shell model demonstrates the most efficient melting process.
Faghani et al. [326]		The enthalpy-porosity method	RT 25	Analyzing melting dynamics in heat exchangers with varied elliptical shapes, determining optimal orientations for improved heat absorption	Optimal heat absorption in elliptical-tube heat exchangers revealed through comprehensive melting process analysis
Pourakabaret et al. [327]		The enthalpy–porosity technique	N-eicosane	Investigating PCM melting and solidification in varied cylindrical containers with different inner tube arrangements, supplemented by copper foam for enhanced phase change rates	Varied tube arrangements influence melting rates, and the introduction of copper foam accelerates phase change rates but suppresses natural convection, particularly during melting
Khedher et al. [328]		The enthalpy-porosity method	RT35	Enhancing TES rates in a vertical latent heat double-pipe heat exchanger through geometry modifications of the PCM container's framed structures	Improved TES rates in vertical heat exchangers via geometry modifications of PCM containers with framed structures

(*continued on next page*)

Table 8 (*continued*)

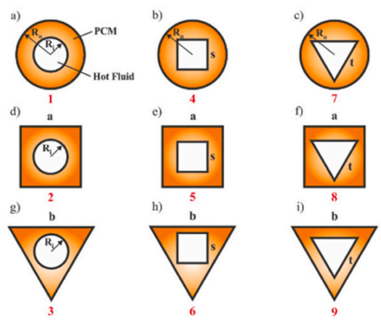
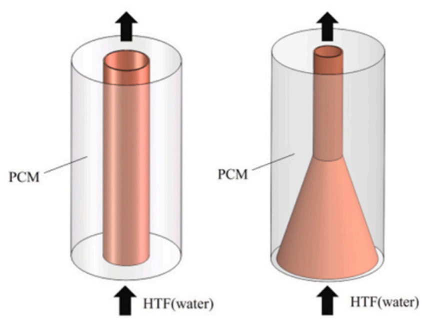
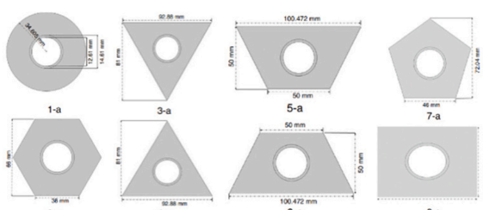
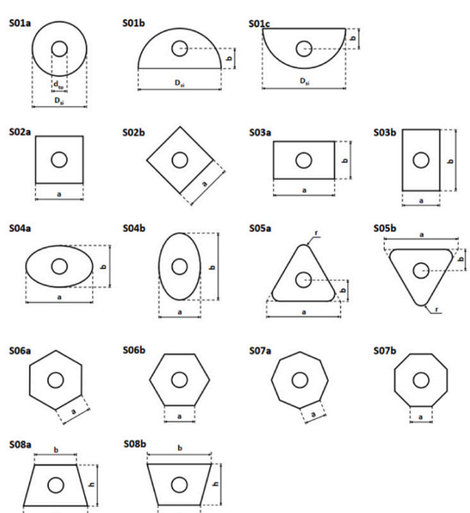
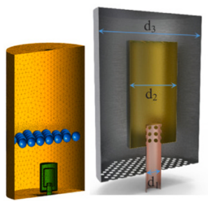
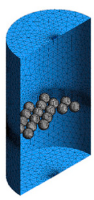
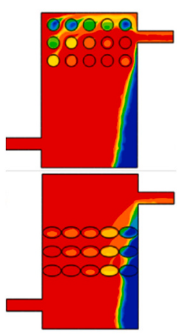
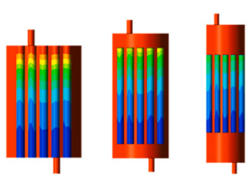
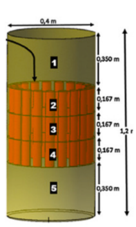
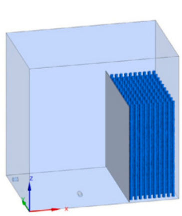
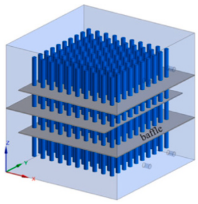
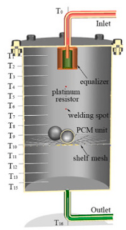
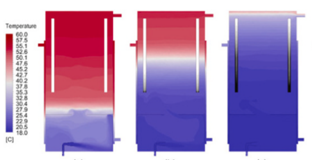
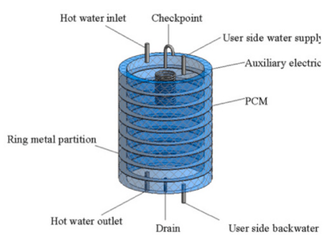
Authors	Geometry	Model Approach	PCM type	Main Research	Main Conclusions
Kursun et al. [329]		The enthalpy-porosity approach	N-eicosane	Optimizing energy storage efficiency in Double-Pipe TES through geometric variations in inner and outer channels	Enhanced energy storage efficiency through optimal inner and outer channel geometries, focusing on melting and solidification performance
Dai et al. [330]		The enthalpy-porosity method	RT35	Enhancing PCM melting and solidification rates in TES units through an improved conical inner tube configuration and a reverse layout technique	the enhanced design accelerates the melting rate but hinders heat transfer during solidification, adopting a reverse technique (180° rotation), resulting in a reduction of 42.63 % in PCM solidification time
Chatterjee et al. [331]		The heat capacity approach	lauric acid	Analyzing the influence of cavity shape and inner tube positioning on PCM melting behavior for enhanced charging rates via numerical simulations	The cavity shape and optimal positioning of the inner tube significantly influence and enhance the melting phenomenon, resulting in a notable improvement in the charging rate.
Wołoszyn et al. [332]		The enthalpy-porosity method	Organic PCM designated as RT54HC	The research investigated shell shape impact on phase-change processes in shell-and-tube structures	optimal shell shapes significantly reduce melting times by 44 % and solidification times, aiding efficiency in such systems

Table 9
Numerical studies on hybrid thermal energy storage tanks.

Authors	Geometry	Dimension type	Solver	Subject
Wang et al. [334]		3D	ANSYS FLUENT 18	Evaluating the impact of the inlet structure on thermal stratification within a heat storage tank containing PCMs.
Qin et al. [4]		3D	ANSYS FLUENT 14.5	Investigating the impact of utilizing PCM balls on the thermal properties within hot water tanks.
Fertahi et al. [335]		2D	OpenFOAM	The study examined the encapsulation of PCM within both cylindrical and elliptical capsules inside the vertical water tank.
Abdellatif et al. [336]		3D	ANSYS FLUENT 22.2	The influence of the (L/D) ratio on the extent of thermal stratification within the TES tank employing PCMs.
Bouhal et al. [337]		1D	The Fortran programming code	Exploring how integrating PCM affects the temperature distribution of water within the tank
Feng et al. [338]		3D	ANSYS FLUENT 19.0	This research involved simulating a novel design for a phase change water tank incorporating a vertical baffle.

(continued on next page)

Table 9 (*continued*)

Authors	Geometry	Dimension type	Solver	Subject
Feng et al. [339]		3D	ANSYS FLUENT	Created a latent heat storage tank by integrating baffles and employing two different PCMs with diverse phase-change temperatures
Wu et al. [340]		3D	ANSYS FLUENT19.2	Focuses on evaluating the thermal stratification performance of a newly designed DHW (Domestic Hot Water) tank featuring a porous equalizer across various operational states.
Izgi et al. [341]		3D	ANSYS FLUENT 19. 18.1	Effect of Cylindrical Encapsulated PCM in Vertical Hot Water Storage Tank on Thermal Stratification in Household Solar Water Heaters
Wu et al. [342]		2D	COMSOL Multiphysics	Structural Optimization of Phase Change Thermal Storage Tanks for Enhanced Thermal Performance

7.4. For porous PCM

Each material possesses distinct thermal properties, which play a crucial role in defining its thermal behavior. Extensive studies have been conducted to investigate the specific characteristics of each PCM or MF material [320,321]. The metal foam structure is influenced by various parameters including porosity (ε), pore density (ω), and ligament diameter (d_l). Porosity (ε) is defined as the ratio of pore volume to the total volume of pores and ligaments in the foam. Pore density (ω) represents the number of pores per linear inch (PPI), while ligament diameter (d_l) is determined based on the pore size (d_p), according to the following relation [322]:

$$\frac{d_l}{d_p} = 1.18\sqrt{\frac{1-\varepsilon}{3\pi}}\left(\frac{1}{1-e^{-\frac{1-\varepsilon}{0.04}}}\right) \tag{98}$$

where;

$$d_p = \frac{0.0254(m)}{\omega(PPI)} \tag{99}$$

The permeability denoted as K, and the inertial coefficient, represented by C_f, for the metal foam was evaluated utilizing the model introduced by Calmidi and Mahajan [322], as follows:

$$\frac{K}{d_p^2} = 0.00073(1-\varepsilon)^{-0.224}\left(\frac{d_l}{d_p}\right)^{-1.11} \tag{100}$$

$$C_f = 0.00212(1-\varepsilon)^{-0.132}\left(\frac{d_l}{d_p}\right)^{-1.63} \tag{101}$$

1. Thermal conductivity-For LTE model

The effective thermal conductivity (k_{eff}) introduced in the LTE model, as described in Eq. (102), is determined by computing the volume-averaged thermal conductivities of the MF and the PCM according to the following expression:

$$k_{eff} = (1-\varepsilon)k_{MF} + \varepsilon k_{PCM} \tag{102}$$

2. Thermal conductivity-For LTNE model

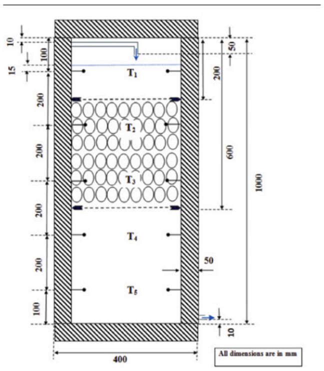
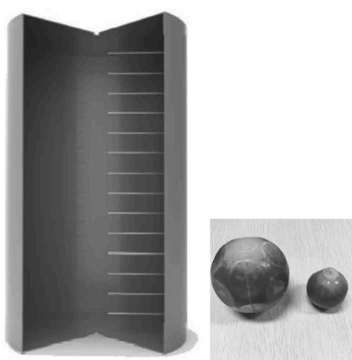
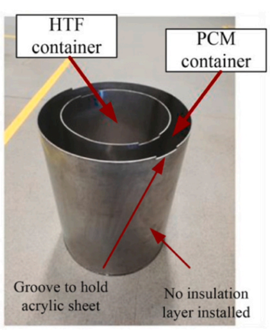
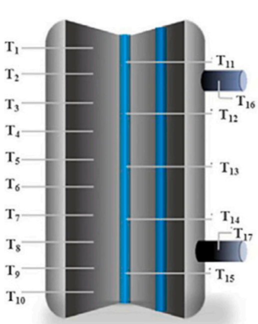
The thermal conductivity of the PCM-metal foam composite was determined using the model outlined by Tian and Zhao [323] as follows:

$$k_{eff} = \left.\frac{\sqrt{2}}{2(M_A + M_B + M_C + M_D)}\right|_{K_S=0} \tag{103}$$

where M_A, M_B, M_C, and M_D represent parameters utilized to simplify the equation. They are defined as:

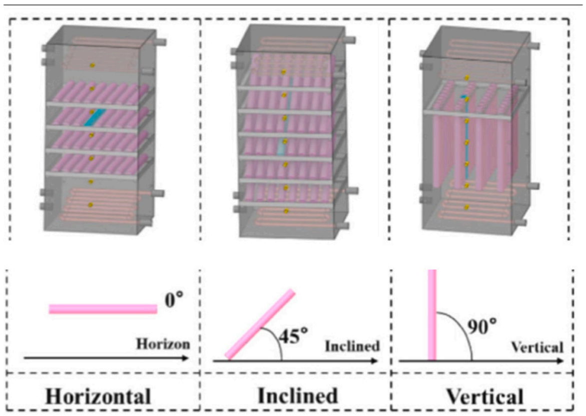
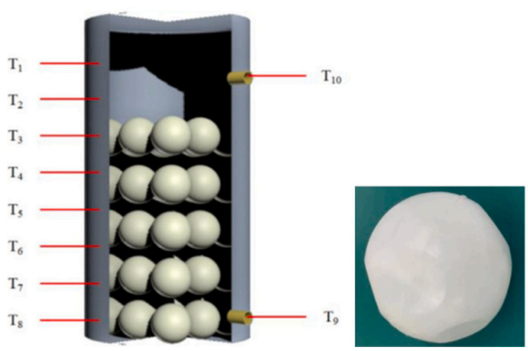
$$M_A = \frac{4\sigma}{(2e^2 + \pi\sigma(1-e))k_s + (4 - 2e^2 - \pi\sigma(1-e))k_{PCM}} \tag{104}$$

Table 10
Experimental studies on hybrid thermal energy storage tanks.

Authors	Geometry	The shape of PCM containers	Research content
Kumar et al. [343]		Elliptical containers	The effect of PCM Capsules on Hot Water Storage Tank Stratification was Analyzed via Experimental Comparison with Sensible TES, revealing shorter charging times in the latter, but improved stratification with PCM capsules at higher temperature differentials.
Wang et al. [344]		Spherical containers	This study investigates thermal stratification in a hot water storage tank utilizing PCM balls, analyzing their position impact under varying flow rates. Results show enhanced thermal stratification when balls are closer to the inlet and have smaller diameters.
Kurnia et al. [345]		The configuration involves PCM placed within the tank wall in a "layered" shape	The study explores the potential of using PCM as insulation and an energy storage medium in a Hybrid TES. PCM is placed in the tank wall, enhancing insulation and boosting heat storage capacity.
Li et al. [346]		Cylindrical containers	This research focuses on improving heat storage tank performance by incorporating phase change material (PCM). Experimental analysis reveals a 12.0 % increase in discharging time and effective water supply, with optimal conditions identified at an inlet water temperature of 70 °C and a flow rate of 15 L/h.

(*continued on next page*)

Table 10 (*continued*)

Authors	Geometry	The shape of PCM containers	Research content
Zhang et al. [347]		Cylindrical containers	The study explores a modular hybrid TES tank, altering PCM tube inclinations. Horizontal tubes demonstrate the shortest melting time, enhancing heat transfer by 7.18 % during charging, signaling the significance of tube inclination in system performance optimization.
Li et al. [348]		Spherical containers	The study examines a sodium ball-filled water tank for PCM heat storage. Optimal inlet temperature and flow rate are identified to achieve peak hot water supply and thermal efficiency. It also introduces a concept to analyze the impact of PCM ball position on heat exchange blind areas.

$$M_B = \frac{(e-2\sigma)^2}{(e-2\sigma)^2 k_s + (2e - 4\sigma - (e-2\sigma)e^2)k_{PCM}} \tag{105}$$

$$M_C = \frac{(\sqrt{2}-2e)^2}{2\pi\sigma^2(1-2e\sqrt{2})k_s + 2(\sqrt{2}-2e-\pi\sigma^2(1-2e\sqrt{2}))k_{PCM}} \tag{106}$$

$$M_D = \frac{2e}{e^2 k_s + (4-e^2)k_{PCM}} \tag{107}$$

$$\sigma = \sqrt{\frac{\sqrt{2}\left(2 - \frac{5}{8e^3\sqrt{2}} - 2\varepsilon\right)}{\pi(3 - 4e\sqrt{2} - e)}} \tag{108}$$

In this context, σ represents a characteristic of the metal foam, while e remains a constant value ($e = 0.339$).

The surface area density of metal foams (a_{sf}) represents the total surface area (m^2) of metal fibers per unit volume (m^3) and can be determined by assuming that all metal fibers possess an ideal cylindrical shape. The formula provided by Calmidi and Mahajan [322] is expressed as follows in Eq. (109):

$$a_{sf} = \frac{3\pi d_f\left[1 - e^{-\left(\frac{1-\varepsilon}{0.04}\right)}\right]}{(0.59 d_P)^2} \tag{109}$$

8. Latent and hybrid thermal energy storage

8.1. Latent heat TES

In the realm of latent heat storage, shell-and-tube heat exchangers play a pivotal role, leveraging PCMs for effective TES and release during the phase transition between solid and liquid states. The melting of PCM is primarily influenced by natural convection. The intricate design of these heat exchangers facilitates efficient heat transfer, making them crucial in applications where precise control and utilization of thermal energy are paramount. Various research studies have explored methods to enhance TES within shell-and-tube configurations, modifying either the shell, tube, or both. Table 8 illustrates some of these studies, categorizing them based on different types of shell and tube configurations.

8.2. Hybrid TES

Hybrid storage systems integrate PCMs into conventional water tanks, enhancing the overall thermal capacity of the system for improved energy storage and management. These systems combine the benefits of sensible and latent heat storage, contributing to optimizing thermal energy utilization across diverse applications, from residential heating to industrial processes [333]. Ongoing research in this field focuses on developing innovative hybrid configurations to further enhance thermal performance. Table 9 presents numerical studies on hybrid systems, while Table 10 outlines experimental studies, providing valuable insights into their design and operation. Additionally, defining and signifying dimensionless numbers in the analysis of LHTES in shell-and-tube heat exchangers and Hybrid tank systems are illustrated in

Table 11
Definitions of key dimensionless parameters used in analyzing phase change problems.

Number.	Symbol	Formula	determination	Ref
Nusselt number	Nu	$Nu = \frac{hd}{k}$	Nu is the ratio of conductive thermal resistance to convective thermal resistance, indicating the proportion of actual heat transferred by a moving fluid compared to the heat transfer that would occur by conduction.	[349,350]
Stefan number	Ste	$Ste = \frac{C_p \Delta T}{L}$	Ste. is the ratio of the thermal capacity of the melted solid to the latent heat.	[351]
Fourier number	Fo	$Fo = \frac{\alpha t}{l^2}$	The Fourier number characterizes the heat flux into a body or system.	[349]
Biot number	Bi	$Bi = \frac{hl}{k}$	The Biot number represents the ratio of conductive to convective heat transfer resistance and determines the uniformity of temperature in a solid.	[352]
Rayleigh number	Ra	$Ra = \frac{g\beta \Delta T l^3}{\alpha \nu}$	The Rayleigh number determines the onset of convection. Below a critical value, heat transfer is primarily driven by conduction.	[350]
Grashof number	Gr	$Gr = \frac{gB\Delta T l^3}{\nu^2}$	Approximates the ratio of buoyancy force to the viscous force.	[353]
Prandtl number	Pr	$Pr = \frac{\nu}{\alpha}$	Prandtl number approximates the ratio of momentum diffusivity to thermal diffusivity. A low Pr indicates effective heat conduction dominated by thermal diffusivity, while a high Pr suggests effective heat convection dominated by momentum diffusivity.	[350]
Reynolds number	Re	$Re = \frac{uL}{\nu}$	Indicating the balance between inertial and viscous forces is crucial for distinguishing between laminar and turbulent flow.	[354]
Stratification number	Str	$Str = \frac{(\partial T/\partial y)_t}{(\partial T/\partial y)_{t=0}}$	The ratio of the mean of the temperature gradients at each time interval to the temperature gradient at the start of the charging process	[355]
Richardson number	Ri	$Ri = \frac{g\beta H(T_{top} - T_{bottom})}{v_{sf}^2}$	The Richardson number serves as a comprehensive indicator for evaluating the stratification performance of storage systems.it is given by the ratio of buoyancy forces to mixing forces.	[343]
Peclet number	Pe	$Pe = \frac{vH}{\alpha}$	The Peclet number establishes a connection between overall heat transfer and conductive heat transfer, considering both convective and conductive mechanisms. It is often utilized alongside the Richardson number to characterize stratification in storage tanks.	[356,357]
Mix number	Mix	$Mix = \frac{M_{stratified} - M_{exp}}{M_{stratified} - M_{full-mixed}}$	The MIX number is employed to depict thermal stratification within a water tank at a specific time. It spans from 0 to 1, signifying varying degrees of thermal stratification. A MIX value of 0 indicates a perfectly stratified water tank, whereas MIX = 1 denotes a fully mixed water tank.	[358]
Charging efficiency	η_{ch}	$\eta_{ch} = \frac{T_i - T_o(t)}{T_i - T_{ini}}$	The TES system's charging efficiency is the proportion of instantaneous heat transfer to the maximum heat transfer at a given temperature and flow rate of the HTF at the inlet	[359]
Thermal storage efficiency	$\eta_{storage}$	$\eta_{storage}(t) = \frac{T_{avg}(t) - T_{ini}}{T_i - T_{ini}}$	thermal storage efficiency of the charging process as the actual energy change at time t divided by the maximum energy change.	[360]

Table 11.

9. Applications of PCMs in various fields

Thermal energy holds significant importance across various domains, spanning from power generation to regulating environmental conditions. TES devices find applicability in systems involved in thermal energy generation or transfer, with storage advantages tailored to specific applications. These advantages encompass enhancements in energy efficiency, reductions in emissions, load management, and thermal regulation, ensuring operational safety and comfort within designated temperature ranges [48]. In response to the projected 28 % increase in global energy demand by 2050 [361], PCMs have garnered significant attention. Several studies [111,362–364] have meticulously examined the merits and demerits of PCMs (Fig. 24), emphasizing their merits, such as compact size, high latent heat, chemical stability, and considerable energy storage density. These qualities have propelled PCMs into a wide array of applications, including building [365], food storage [366], energy storage [367,368], agriculture greenhouse [369], cooling and refrigeration [370], smart textiles [371], solar systems [372], heat sinks [278,373,374], batteries [375], and waste heat recovery [376], as illustrated in Fig. 25. This diversified utilization underscores the potential of PCMs in addressing the multifaceted challenges associated with energy conservation, sustainability, and climate adaptation. The thermal energy generated in industrial processes is often localized and time-sensitive, necessitating efficient storage and distribution methods [377]. For nearby consumers, integrating a district heating system with LHTES offers a reliable and cost-effective solution for heat utilization [378]. Conversely, for distant users, a mobile LHTES system presents an innovative approach for capturing and transporting low-temperature industrial waste heat via vehicles [379]. Therefore, leveraging PCMs with high thermal energy density enables the rapid establishment of district heating and network-based energy distribution systems for long-term storage and distribution.

10. Summary

In conclusion, this comprehensive review underscores the critical role of PCMs in efficient energy storage, emphasizing the need for innovative solutions to address their unsteady nature. PCMs offer a promising avenue for energy storage, particularly in renewable energy systems where intermittent availability poses significant challenges. By effectively storing surplus energy during peak production periods and releasing it when needed, PCMs contribute to stabilizing energy supply and demand dynamics, thereby enhancing overall system reliability and resilience. The study thoroughly explores TES technologies, with a particular focus on latent heat storage and hybrid systems. Latent heat storage, in which energy is absorbed or released during phase transitions, offers a high energy density solution with minimal volume variation, making it ideal for various thermal applications. Additionally, hybrid systems that integrate PCMs with other storage technologies provide enhanced flexibility and performance, further expanding the applicability of PCM-based energy storage solutions. The review provides valuable insights into the classification, selection criteria, and applications of PCMs across different temperature ranges. Understanding these aspects is crucial for effectively leveraging PCM technology in diverse contexts, ranging from residential heating to industrial processes. Moreover, the study delves into innovative methods aimed at enhancing PCM performance, including the use of fins, nanoparticles, and encapsulation techniques. These approaches not only improve energy storage efficiency but also address practical challenges such as thermal stability and scalability. Numerical simulation methods, particularly the enthalpy, enthalpy-porosity, and heat capacity methods, are identified as pivotal tools for understanding and optimizing TES systems. By employing these simulation techniques, scientists and engineers can model complex thermal processes, optimize system design, and predict performance under various operating conditions. Furthermore, the emphasis on mesh quality parameters highlights the importance of precise meshing for accurate and reliable simulation results, ensuring confidence in the predictive capabilities of numerical models. The inclusion of summarizing tables consolidates key studies and findings, enhancing the practical utility of the review for scientists and practitioners in sustainable energy systems. These tables serve as valuable reference resources, facilitating easy access to relevant literature and enabling informed decision-making in the design and implementation of PCM-based energy storage systems. Overall, this study provides a comprehensive overview of PCM-based energy storage technologies and offers a roadmap for future advancements in efficient energy storage. By addressing critical challenges and highlighting emerging opportunities, the review contributes to advancing the state-of-the-art in sustainable energy systems and accelerating the transition towards a more resilient and renewable energy future.

11. Conclusion

1. The role of PCMs in energy storage is crucial, necessitating innovative solutions for their inherent unsteadiness and nonlinearity.
2. A comprehensive exploration of TES technologies, particularly in latent heat storage and hybrid systems, contributes to optimizing PCM-based energy storage.
3. Innovative methods to enhance PCM performance, including the use of fins, multiple PCM arrangements, nanoparticles, porous metal matrices, encapsulation, and shape stabilization, demonstrate a holistic approach to addressing PCM challenges.
4. Numerical simulation methods play a pivotal role in the modeling of mushy zones. Specific methods detailed for this purpose include the Enthalpy, Enthalpy-Porosity, LBM method, and Heat Capacity methods.
5. The critical importance of mesh quality, considering parameters like skewness, orthogonal quality, and element quality, underscores the need for accuracy in numerical simulations.
6. Integrating nanoparticles into PCMs enhances thermal efficiency, altering properties such as density, specific heat, latent heat, and viscosity.
7. Hybrid NePCM exhibits altered thermophysical properties due to incorporating hybrid nanoparticle additives, impacting density, specific heat, viscosity, latent heat, thermal conductivity, and thermal expansion coefficient.
8. Understanding the thermophysical properties of hybrid NePCM is essential for designing and optimizing efficient energy storage systems.
9. EPCM exhibits distinct thermophysical properties due to its core-shell configuration, influencing characteristics such as density, specific heat, and thermal conductivity.
10. Understanding the thermophysical properties of the porous medium is essential for designing and optimizing energy storage systems that utilize PCMs for enhanced heat transfer and thermal management.

The originality of this review lies in its unique integration of theoretical, numerical, and experimental perspectives to provide a comprehensive framework for PCM-based TES optimization. Unlike existing reviews, this study not only categorizes enhancement methods but also evaluates their applicability in hybrid TES configurations. Moreover, it presents a structured comparison of various numerical modeling techniques, offering engineers a direct reference for selecting appropriate simulation tools. By addressing the limitations of previous research and incorporating key experimental insights, this review bridges the gap between laboratory-scale studies and real-world engineering applications, making it a valuable resource for researchers, industry professionals, and policymakers.

Based on the limitations identified in various sections of this review, further research should prioritize several key areas to enhance TES technologies. For sensible heat storage (SHS), future studies should explore innovative methods to increase energy storage capacity without relying solely on temperature rise. In the context of PCM phase transitions, optimizing fin design and configuration under different operational conditions remains a critical area for improvement. Additionally, while employing multiple PCM stages enhances TES system performance, research should determine the optimal number of stages to maximize efficiency gains. Addressing the reduction in latent heat due to nanoparticle incorporation and improving the realism of numerical simulations for complex structures like metal foams are also essential. Moreover, developing robust, cost-effective, and long-lasting encapsulation materials for PCMs will significantly enhance practical applications. Finally, refining numerical methods to accurately model phase changes and account for the flow behavior of melted PCM will improve the precision and reliability of simulations, contributing to better system designs and performance predictions.

The current review of advanced TES technologies, particularly latent heat storage and hybrid systems, has significant implications for industry, policymakers, and researchers. The findings can guide the development of more efficient TES systems, reduce operational costs, and enhance energy security. For countries and regions, adopting advanced TES technologies aids in achieving climate change targets by improving energy efficiency and reducing greenhouse gas emissions. Policymakers can use these insights to develop regulations and standards for safe and effective TES implementation. Economically and environmentally, these advancements support renewable energy integration and contribute to international climate targets, aligning with UN SDGs related to affordable and clean energy, industry innovation, and climate action.

12. Future prospects

1. Explore emerging numerical simulation techniques that go beyond traditional methods, such as machine learning algorithms or artificial intelligence, to enhance accuracy and efficiency in modeling mushy zones. This includes investigating the potential of incorporating data-driven approaches to improve predictive capabilities and reduce computational costs.
2. Investigate the potential of integrating PCMs with other advanced energy storage technologies, creating hybrid systems that could offer improved overall performance and versatility. This involves

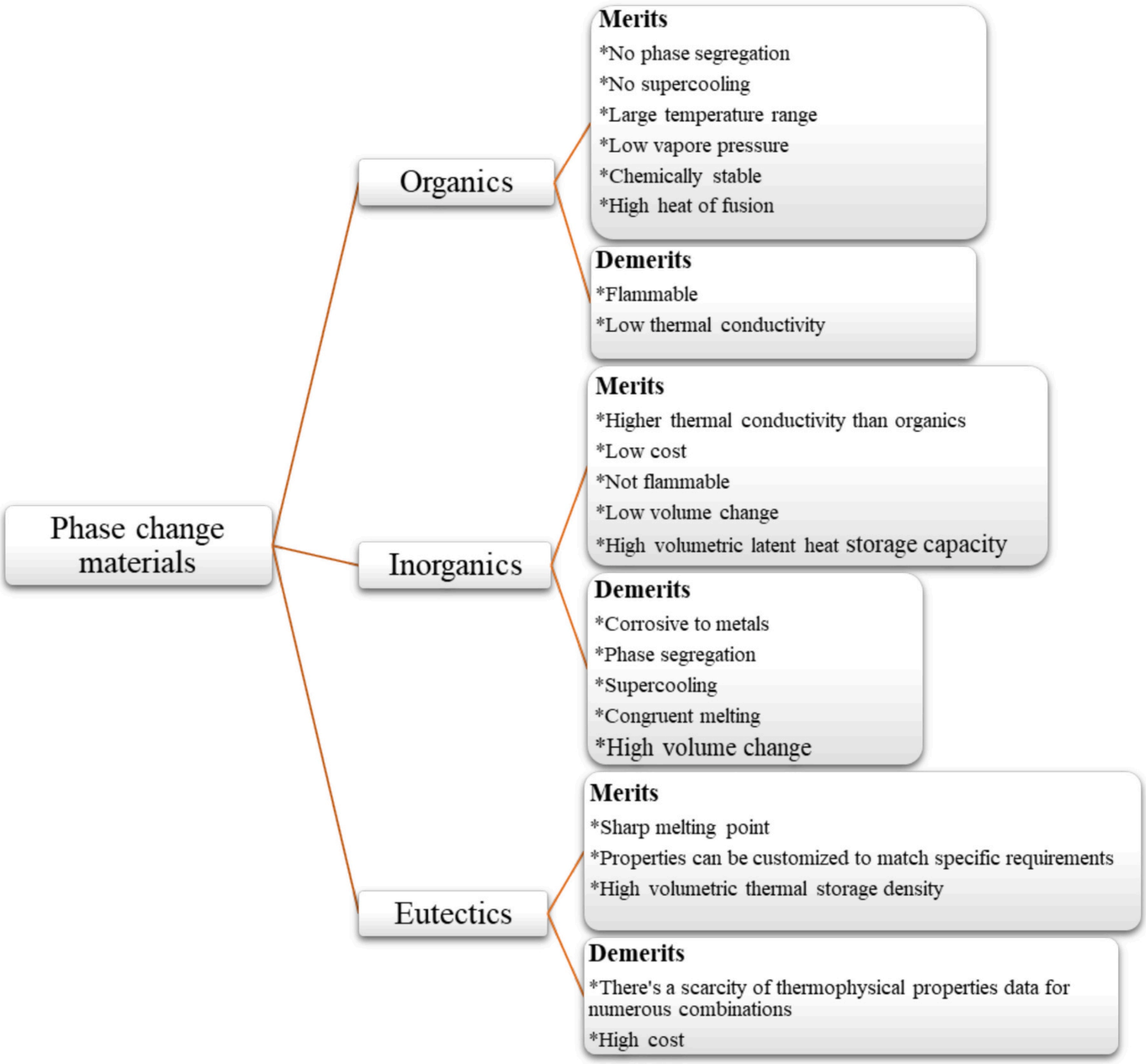


Fig. 24. Overview of the merits and demerits of three distinct PCM categories [111,362–364].

exploring synergies between PCM-based storage and complementary technologies like TES, batteries, or renewable energy sources to optimize energy management and grid stability.

3. Develop and assess optimization strategies for PCM-based energy storage systems, considering factors like material composition, system design, and operational parameters to maximize efficiency. This includes conducting comprehensive techno-economic analyses to identify the most cost-effective and sustainable solutions for various applications and geographical locations.
4. Conduct practical experiments and case studies to validate numerical models and assess the real-world performance of PCM systems in diverse applications, from residential to industrial settings. This involves collaborating with industry partners to deploy pilot projects and gather empirical data on system performance under different operating conditions and environmental factors.
5. Evaluate the environmental impact of PCM-based systems, considering factors such as material sourcing, manufacturing processes, and end-of-life disposal, to ensure sustainable and eco-friendly energy storage solutions. This requires conducting life cycle assessments (LCAs) to quantify the environmental footprint of PCM technologies and identify opportunities for improvement in terms of resource efficiency and waste management.
6. Investigate methods for scaling up PCM-based energy storage systems to meet the demands of commercial applications, addressing challenges related to cost-effectiveness, scalability, and reliability. This includes exploring innovative manufacturing techniques, modular system designs, and deployment strategies to facilitate the widespread adoption of PCM technologies in mainstream energy markets.

By delving into these prospective avenues, future research can contribute to the continual evolution of PCM-based energy storage, fostering innovation and addressing practical challenges for broader and more impactful applications. However, key challenges remain, including optimizing PCM properties for specific applications, overcoming thermal stability issues, and integrating PCM systems into existing infrastructure seamlessly. Collaboration between scientists, industry stakeholders, and policymakers will be crucial in overcoming these challenges and unlocking the full potential of PCM-based energy storage. Fig. 26 illustrates future studies and some recommendations.

The findings of this review have direct engineering applications in optimizing PCM-based TES systems across various industries. The insights into enhancement techniques (e.g., fins, nanoparticles, porous matrices, encapsulation) provide engineers with practical solutions to improve thermal performance and reliability in HVAC, industrial waste heat recovery, and renewable energy storage. Additionally, the comparative evaluation of numerical simulation methods offers a guideline for selecting appropriate modeling tools, reducing

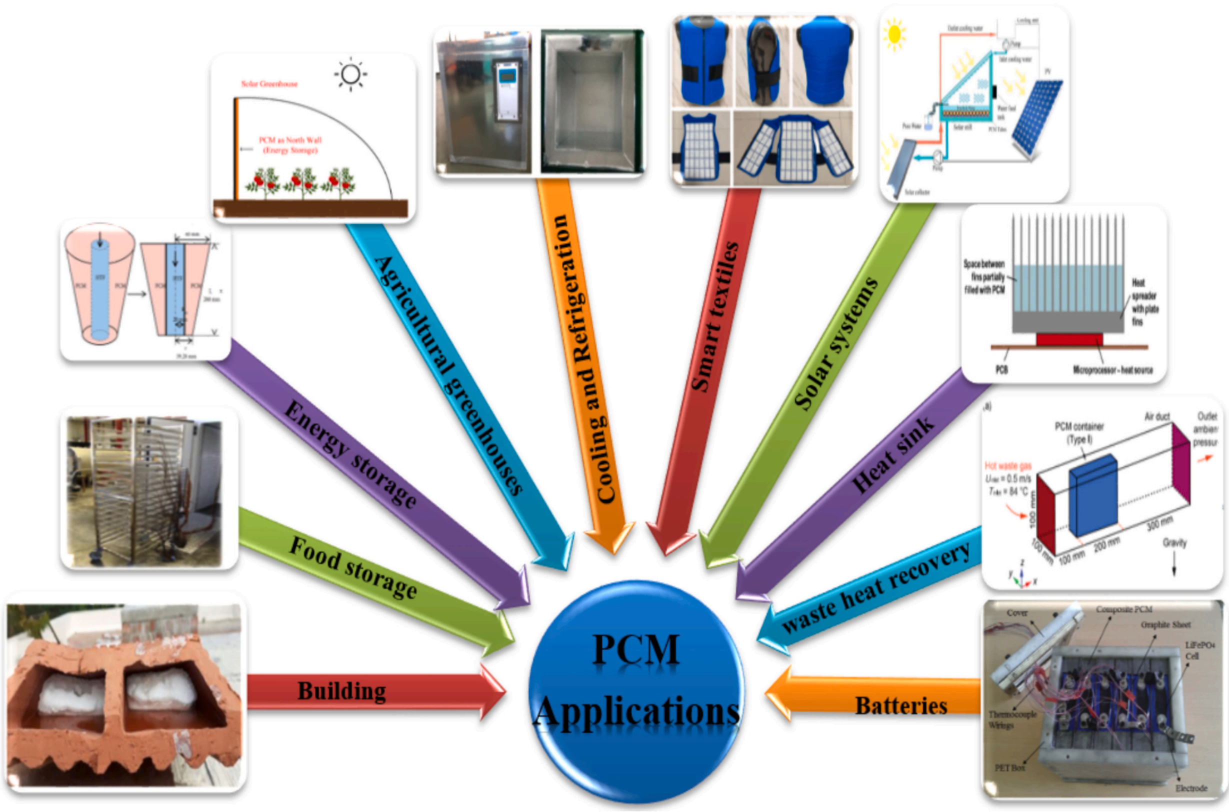


Fig. 25. Applications of phase change materials.

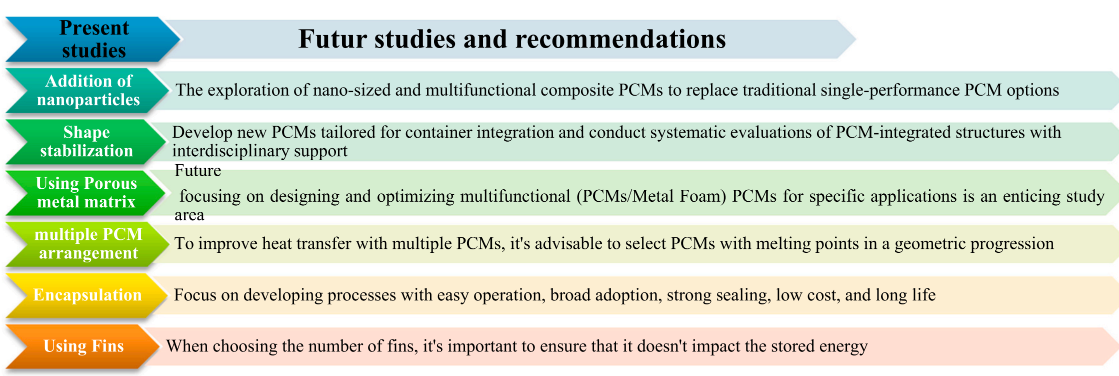


Fig. 26. Future research directions in PCM Integration.

computational costs while ensuring accurate system design.

From an engineering perspective, integrating PCM-based TES with existing energy systems remains a key challenge, particularly in terms of scalability, cost-effectiveness, and thermal stability. This review addresses these challenges by outlining optimization strategies that enhance system performance while ensuring feasibility for commercial and industrial applications. Furthermore, the study highlights the importance of life cycle assessment (LCA) to help engineers and decision-makers assess the sustainability and economic viability of PCM technologies before large-scale implementation.

To bridge the gap between theory and application, future research should focus on real-world pilot projects that validate the numerical findings presented in this review. Collaborations between research institutions, industry stakeholders, and policymakers will be essential to accelerate adoption and standardize PCM-based TES systems for broader engineering applications.

CRediT authorship contribution statement

Houssam Eddine Abdellatif: Writing – original draft, Resources, Investigation, Formal analysis, Data curation, Conceptualization, Visualization, Writing – review & editing. **Ahmed Belaadi:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Visualization, Writing – original draft, Resources. **Adeel Arshad:** Writing – review & editing, Visualization, Supervision, Resources, Project administration, Conceptualization, Writing – original draft. **Mostefa Bourchak:** Writing – review & editing, Funding acquisition, Project administration, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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